

3D CAD 2

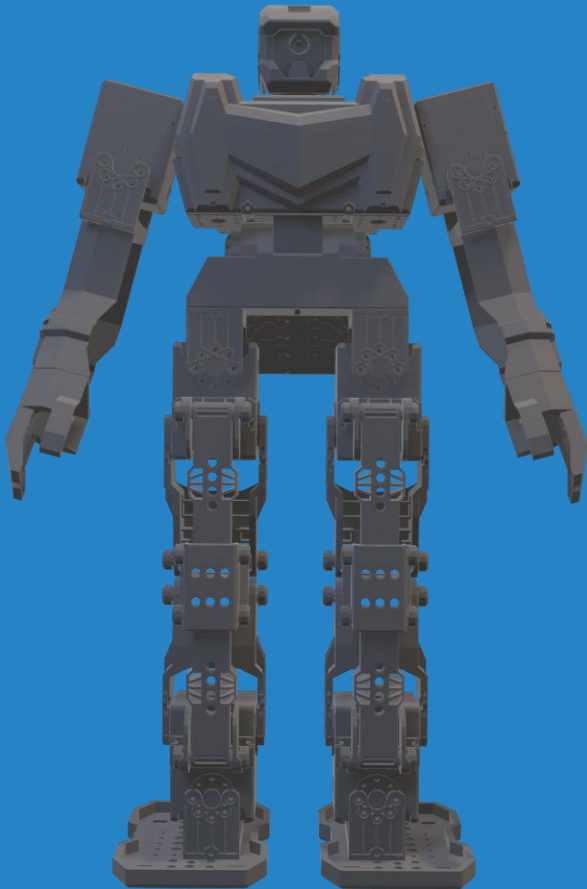
Final Project

- Reverse Engineering -

2019971056 유찬영

2021959048 양민석

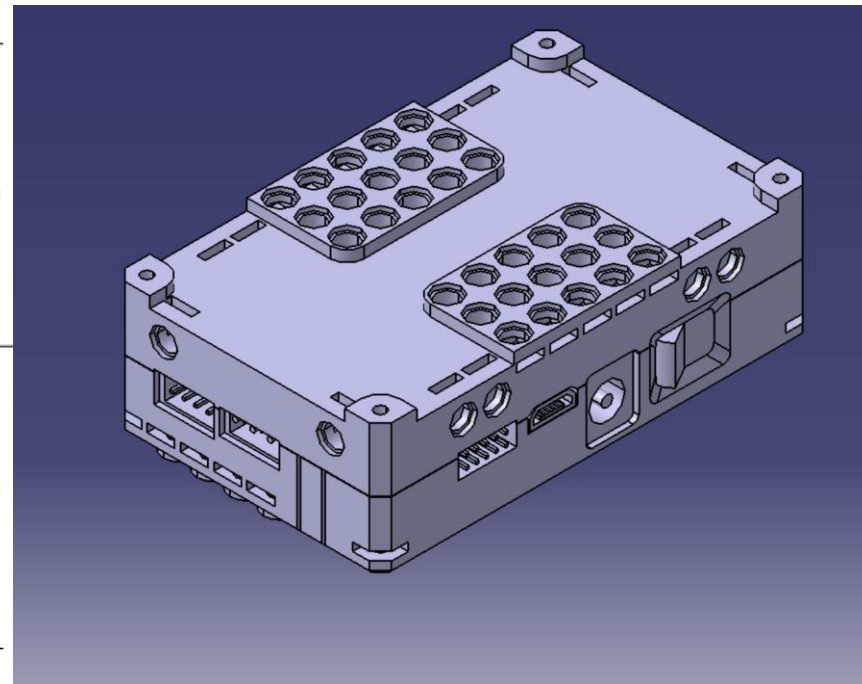
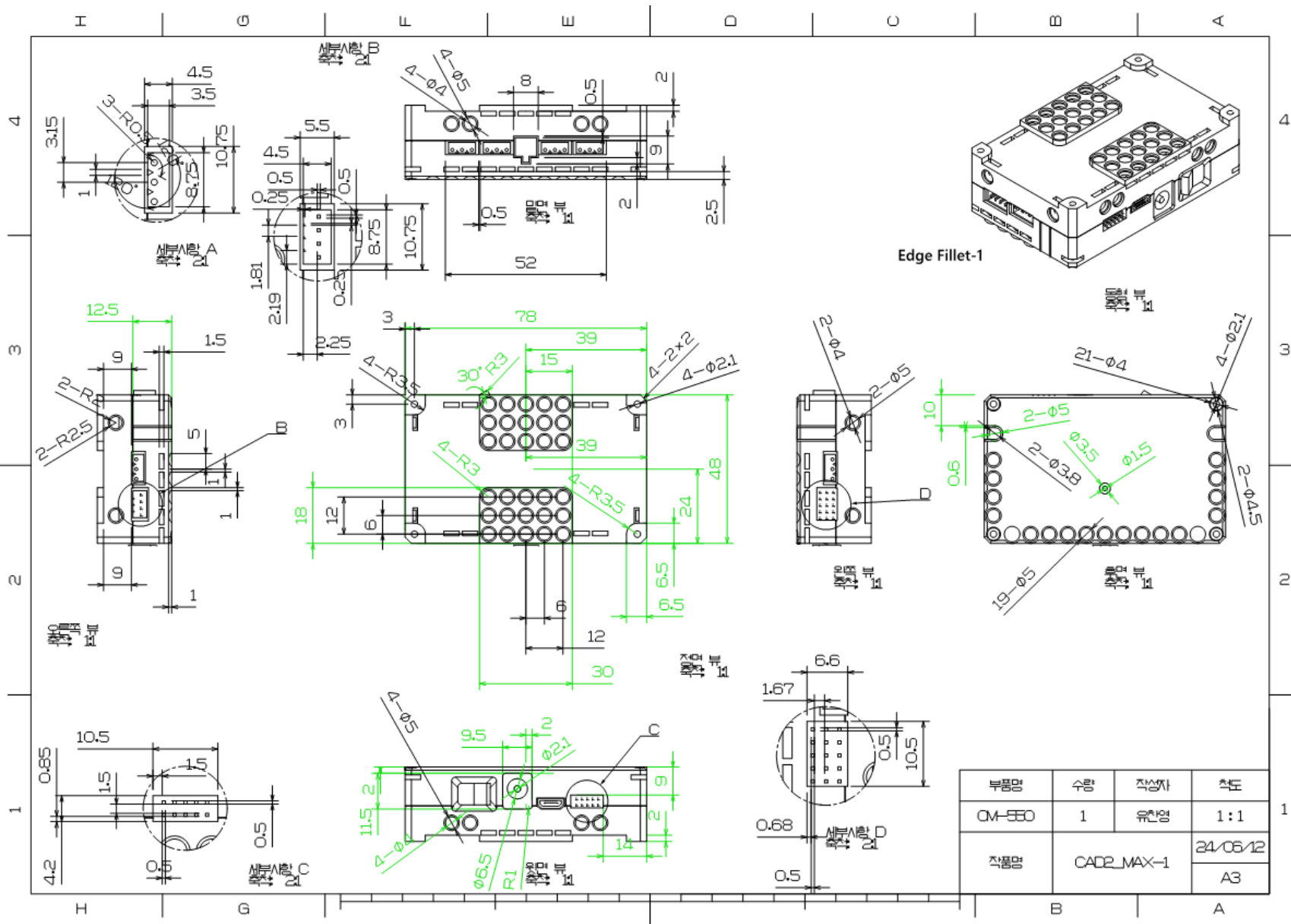
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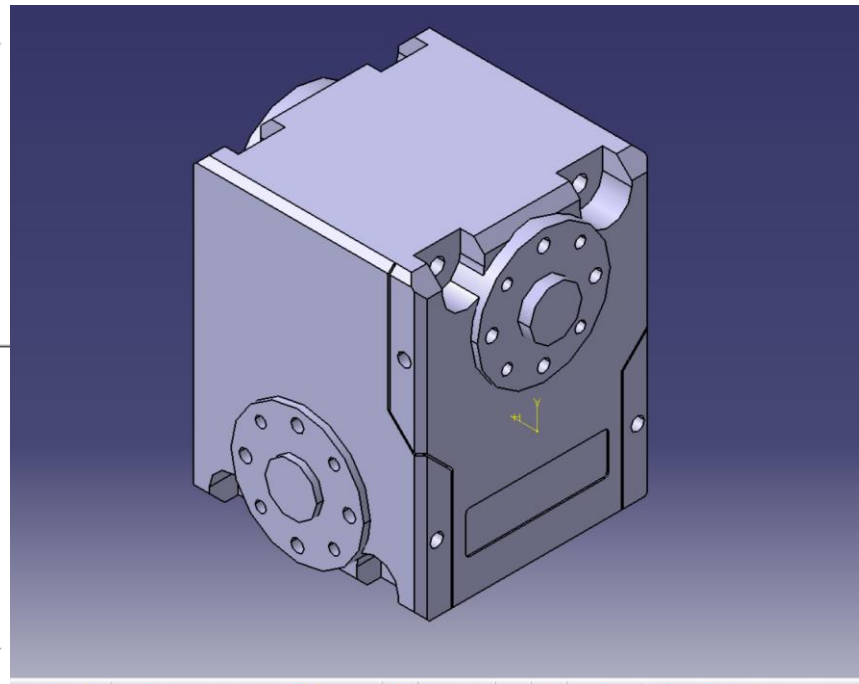
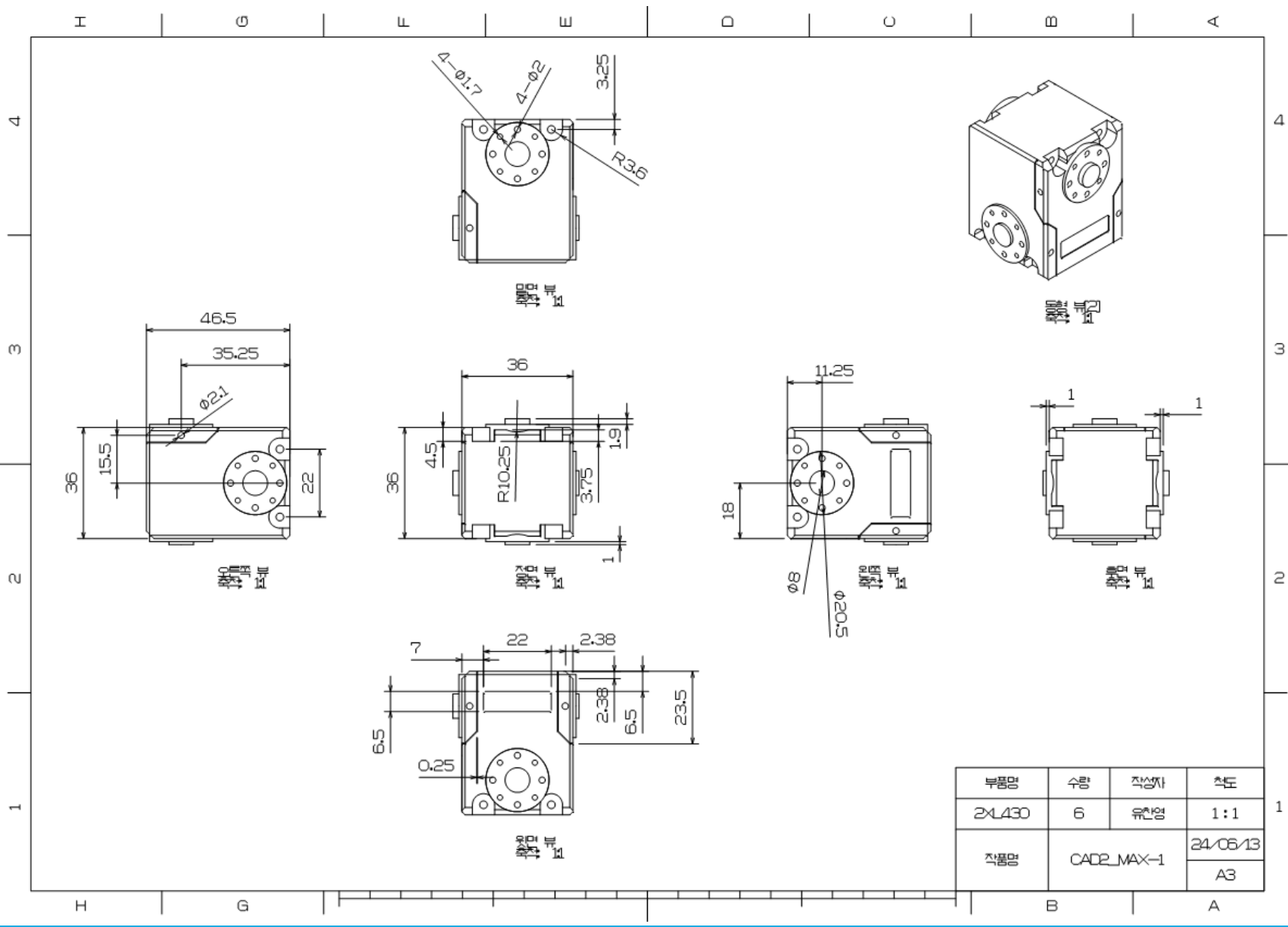


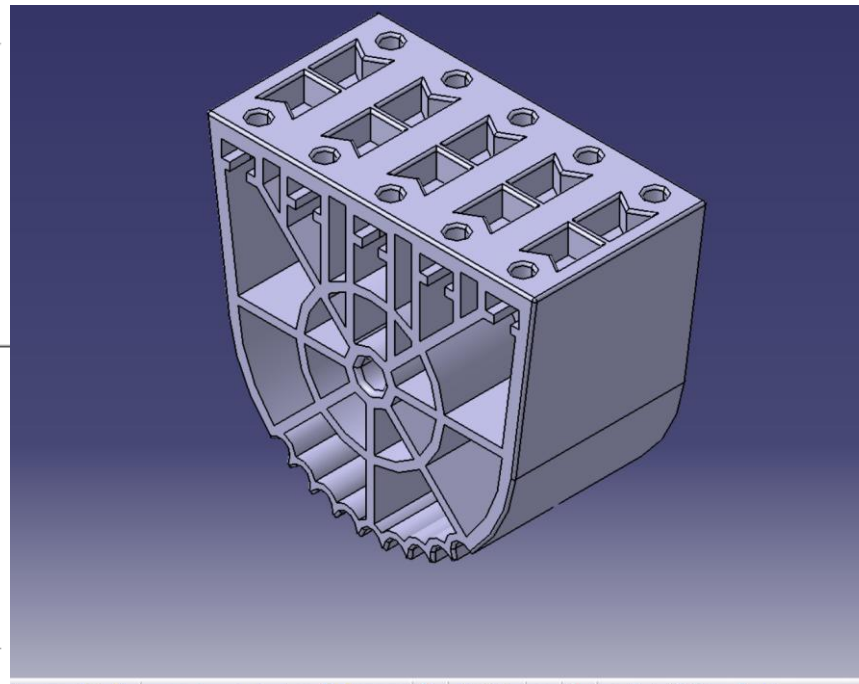
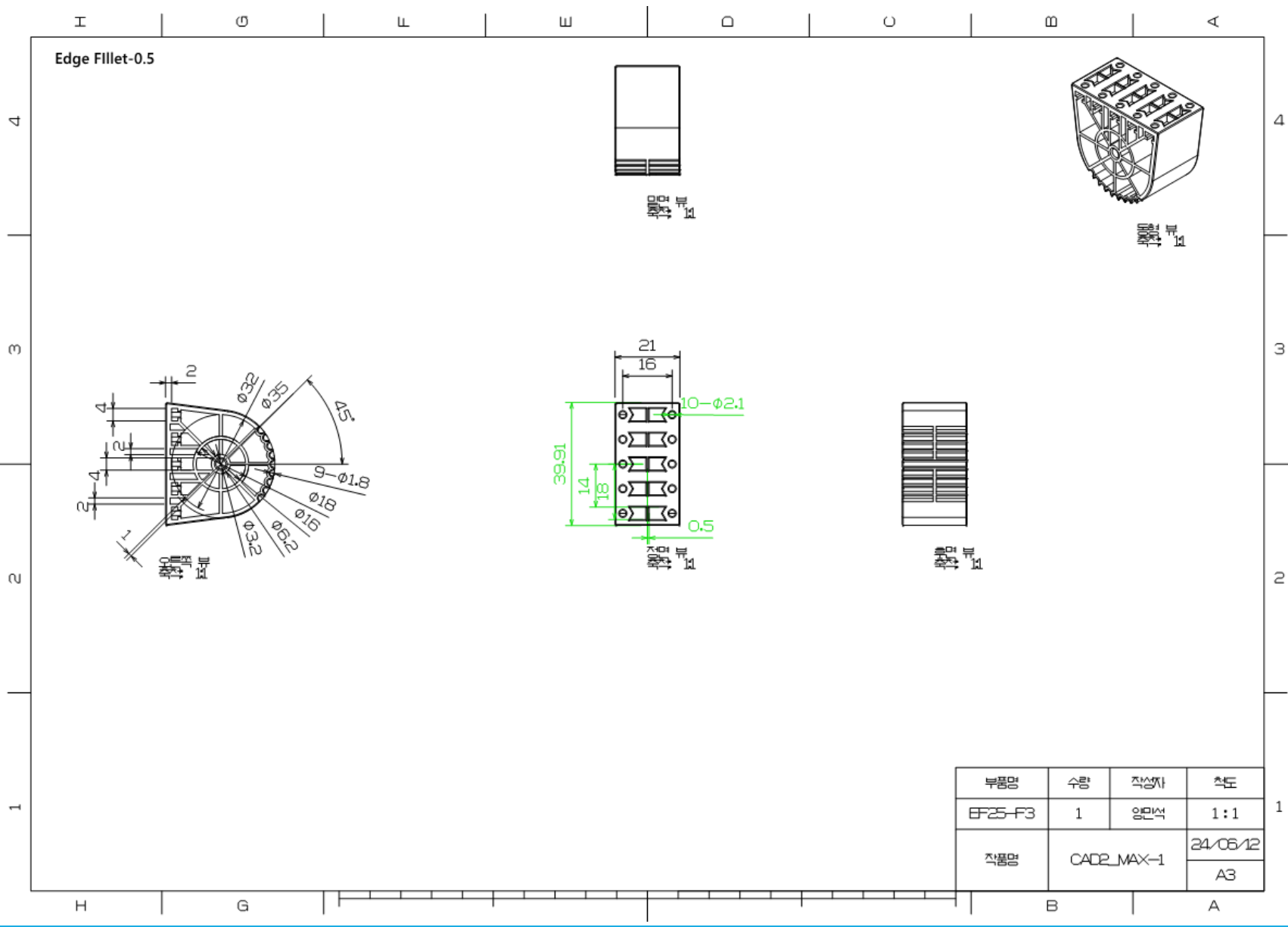
1. Parts Introduction
2. Comparison
3. Results
(CATIA & Rendering)
4. Crisis
5. Takeaway
6. Conclusion
7. Q&A

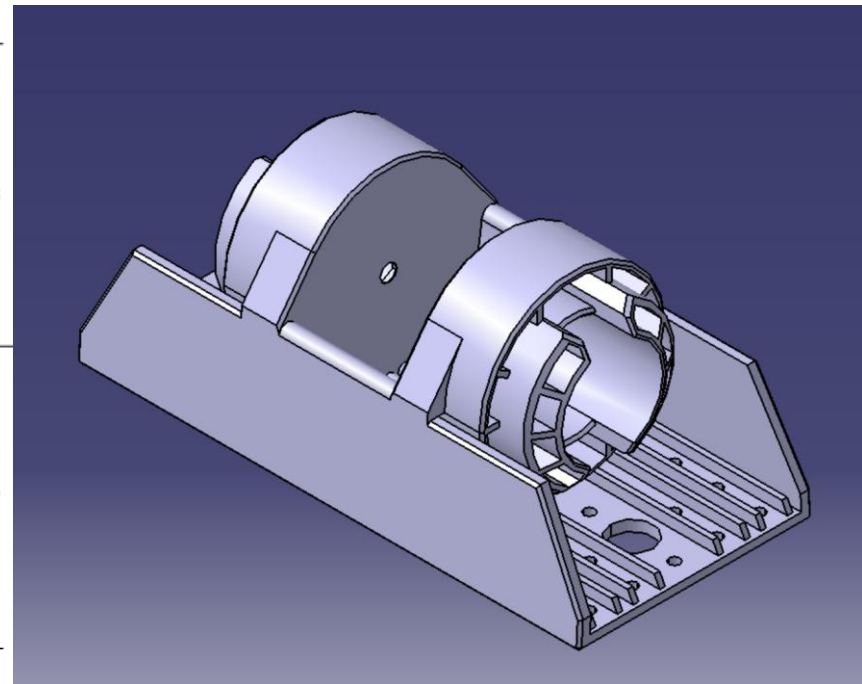
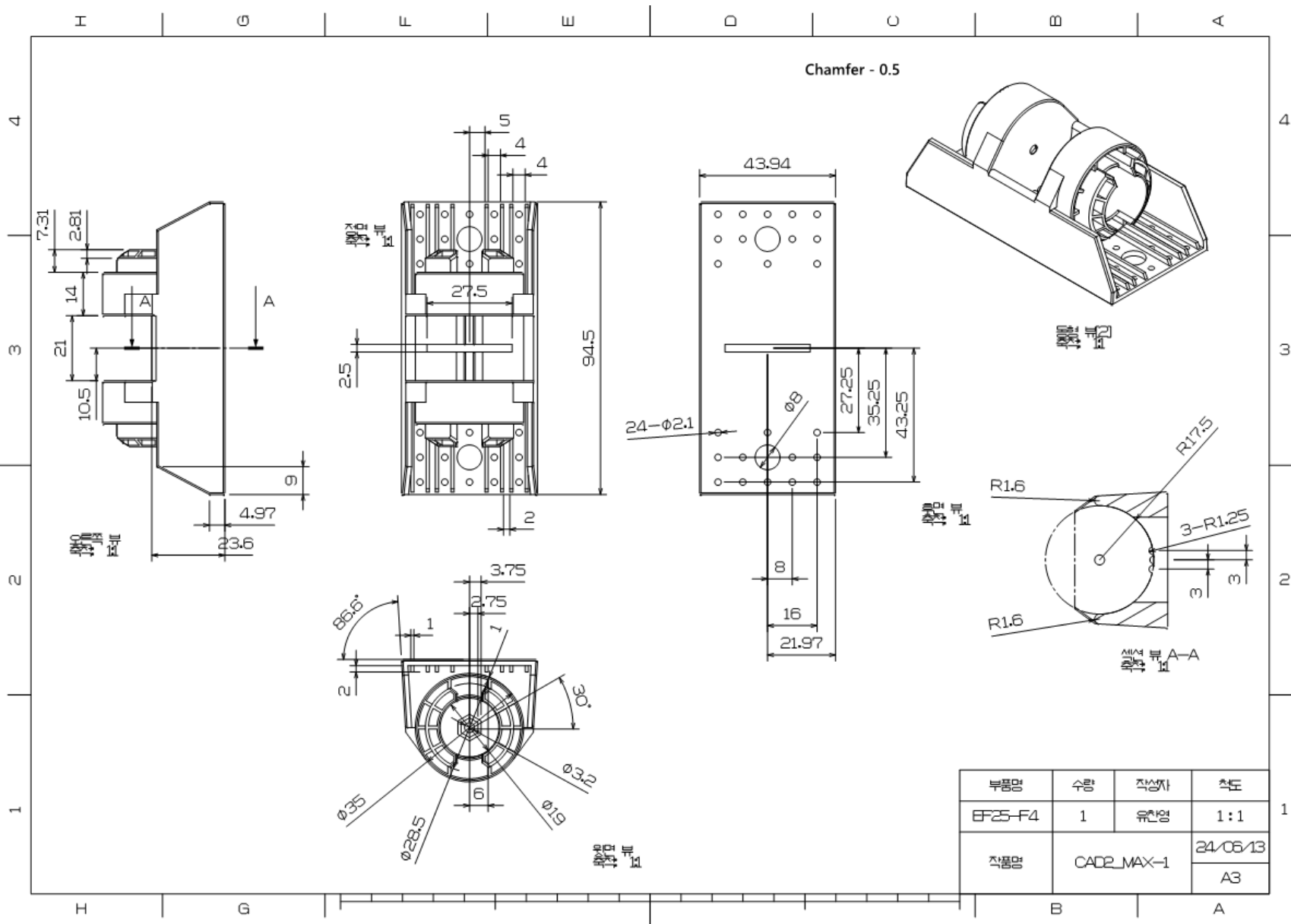
ROBOTIS ENGINEER MAX-E1 Parts Lists

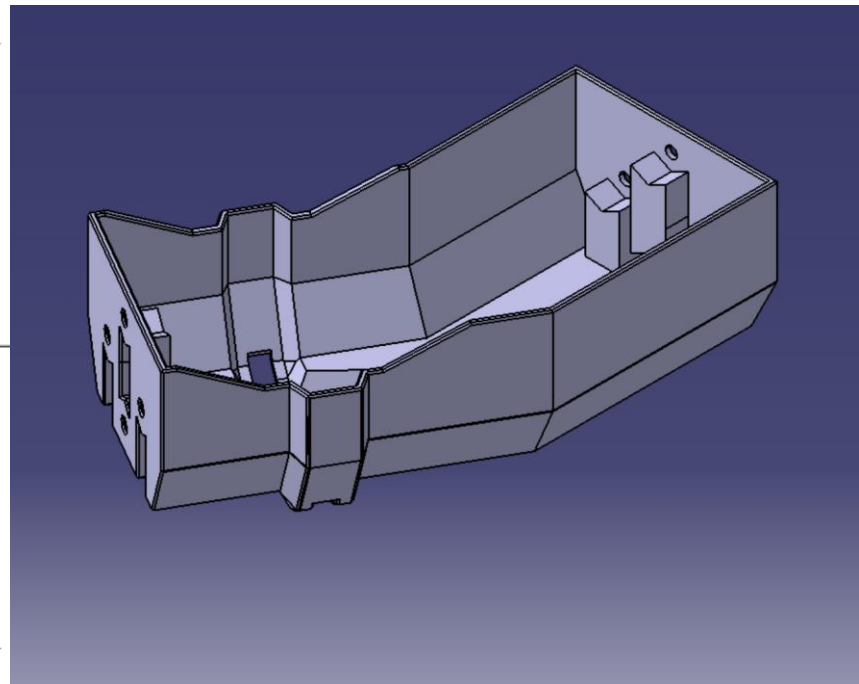
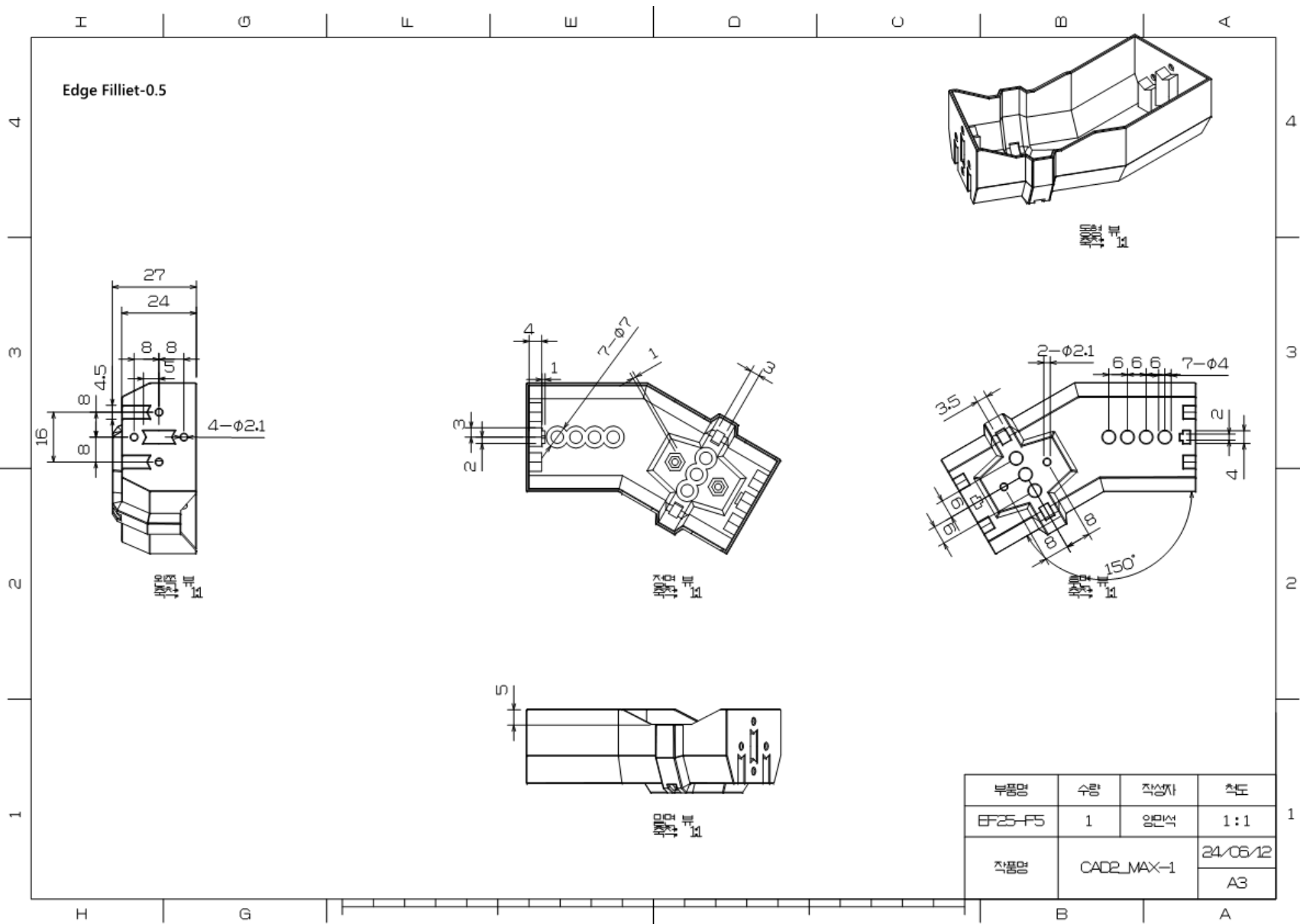
Description(품명)	Quantity(수량)	Remarks(비고)	Description(품명)	Quantity(수량)	Remarks(비고)
2XL430-W250-T	6	Right Arm x 1 Left Arm x 1 Right Leg x 2 Left Leg x2	EF25-F9	4	Right Leg x 2 Left Leg x 2
XL-320D	1	Head x 1	EF25-F10	2	Right Leg x 1 Left Leg x 1
CM-550	1	Back x 1	EF25-F11	4	Right Leg x 2 Left Leg x 2
EF25-F1	1	Head x 1	EF25-F12	4	Right Leg x 2 Left Leg x 2
EF25-F2	2	Chest x 1 Back x 1	EF25-F13	24	Right Leg x 12 Left Leg x 12
EF25-F3	1	Back x 1	EF25-F14	6	Arms x 2 Body x 2 Legs x 2
EF25-F4	1	Body x 1	EF25-F15	2	Right Leg x 1 Left Leg x 1
EF25-F5	1	Right Arm x 1	EF25-F26	1	Back x 1
EF25-F6	1	Left Arm x 1	EF25-F27	1	Back x 1
EF25-F7	1	Right Arm x 1	EF25-F28	1	Chest x 1
EF25-F8	1	Left Arm x 1	Total - 21	66	

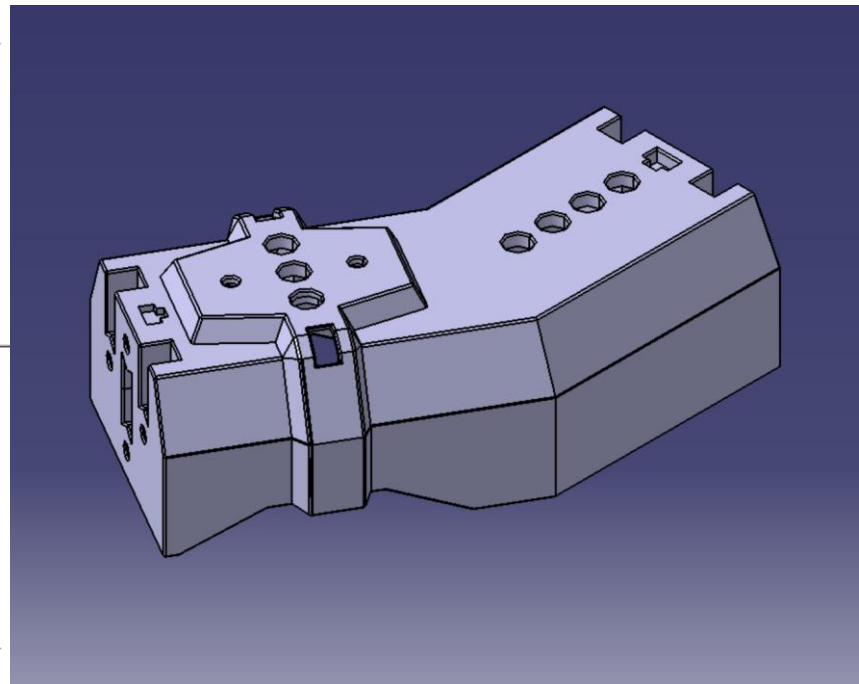
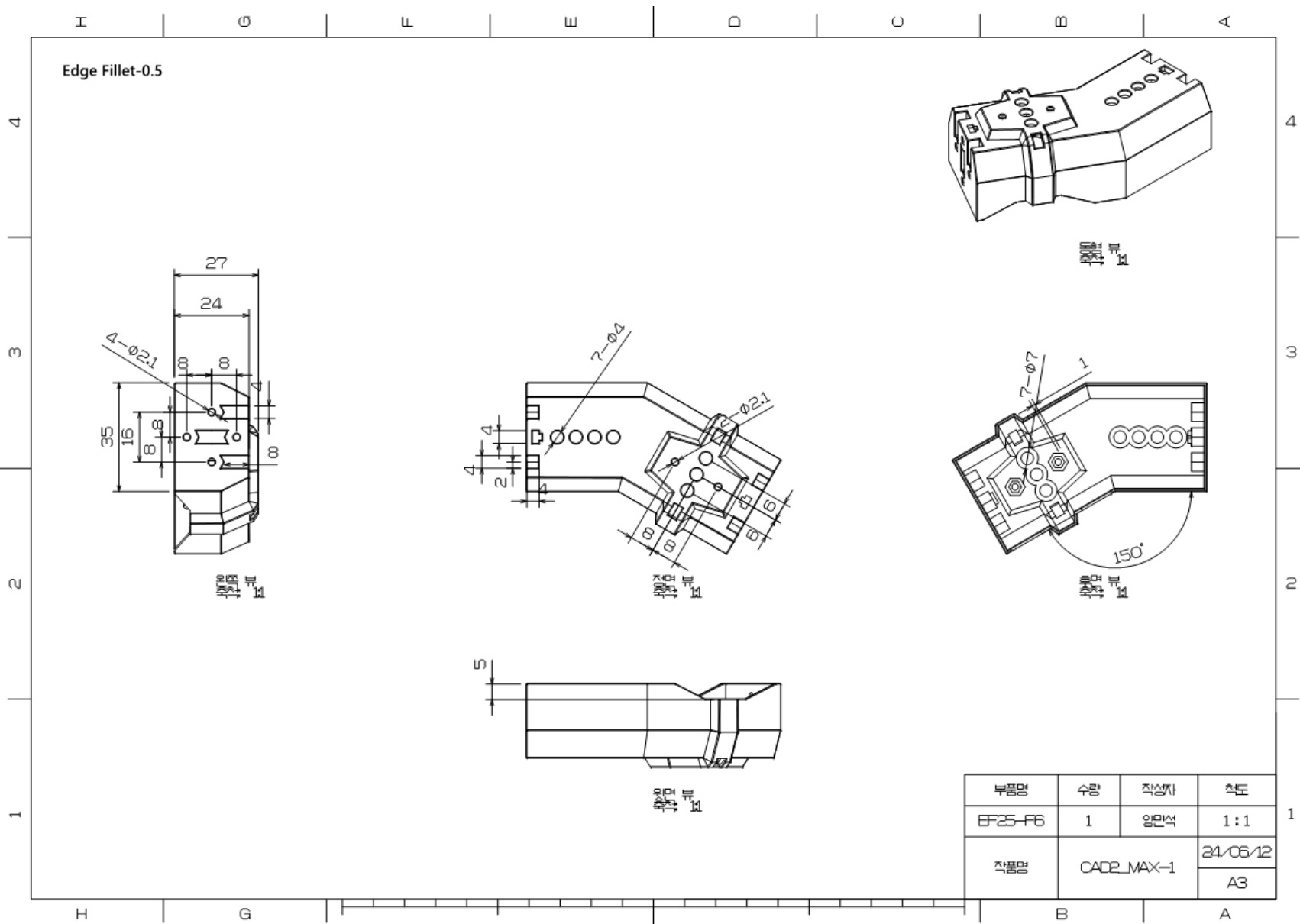


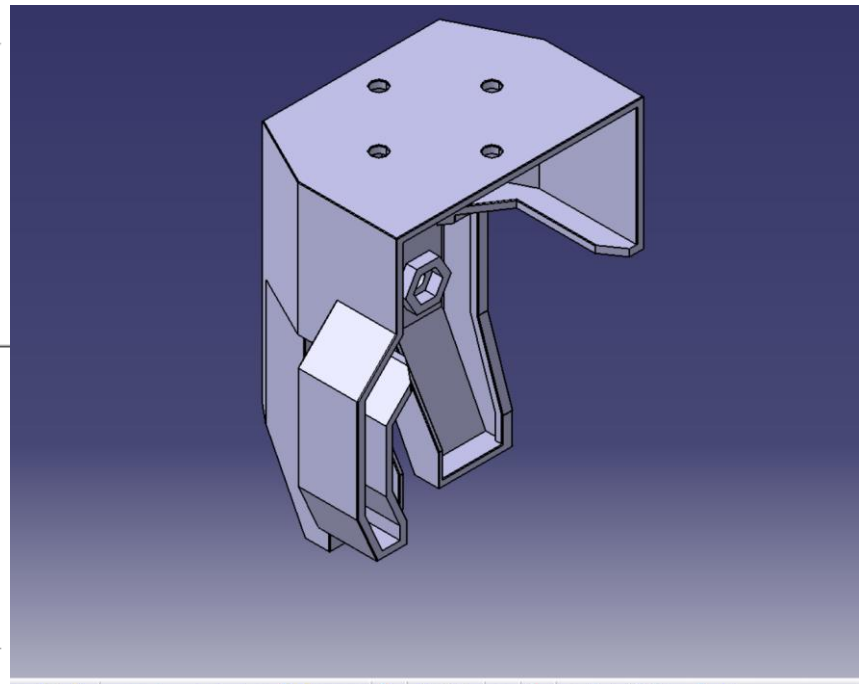
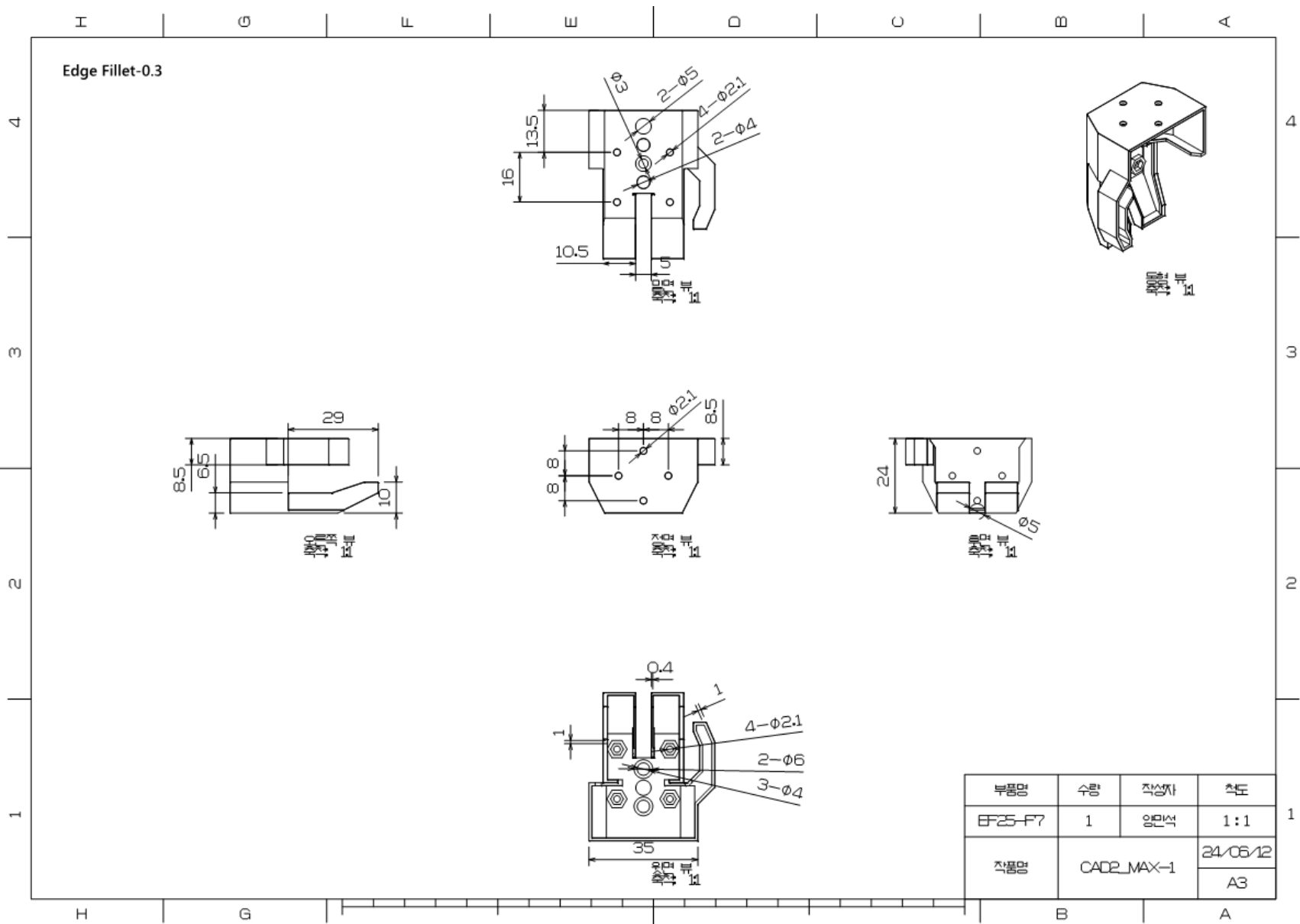


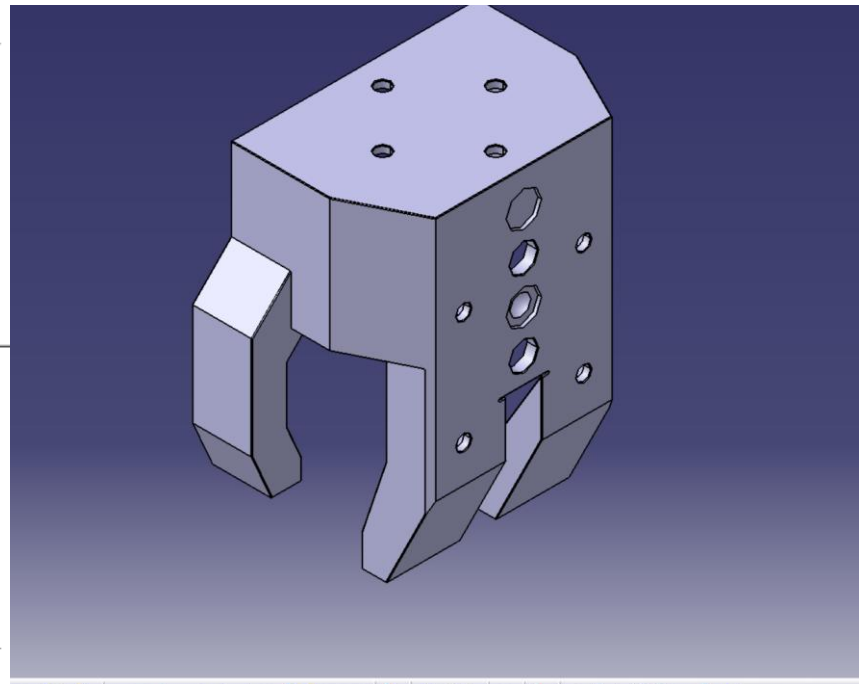
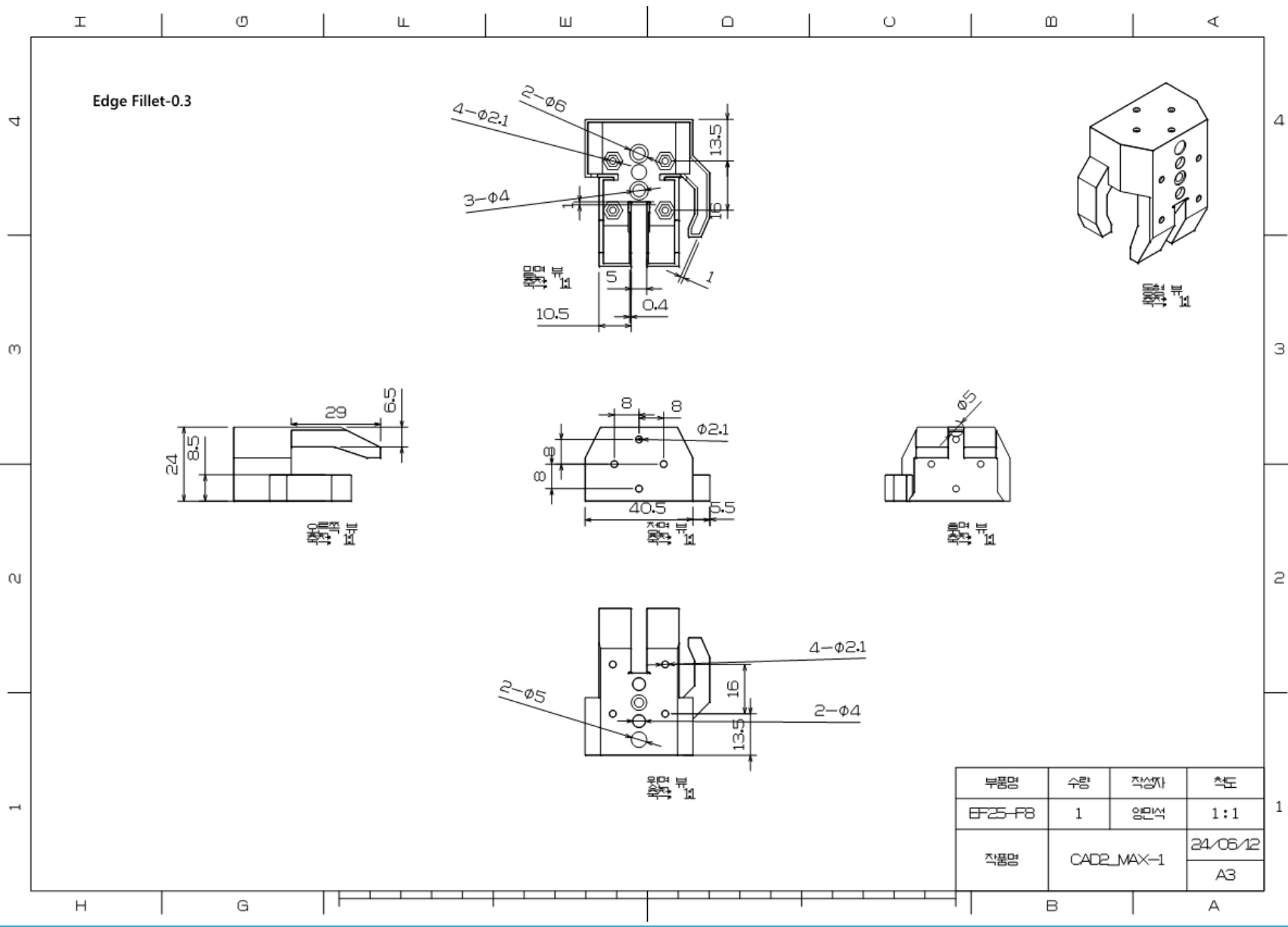


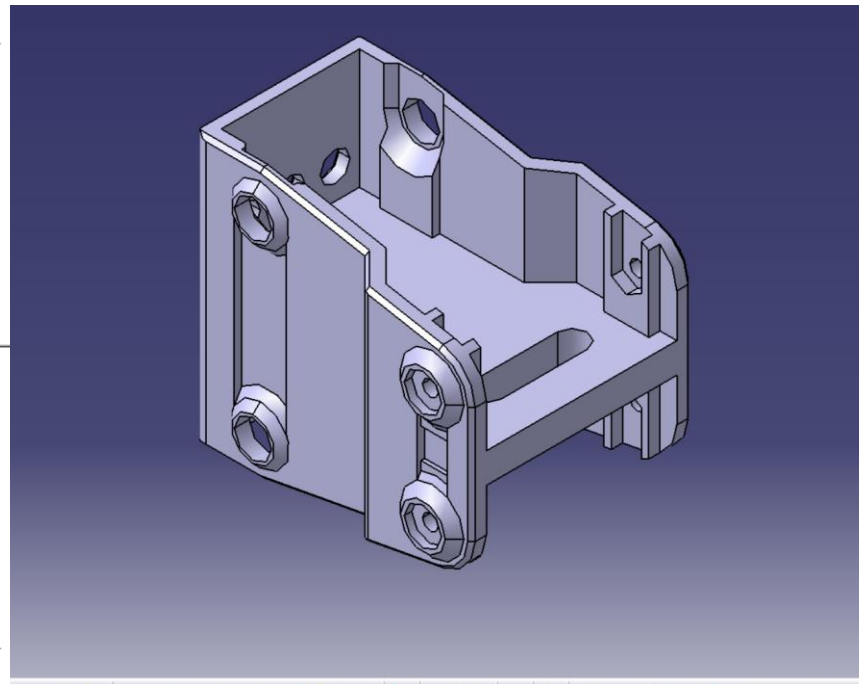
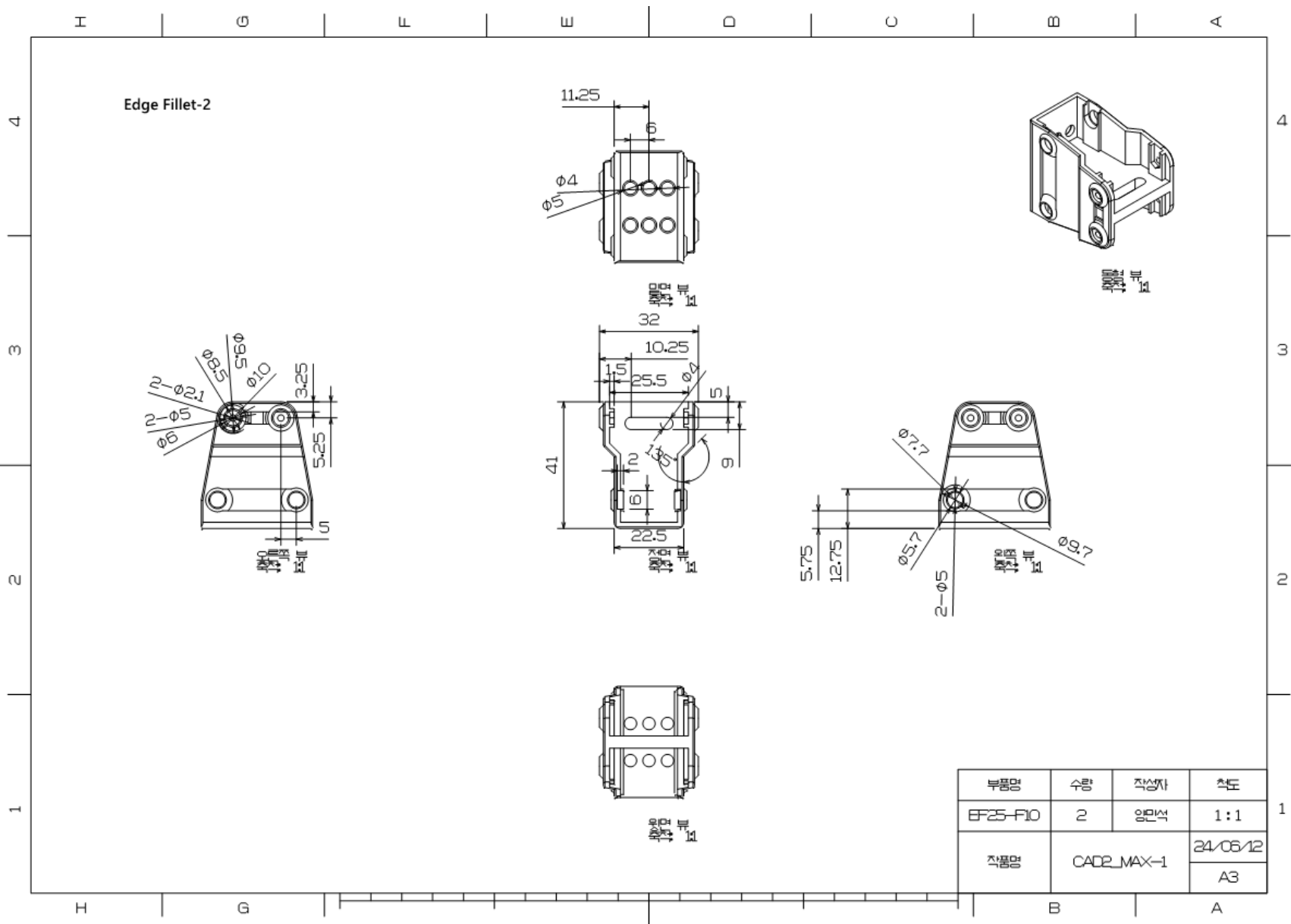


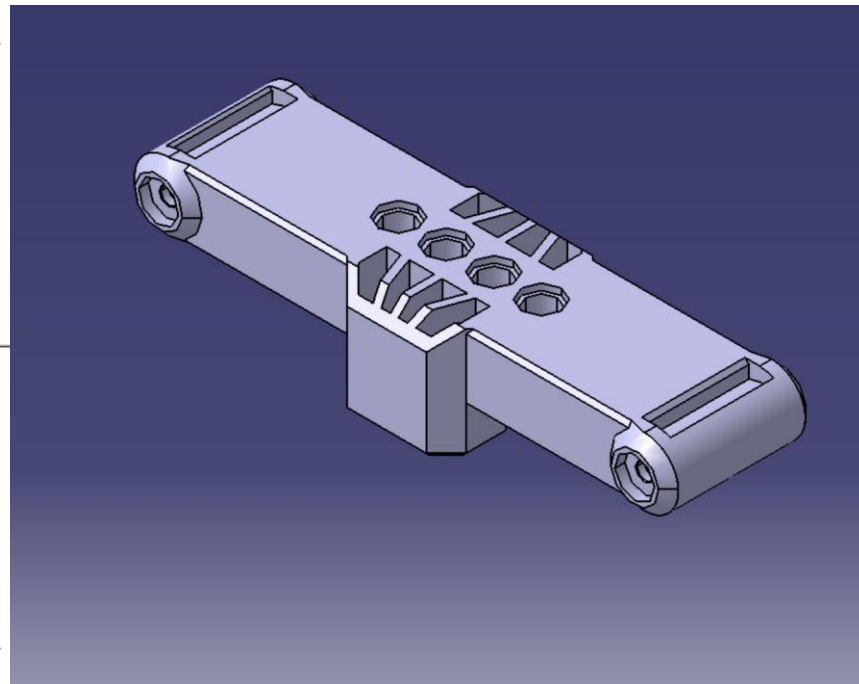
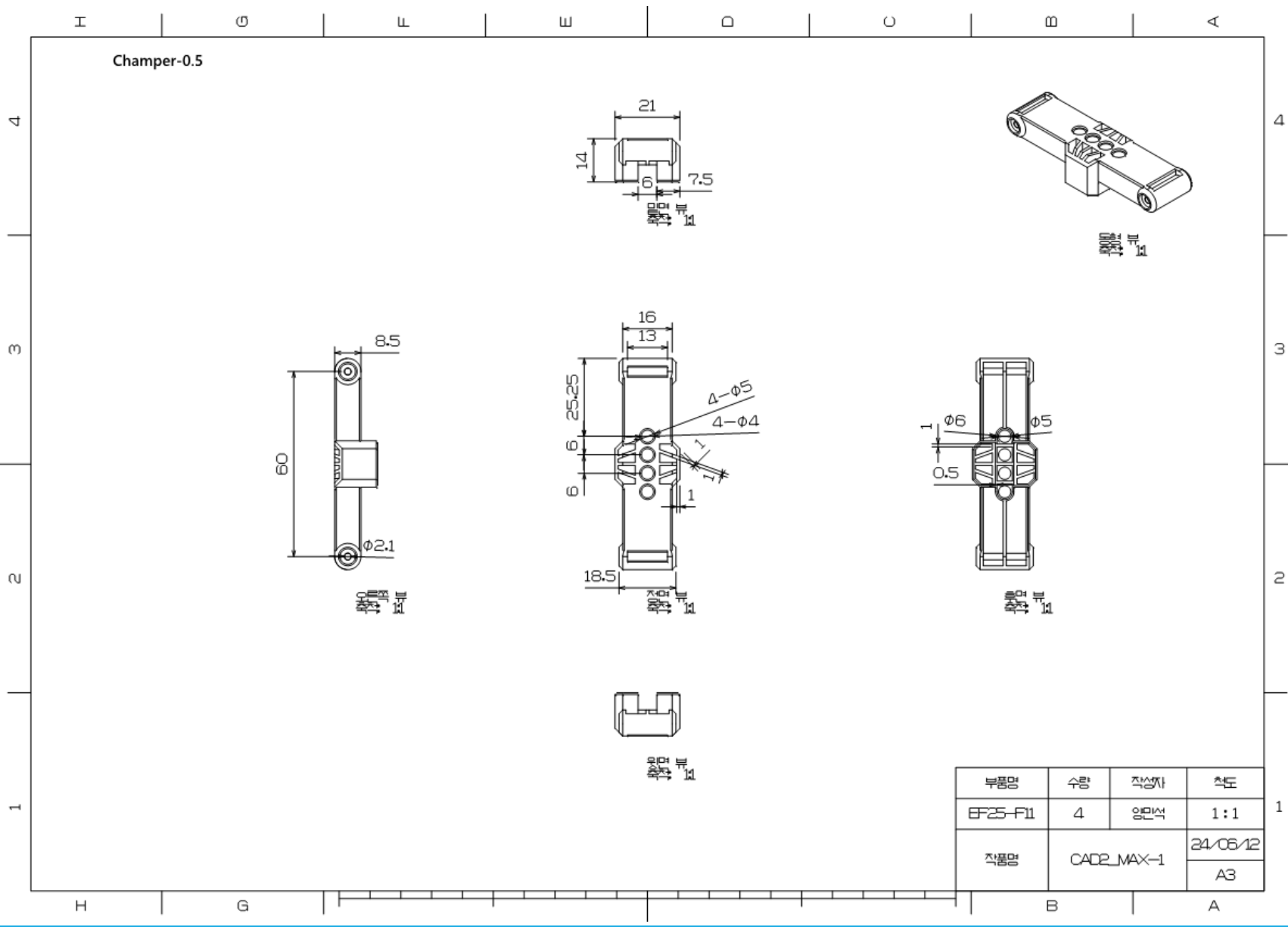


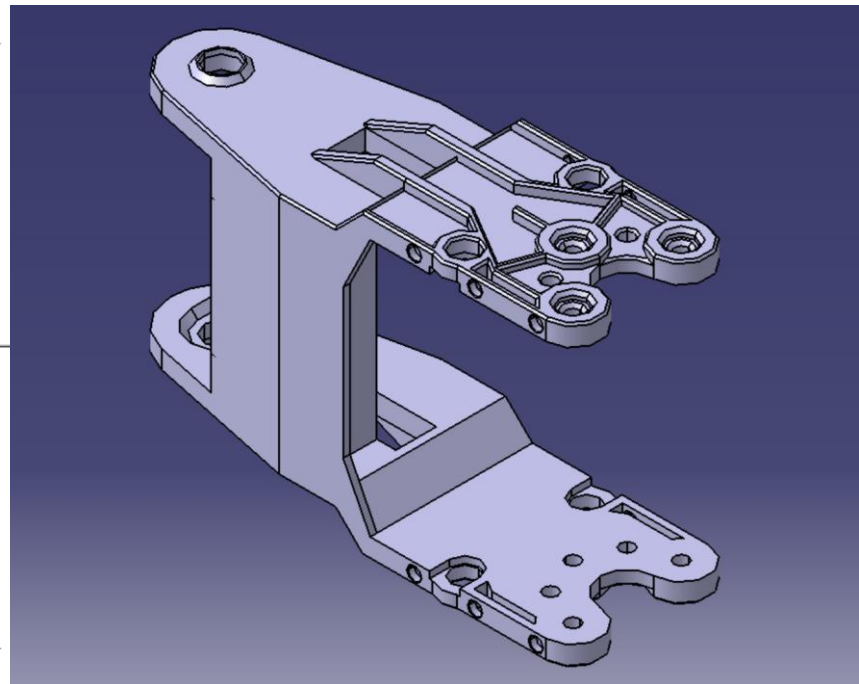
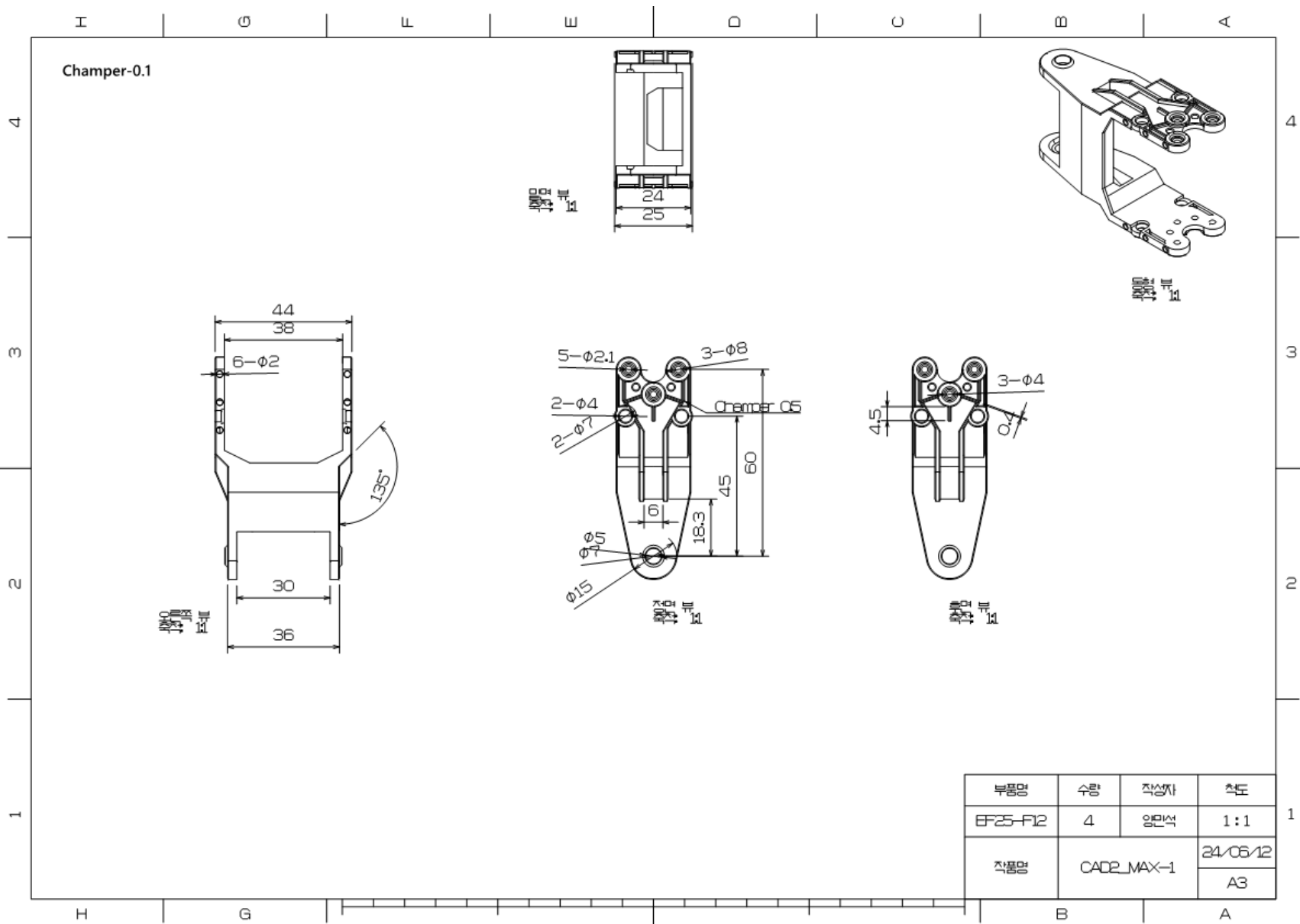


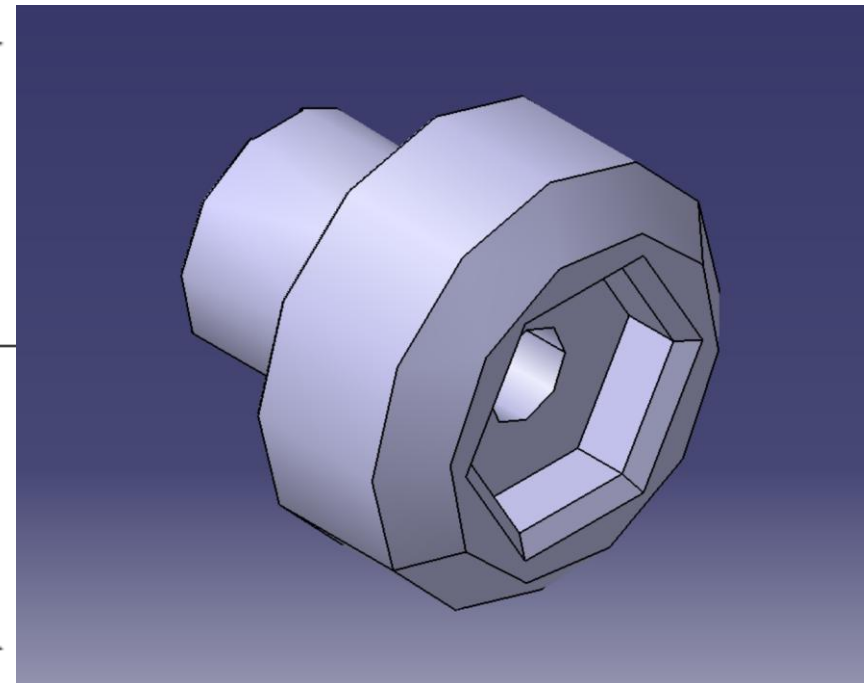
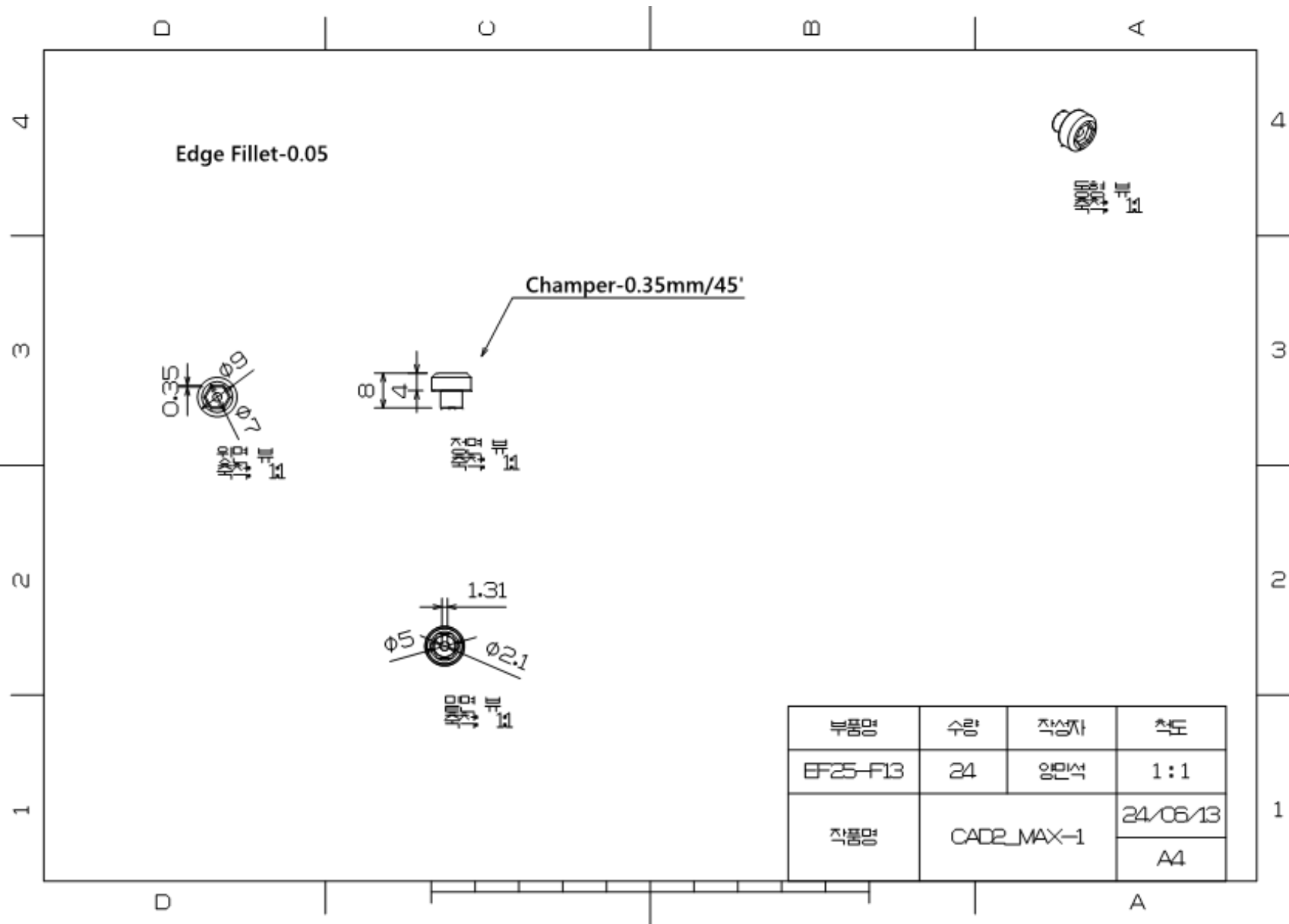


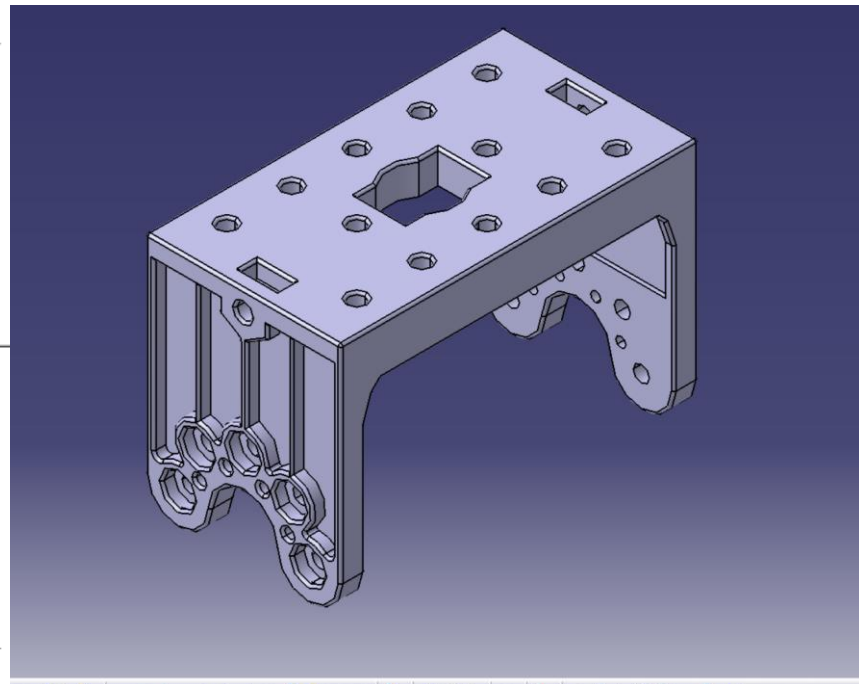
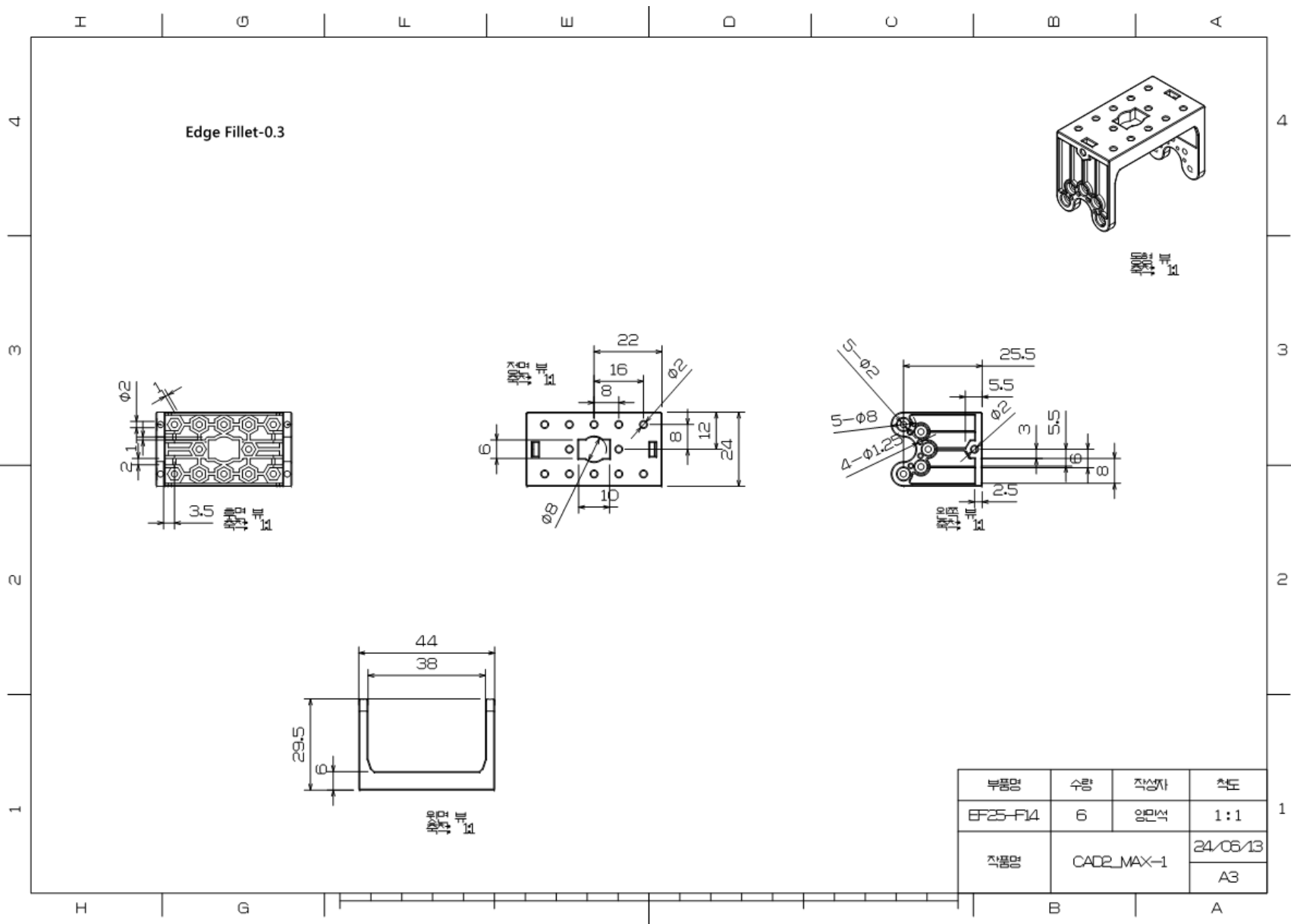


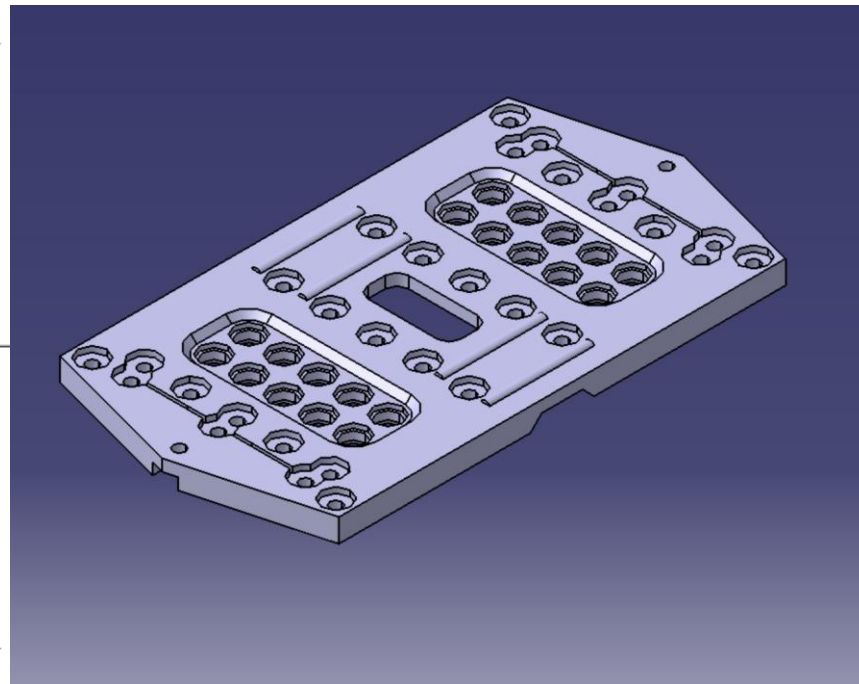
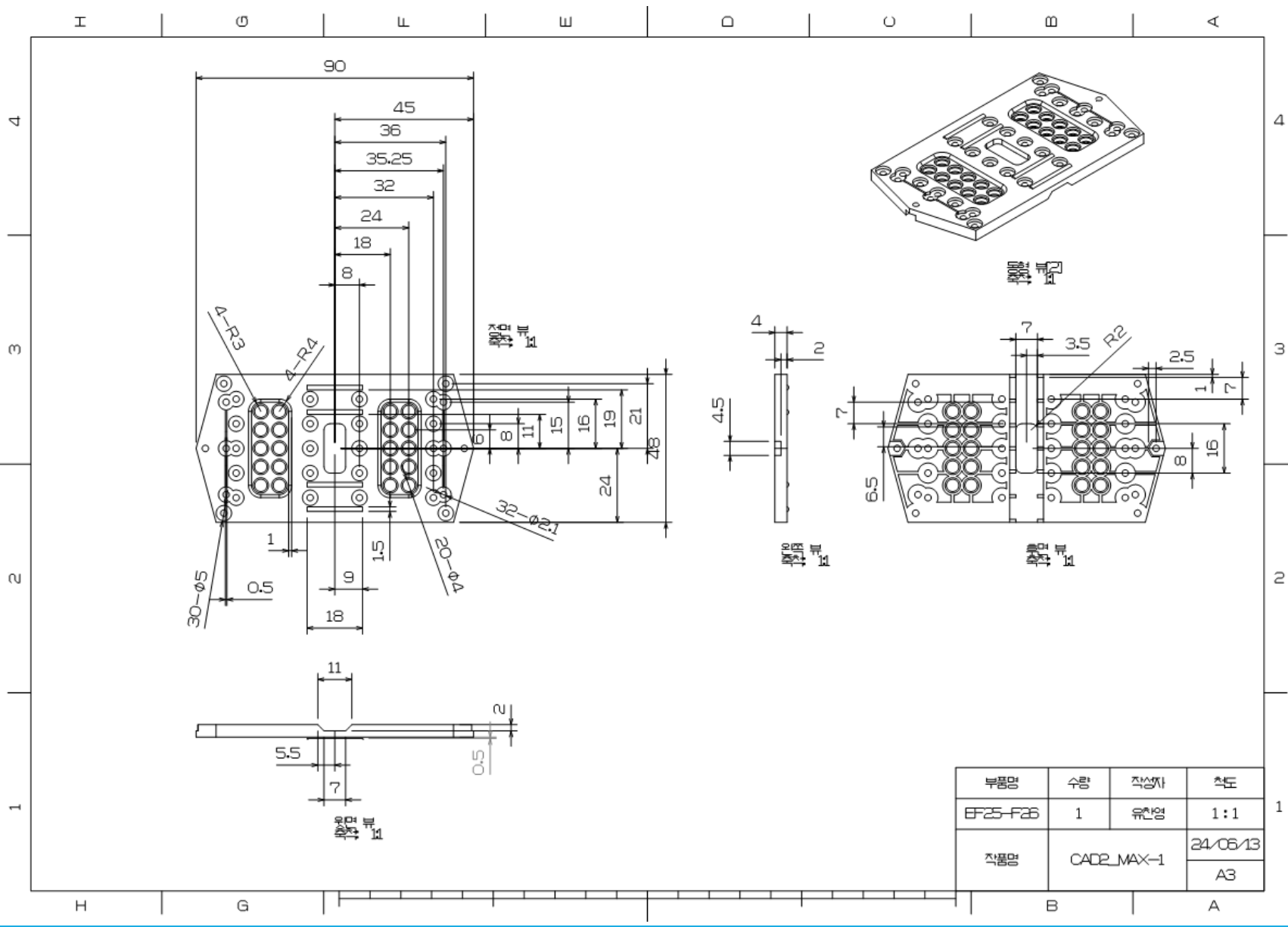


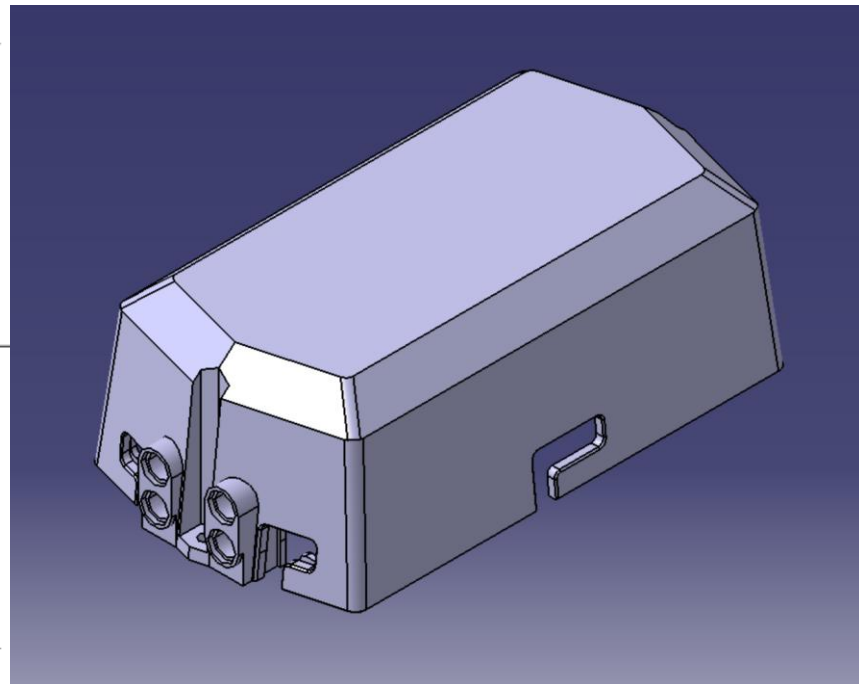
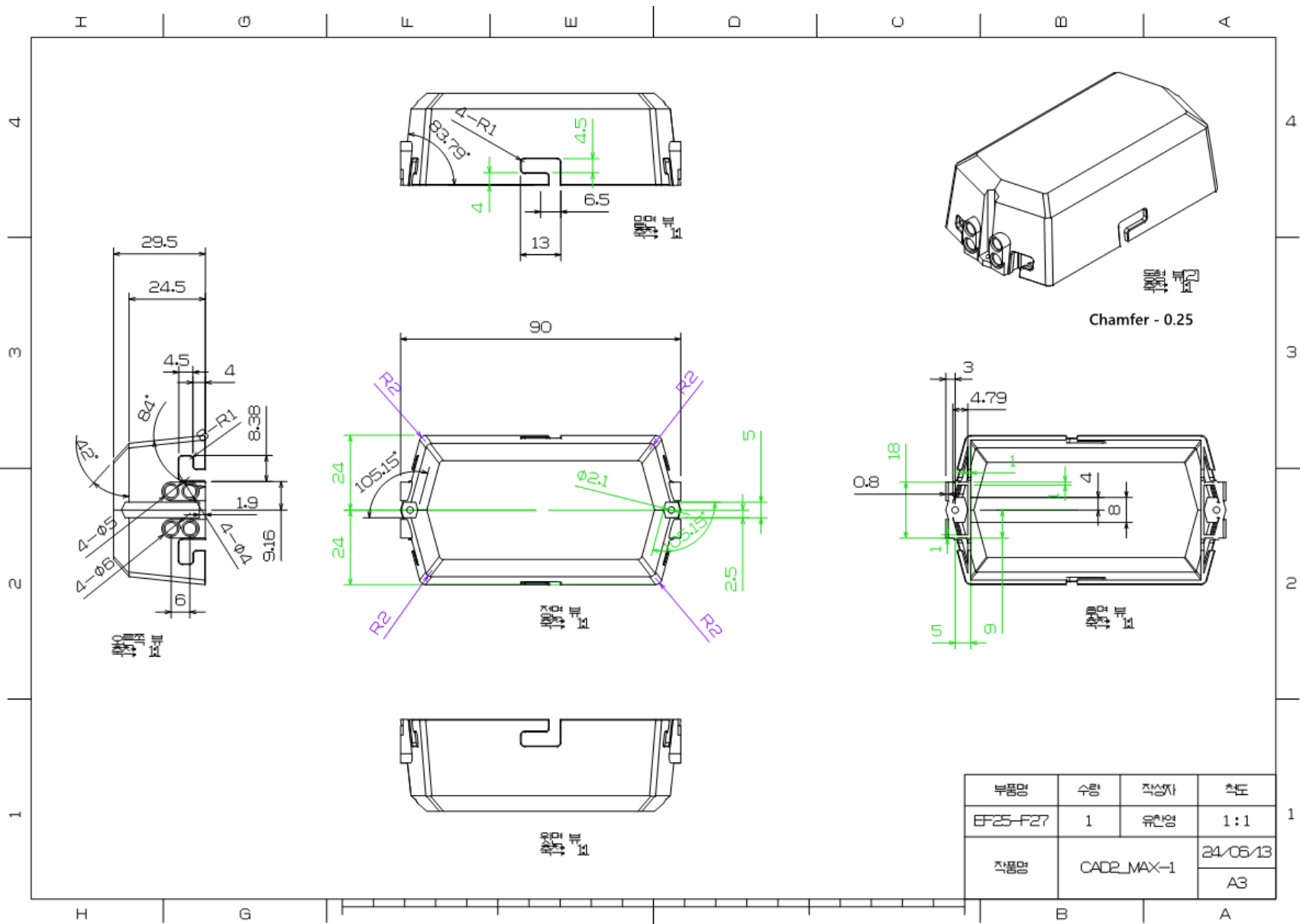


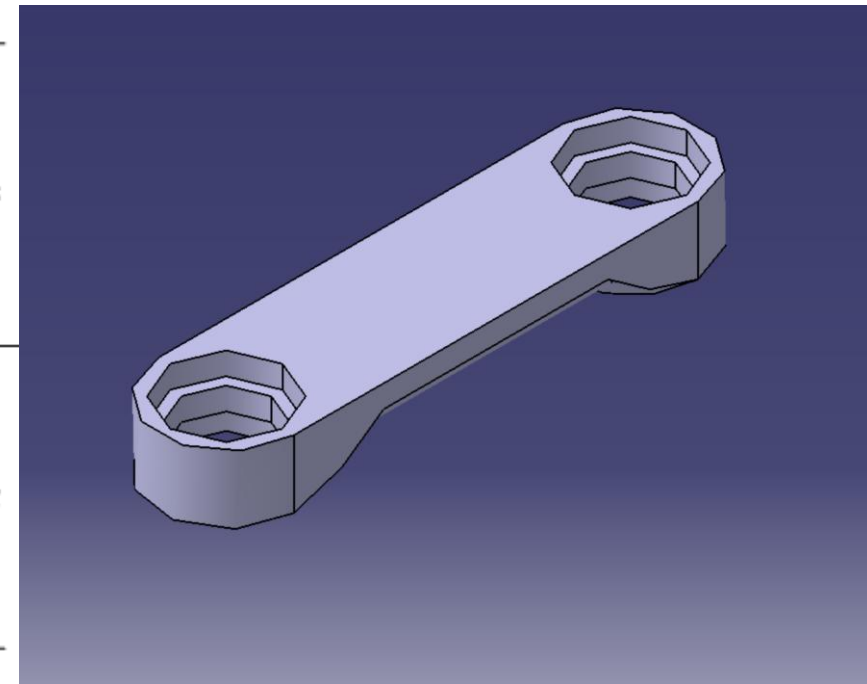
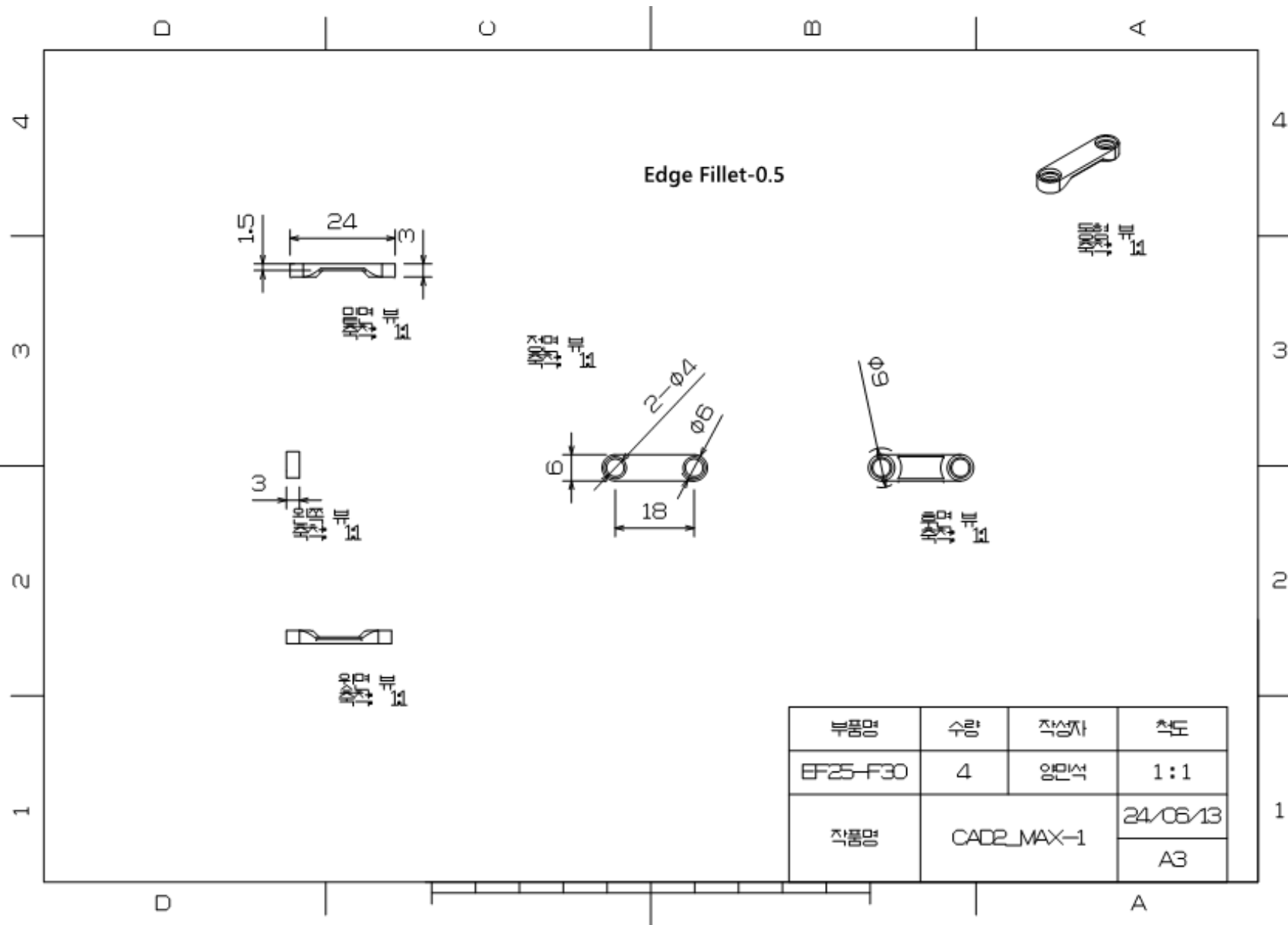


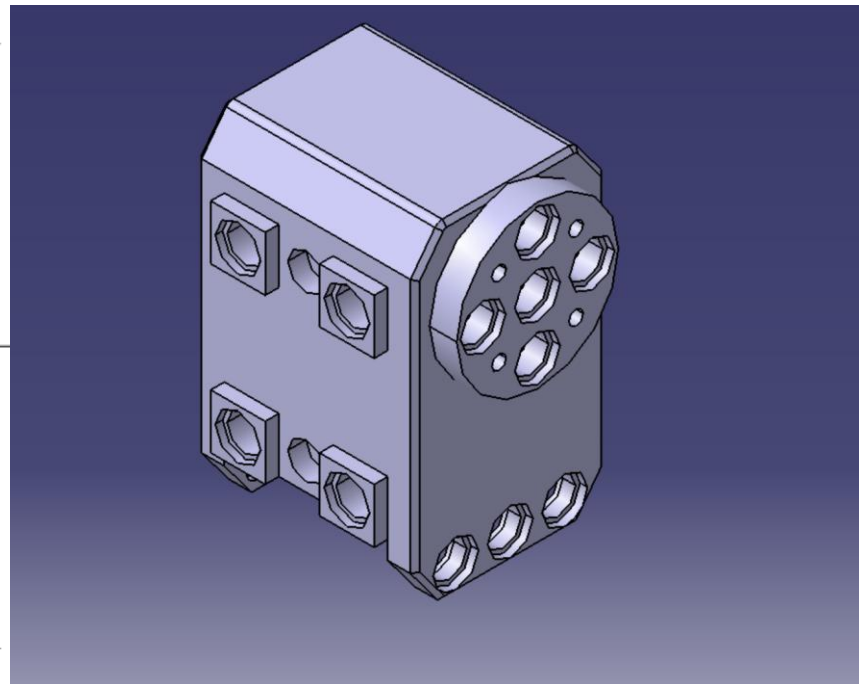
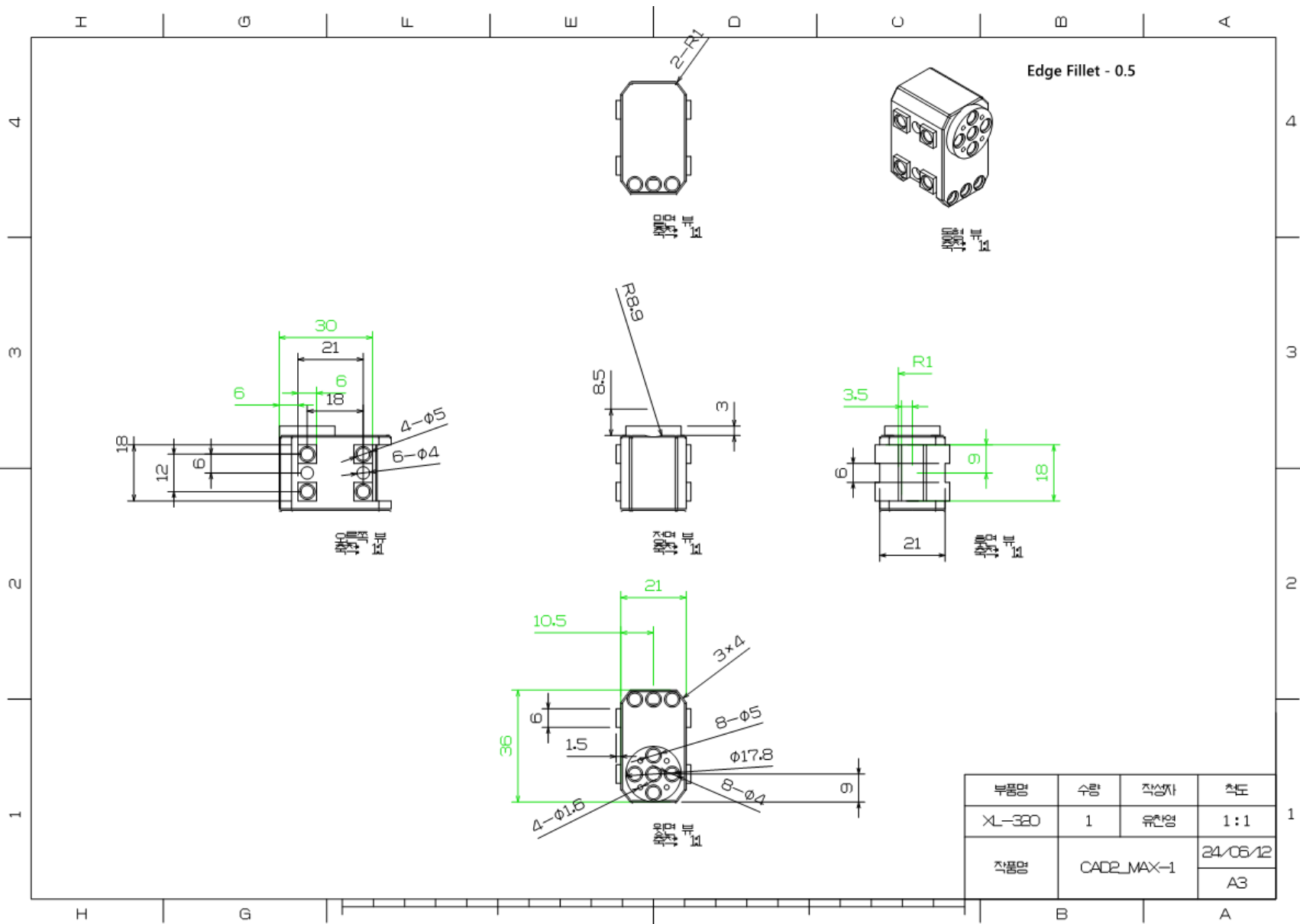






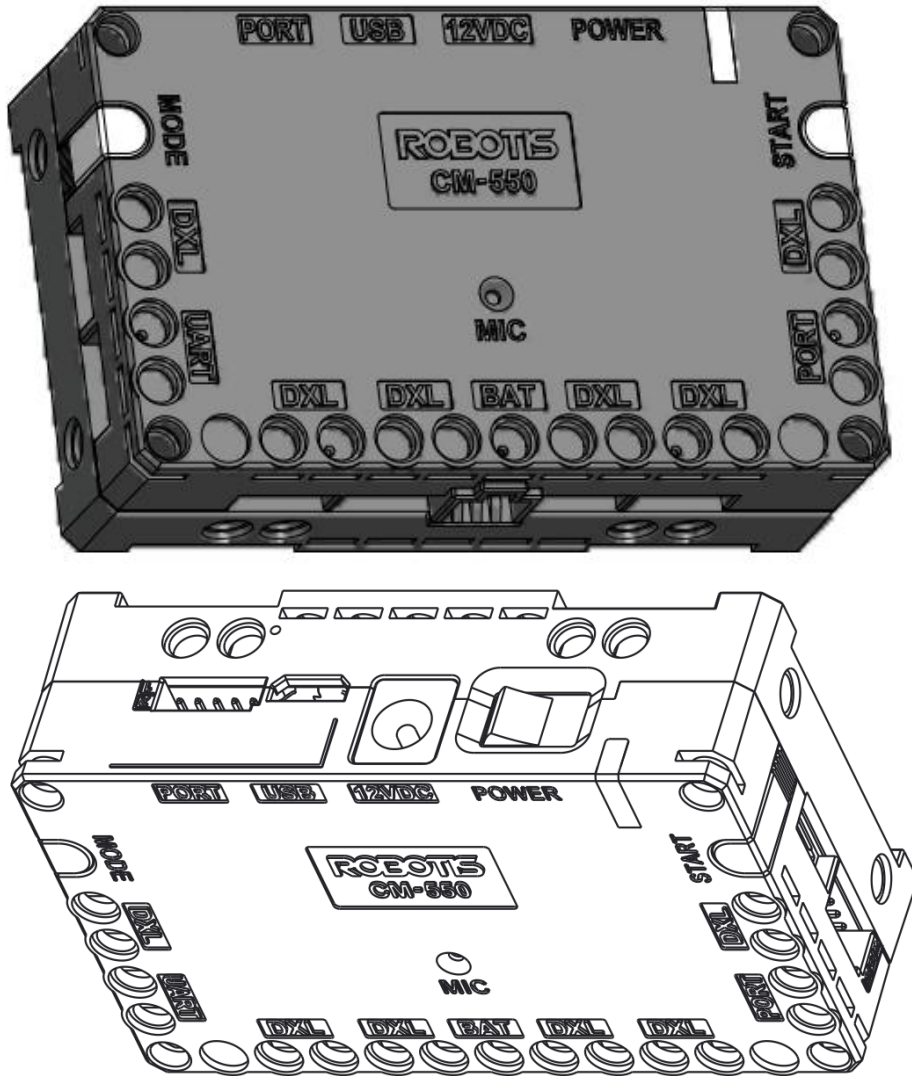




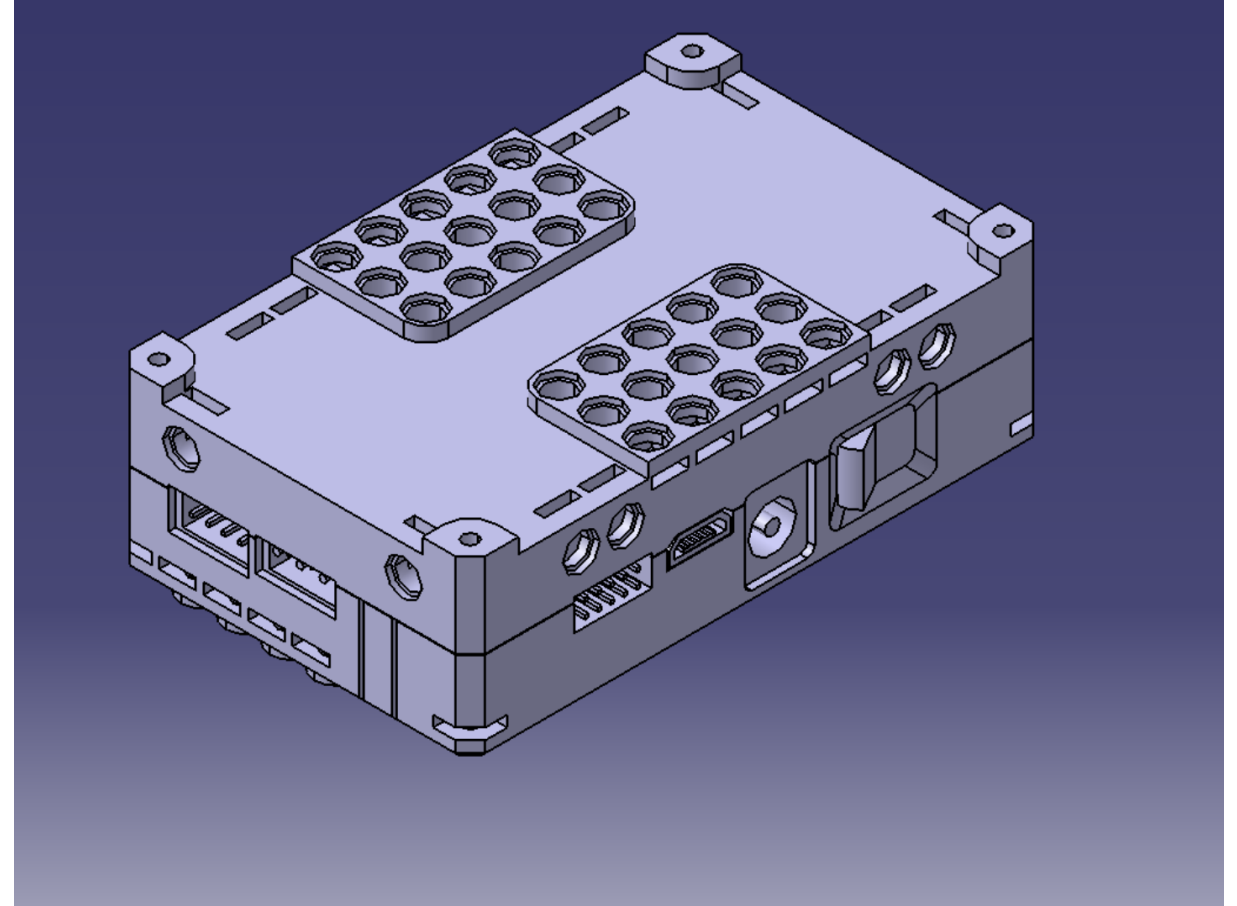


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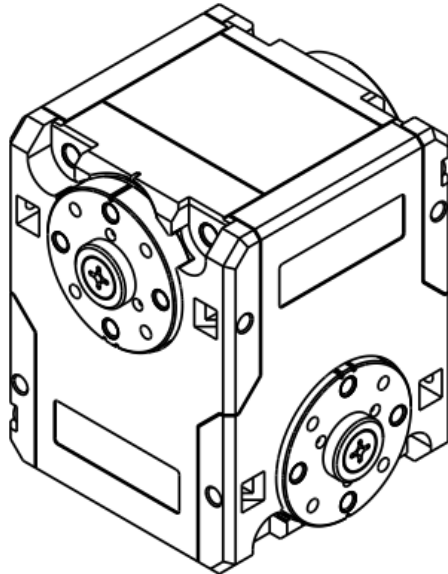
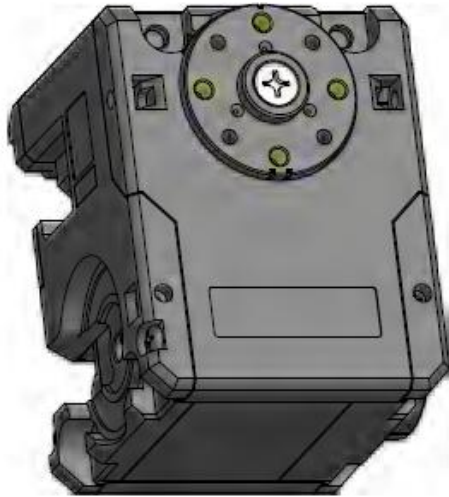
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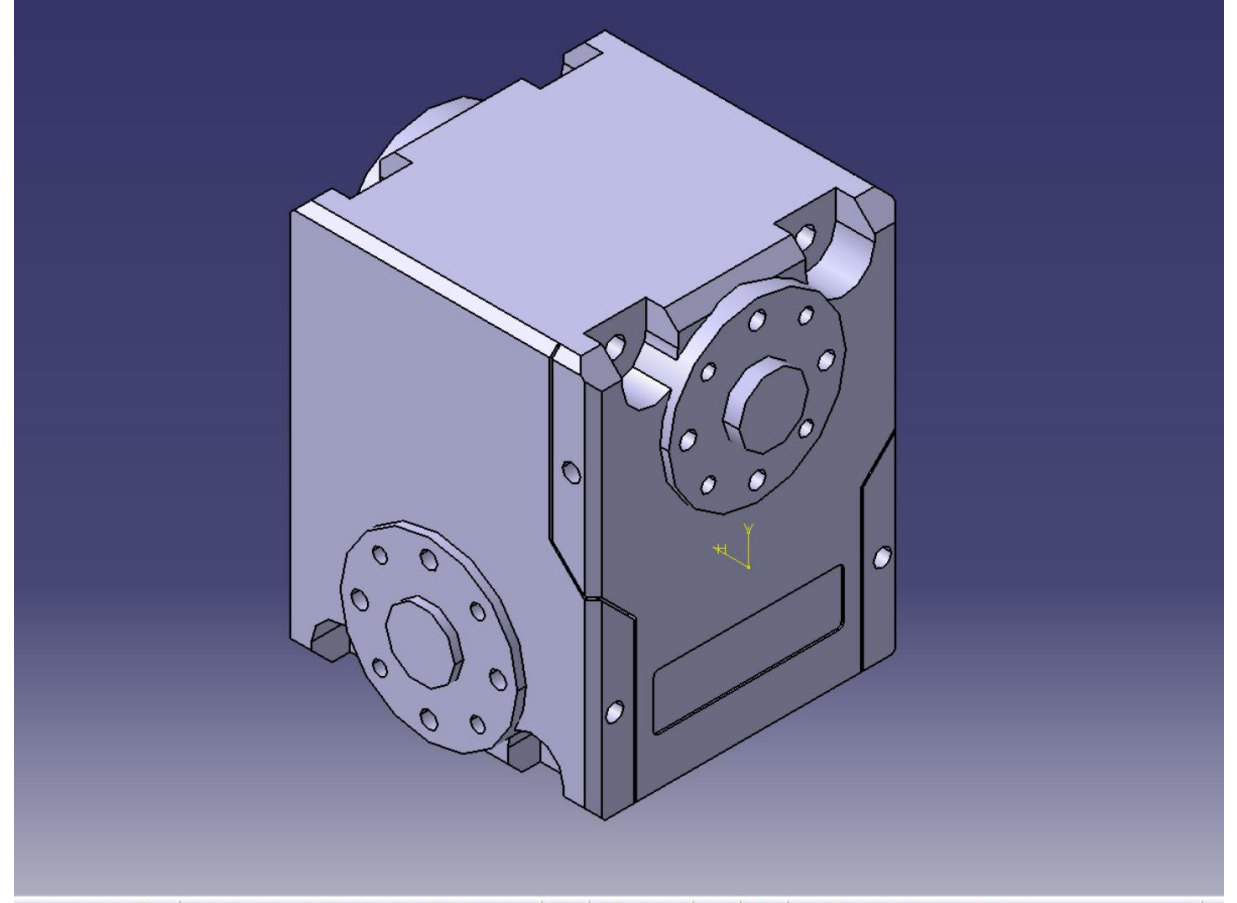
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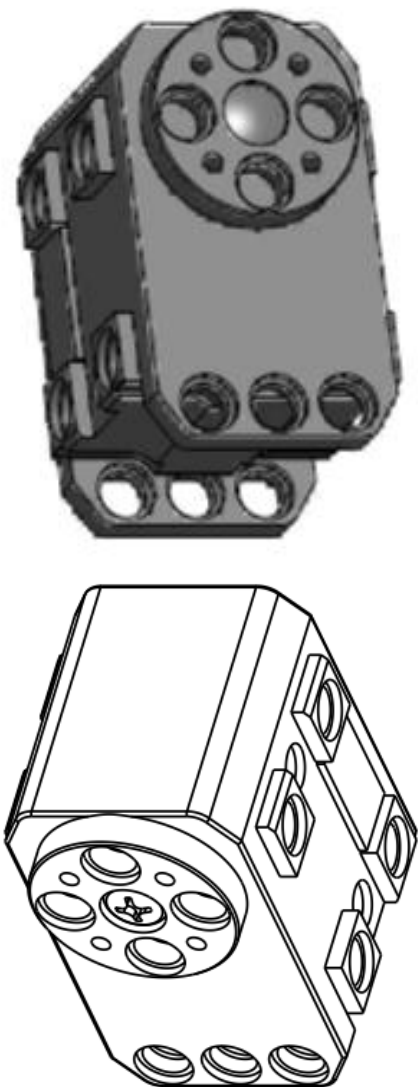
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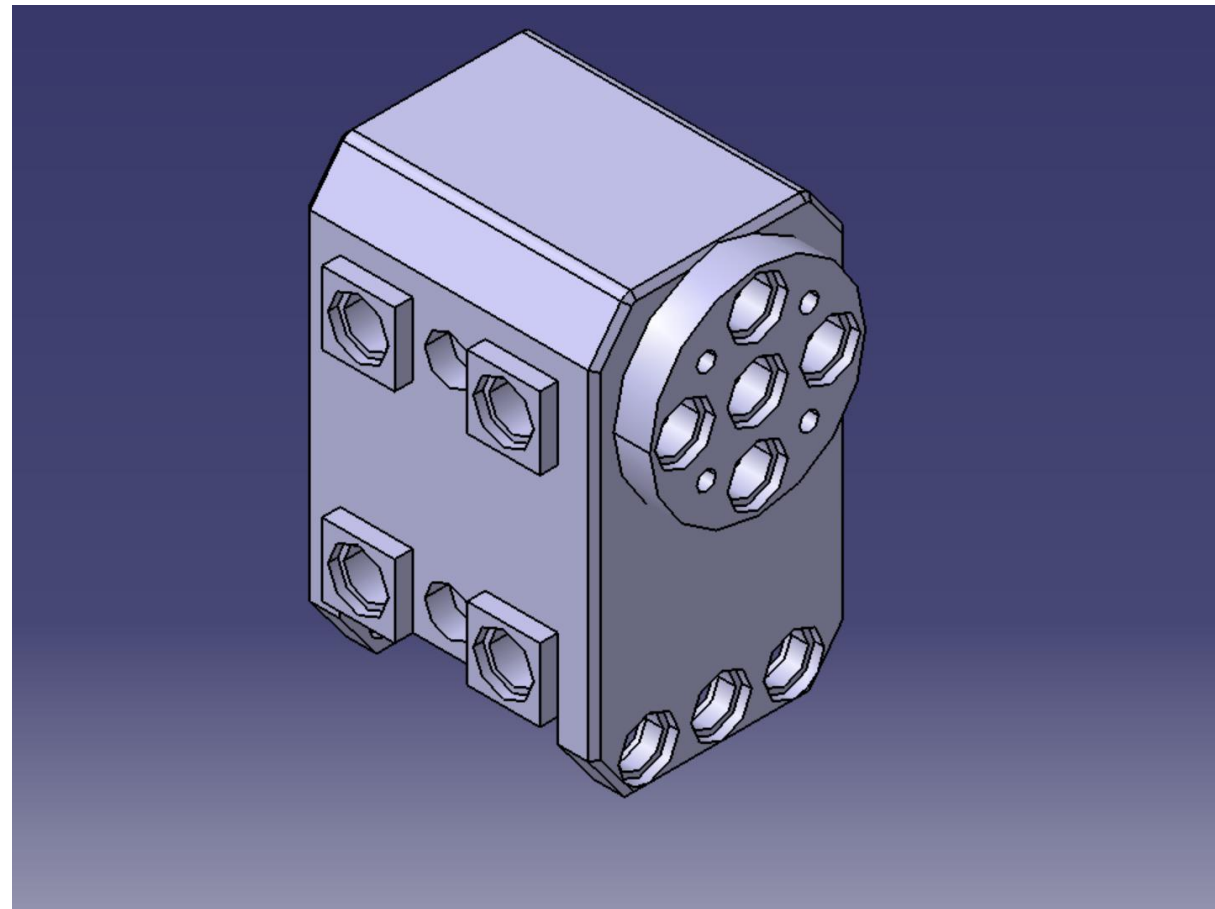
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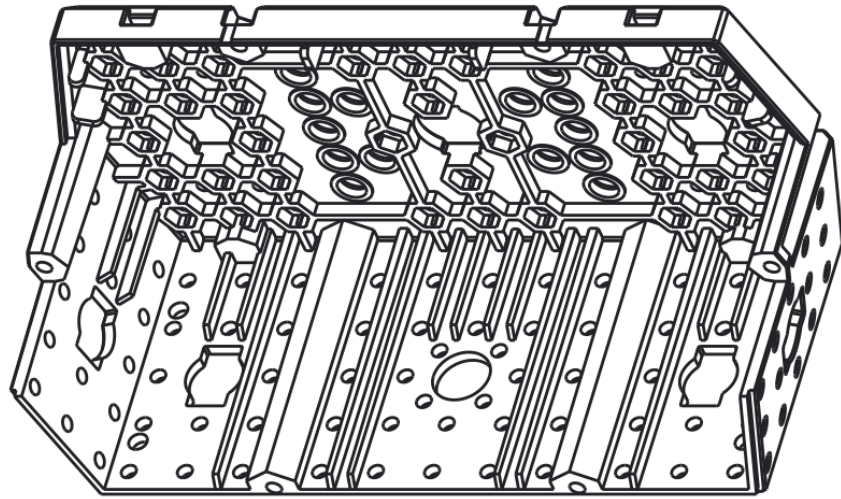
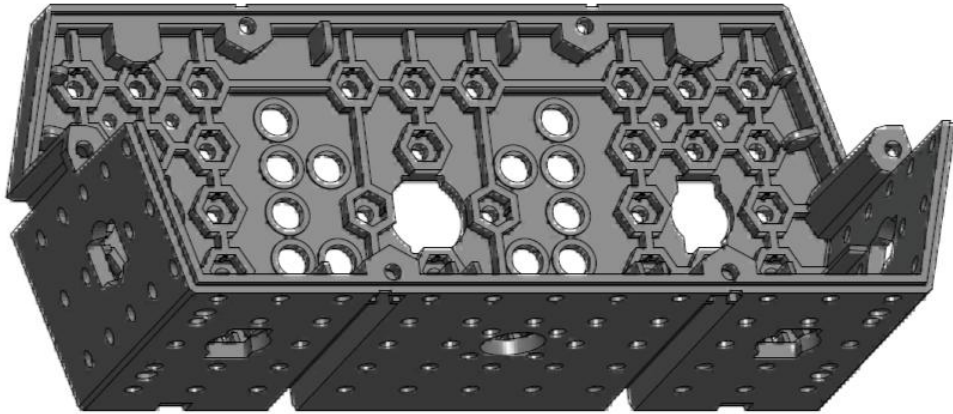
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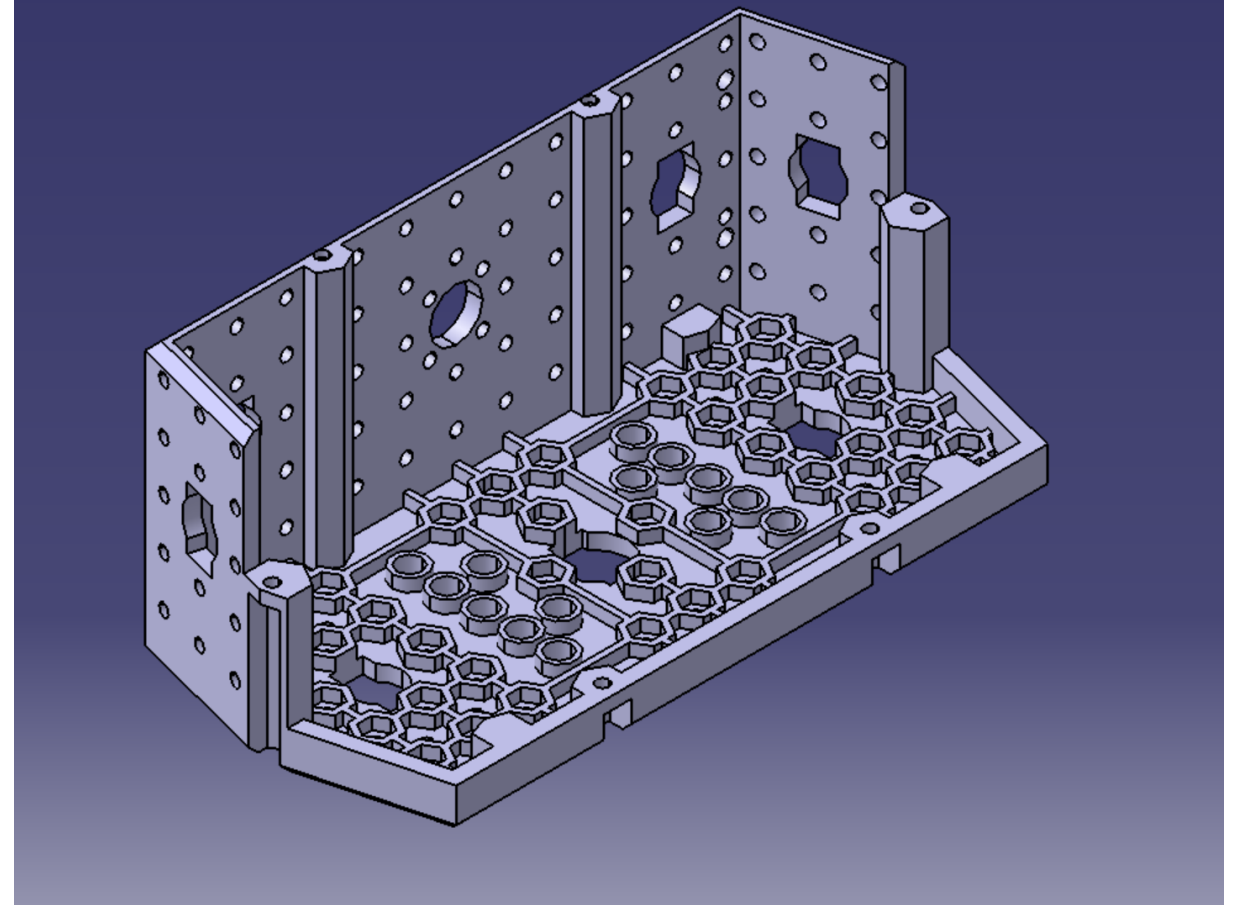
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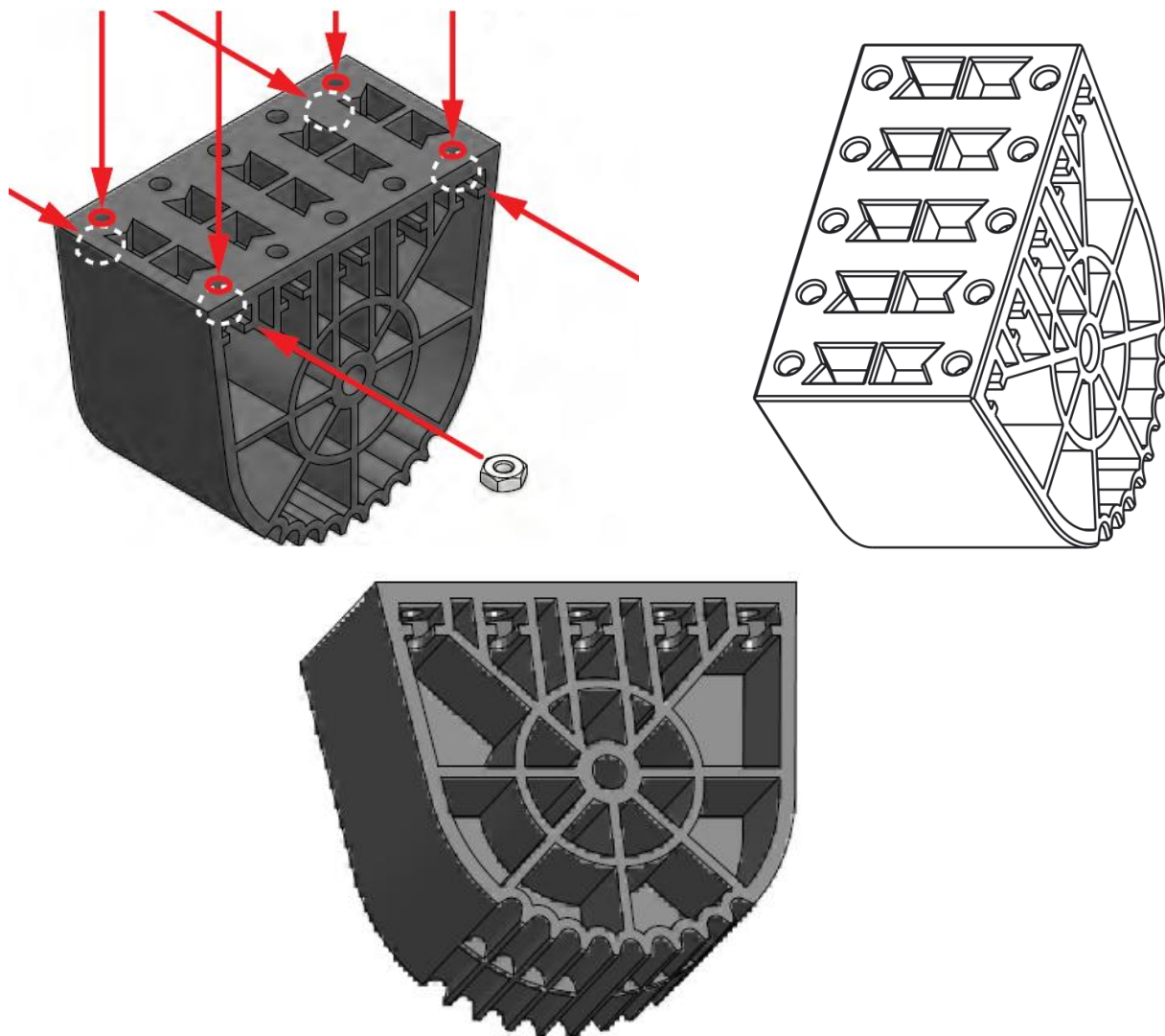
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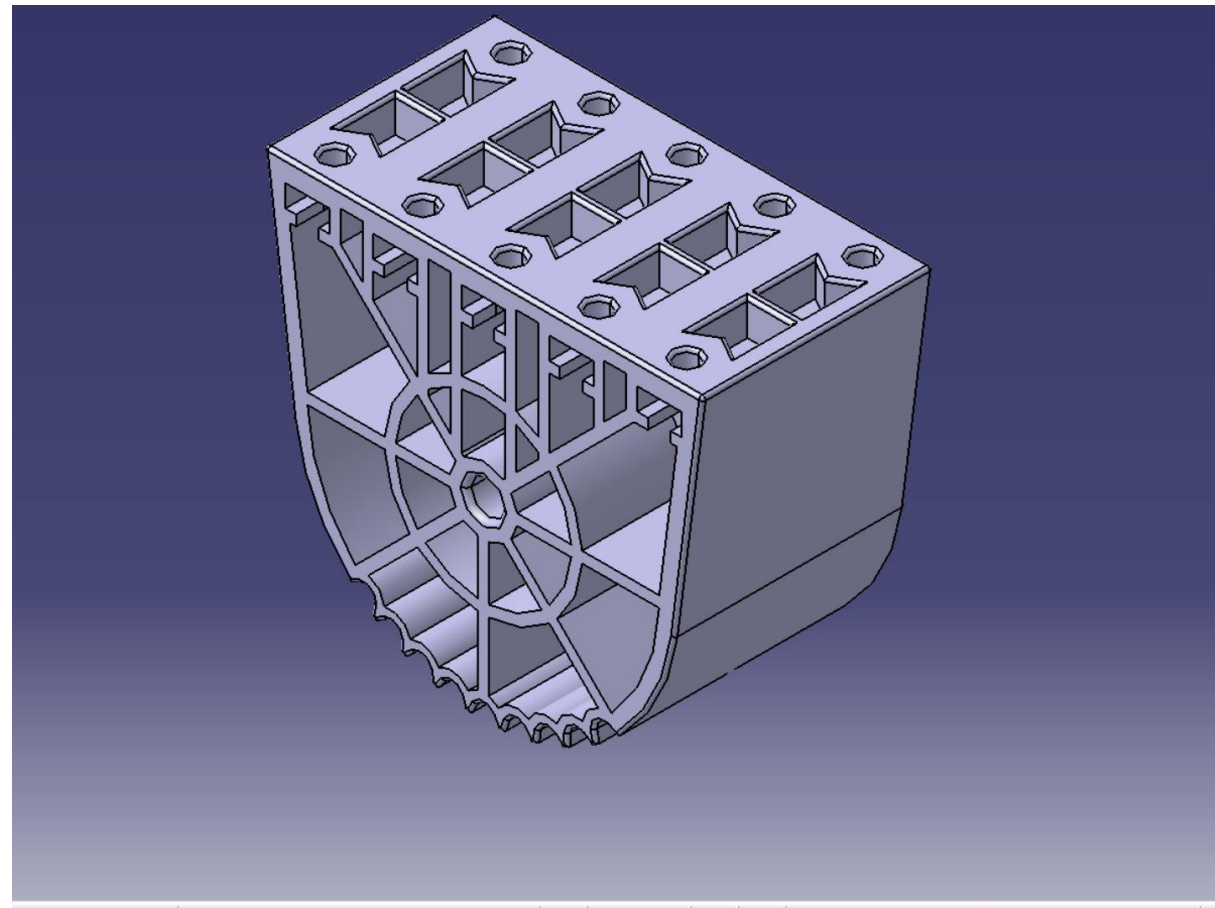
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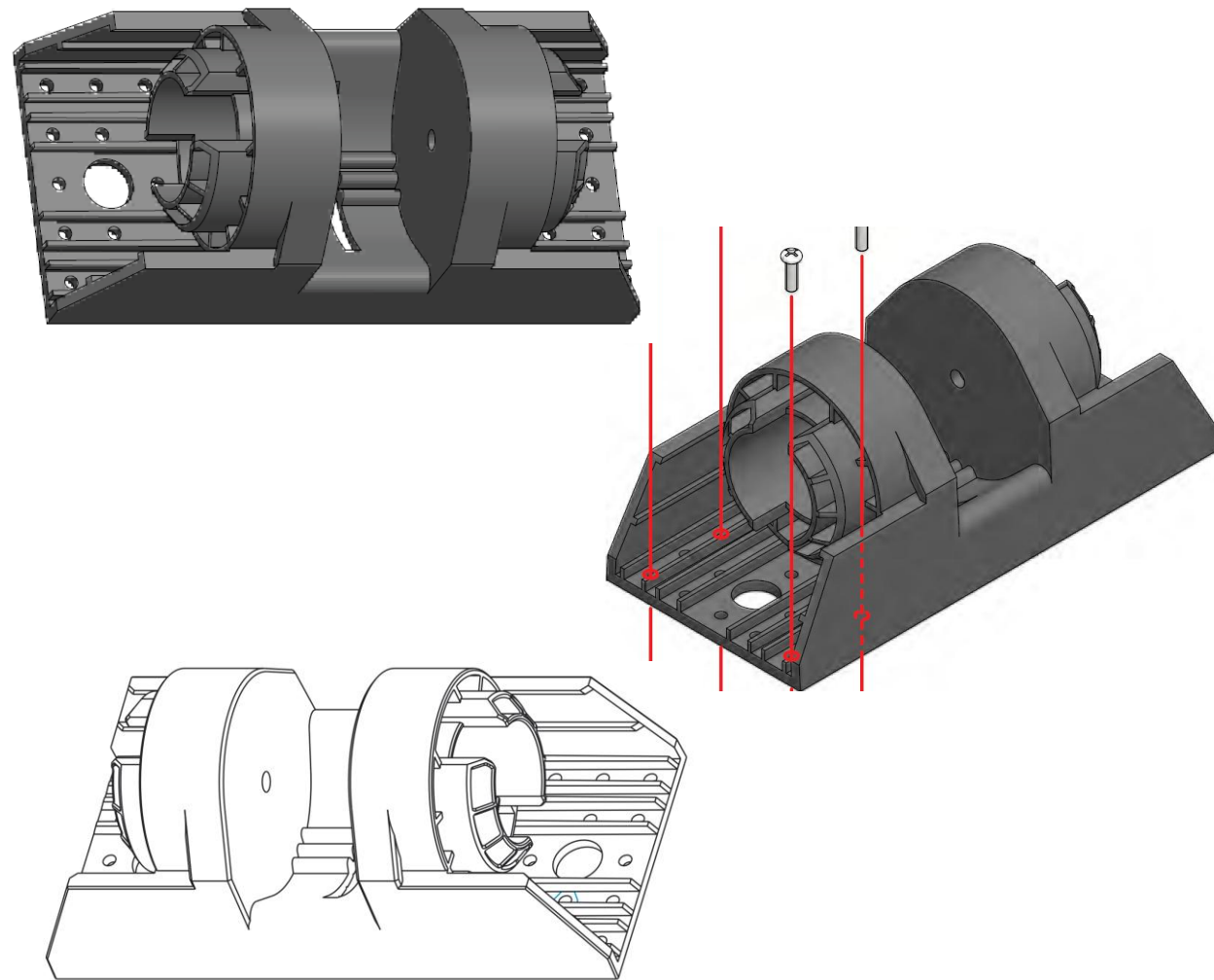
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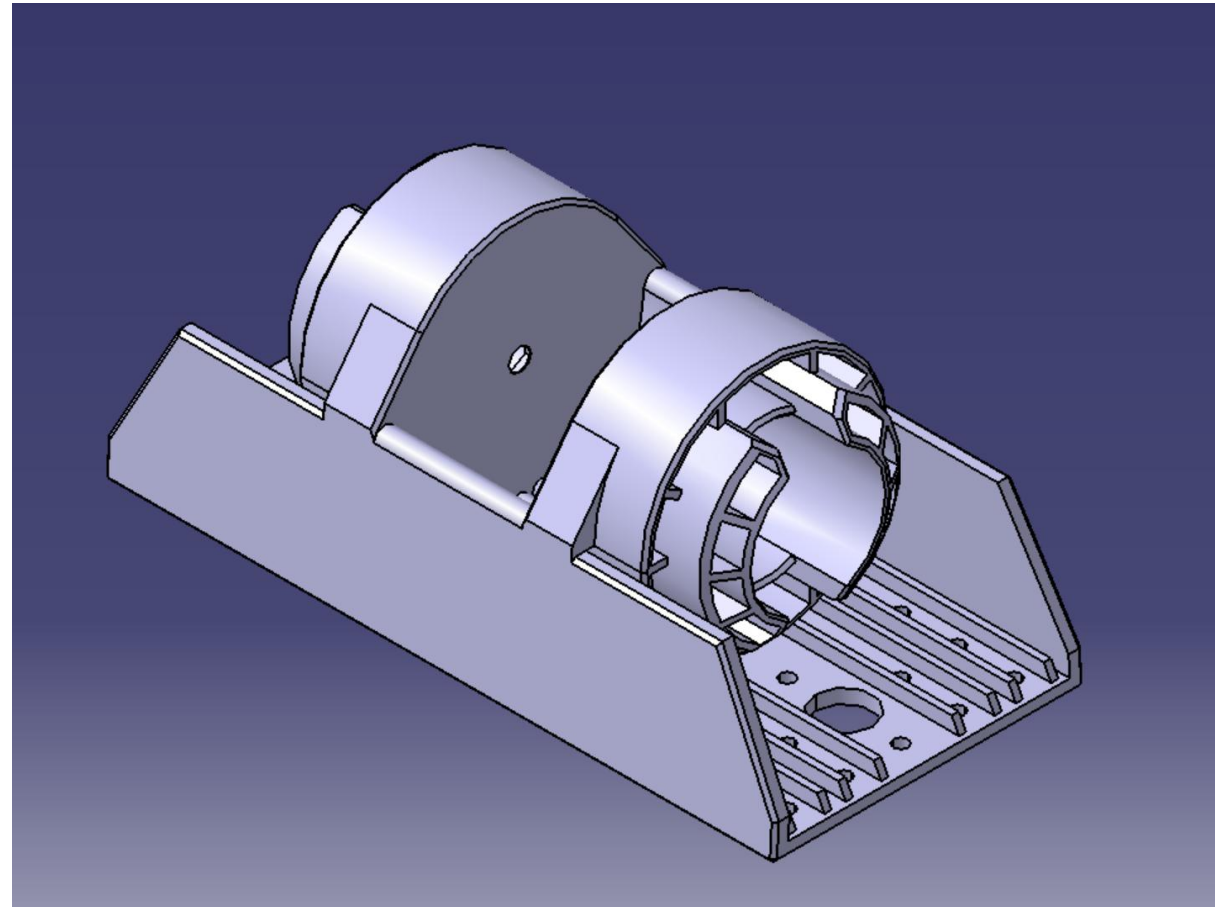
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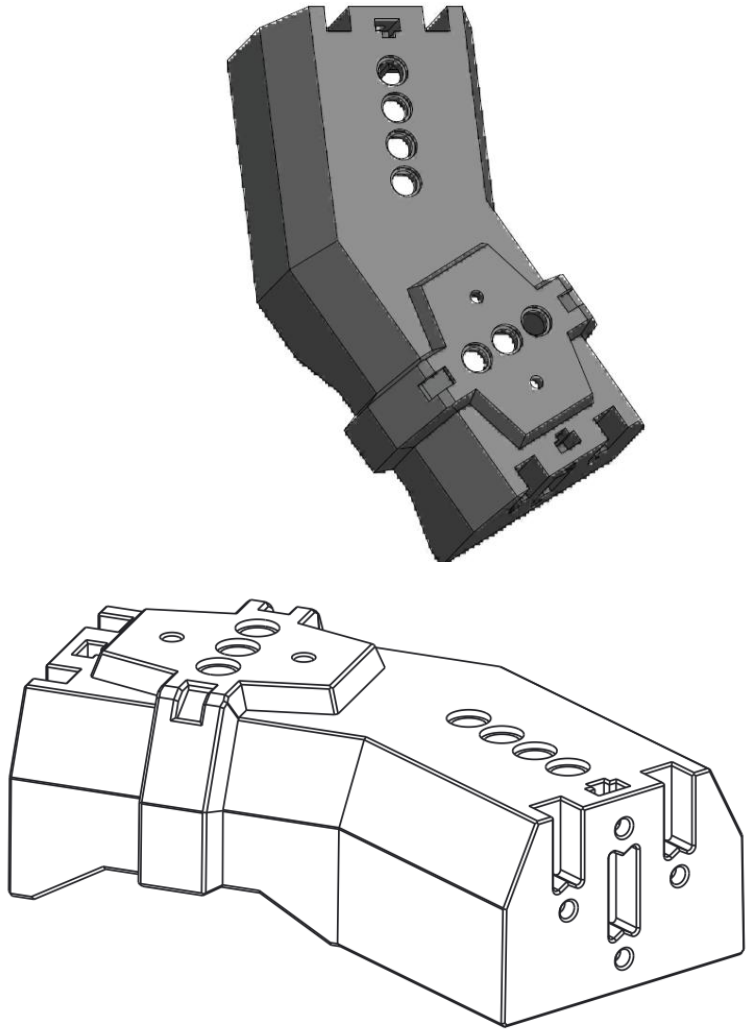
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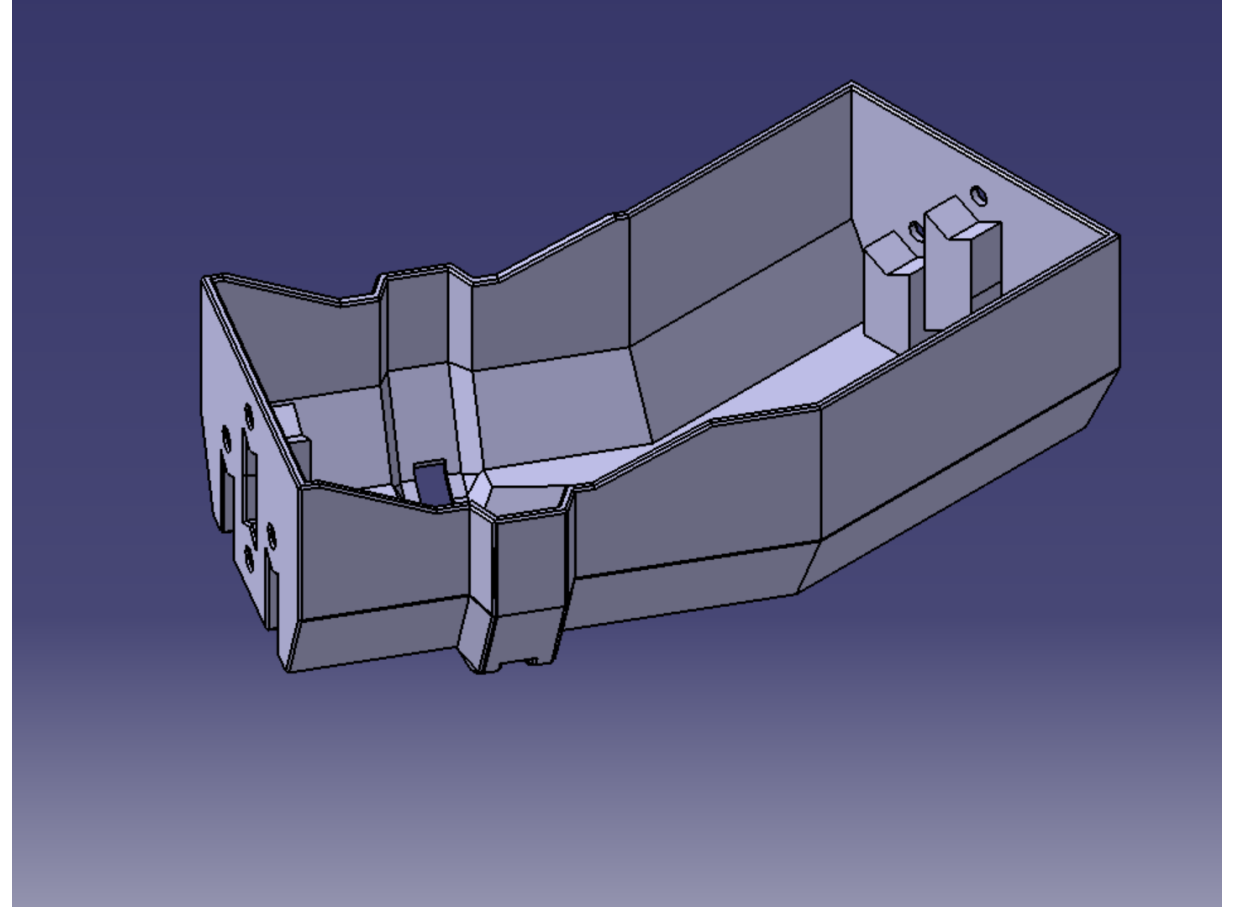
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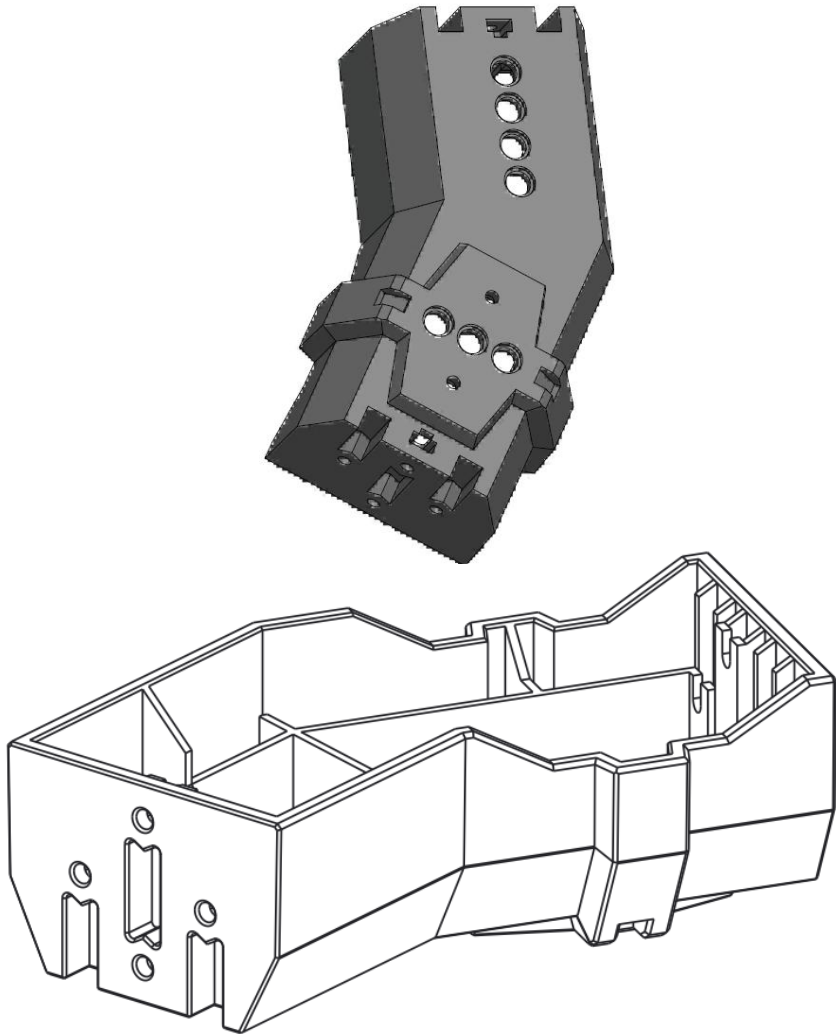
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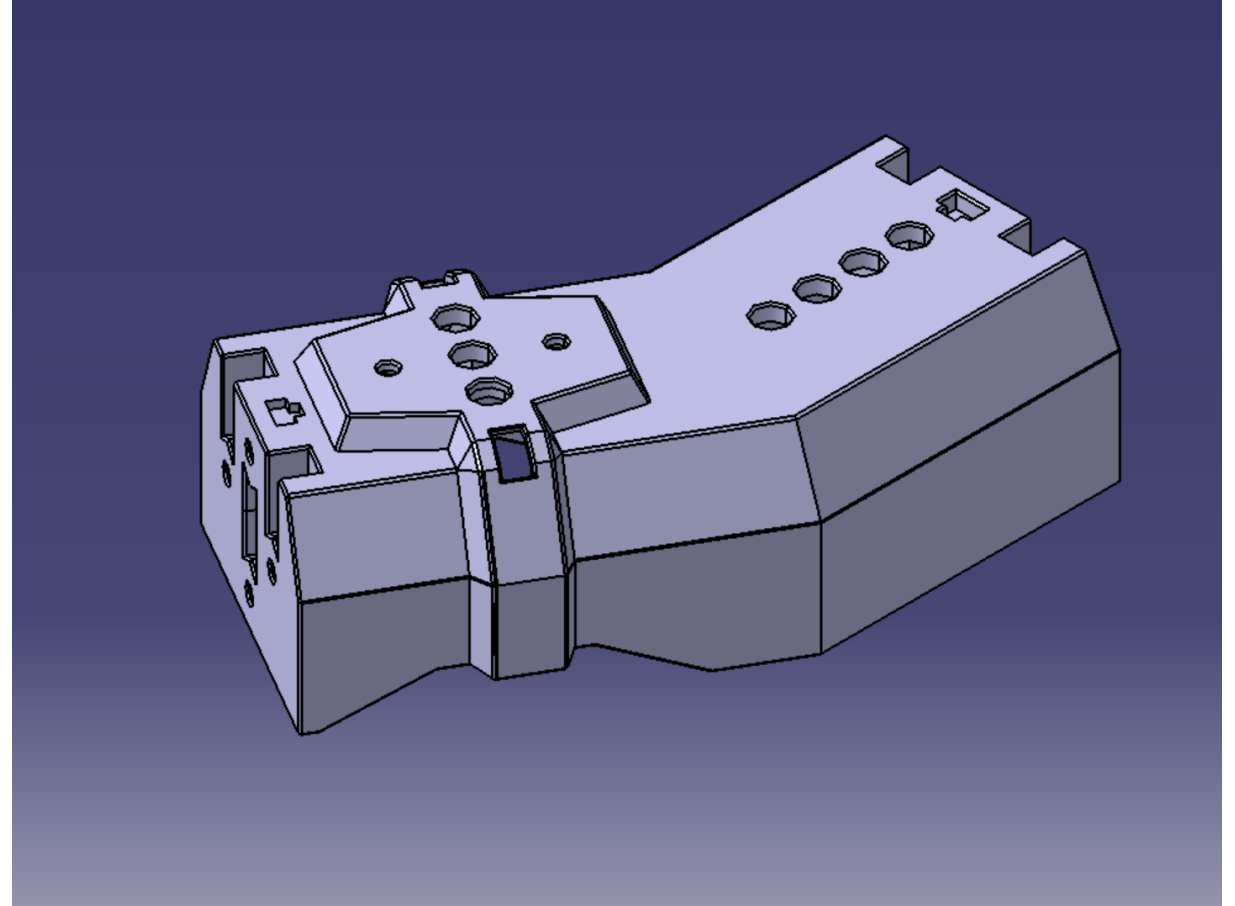
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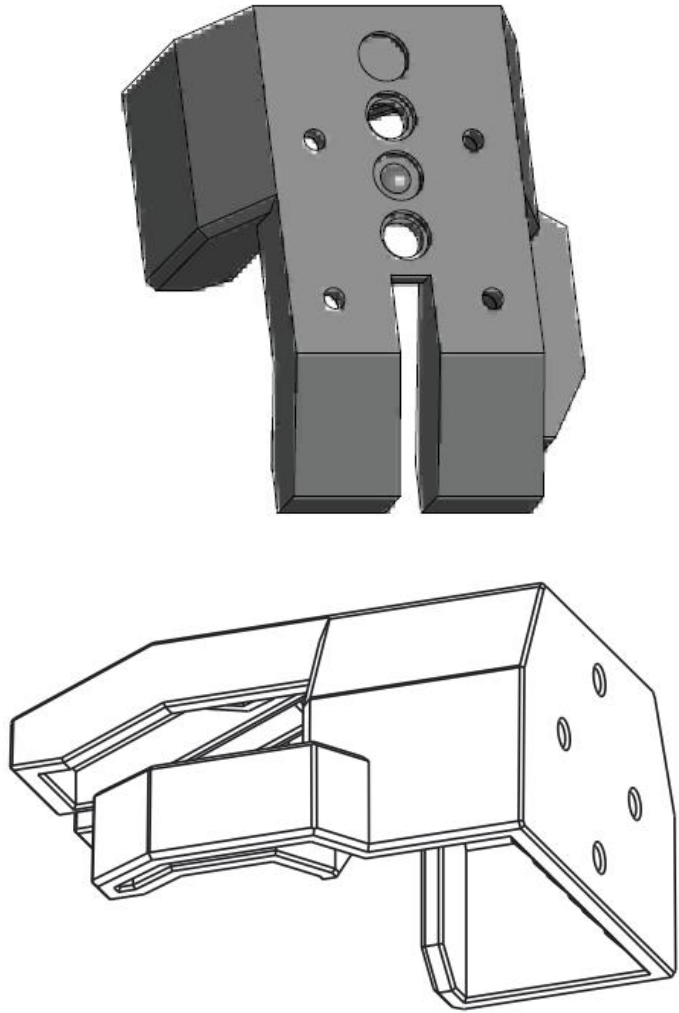
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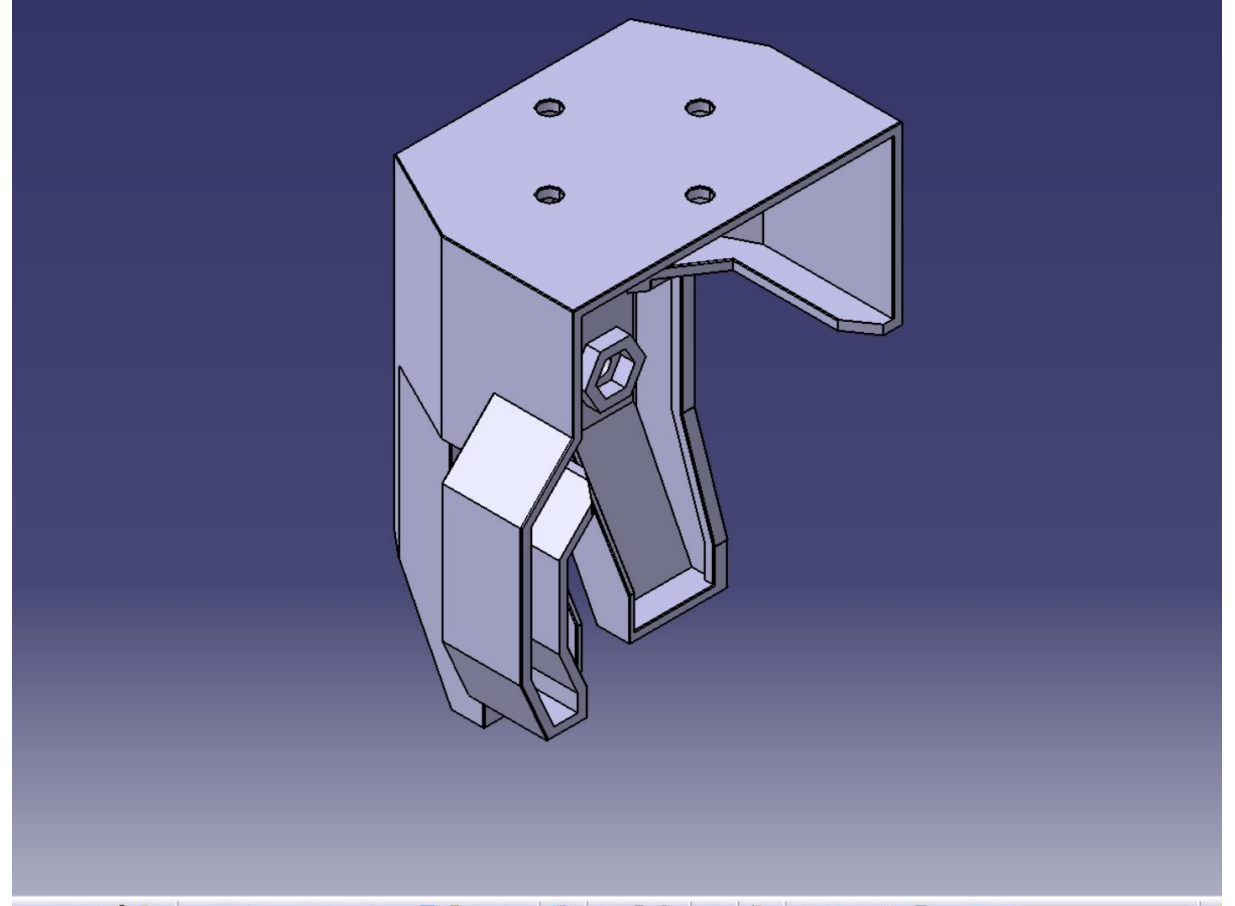
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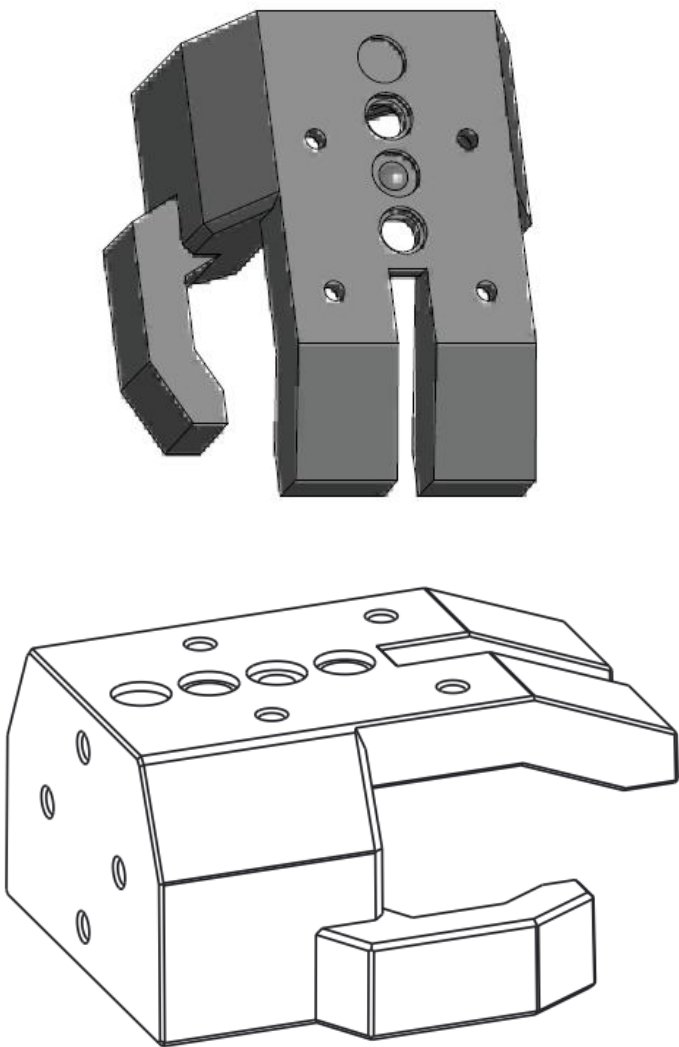
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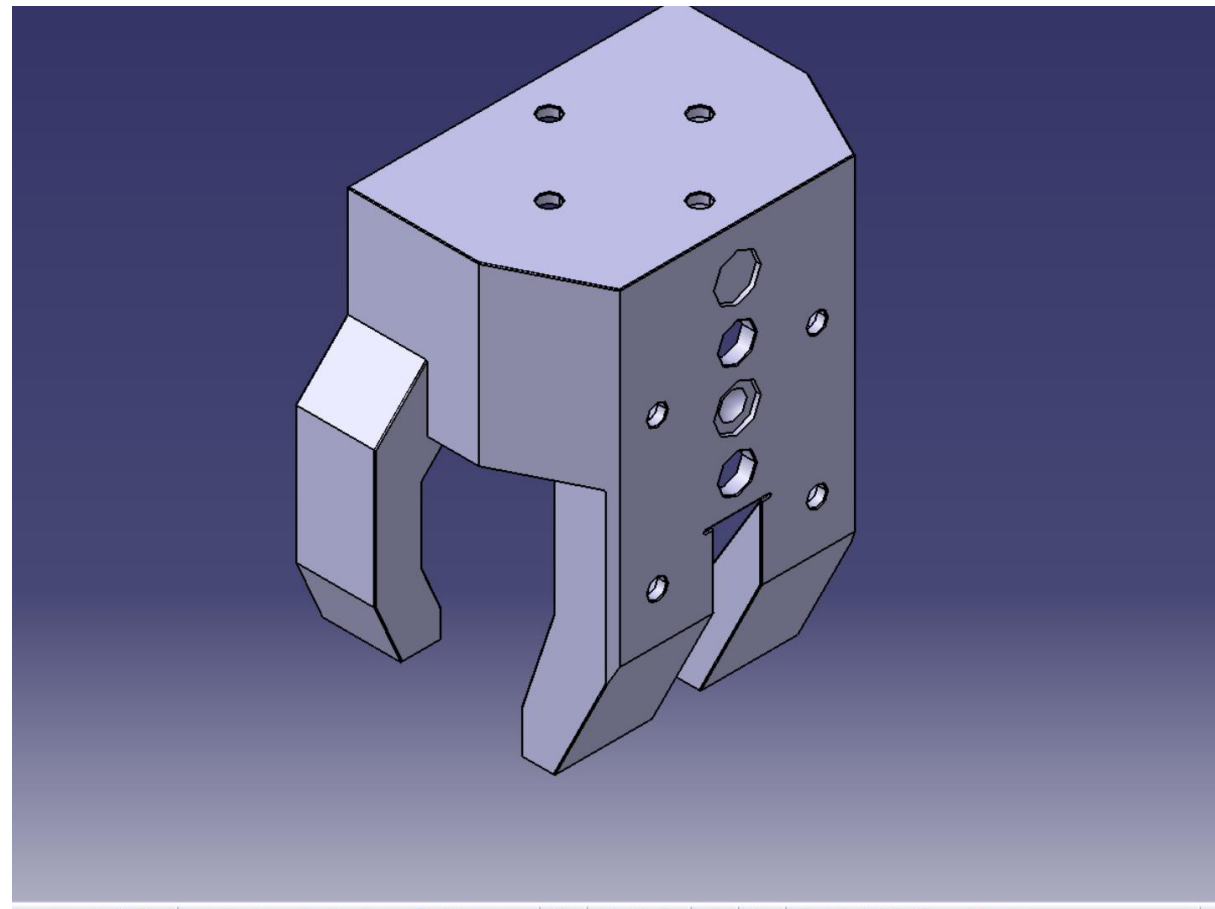
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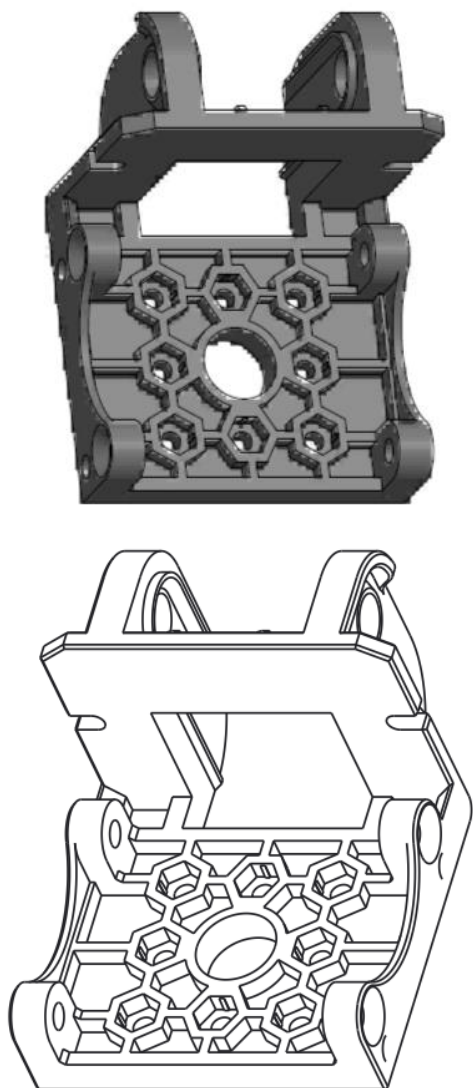
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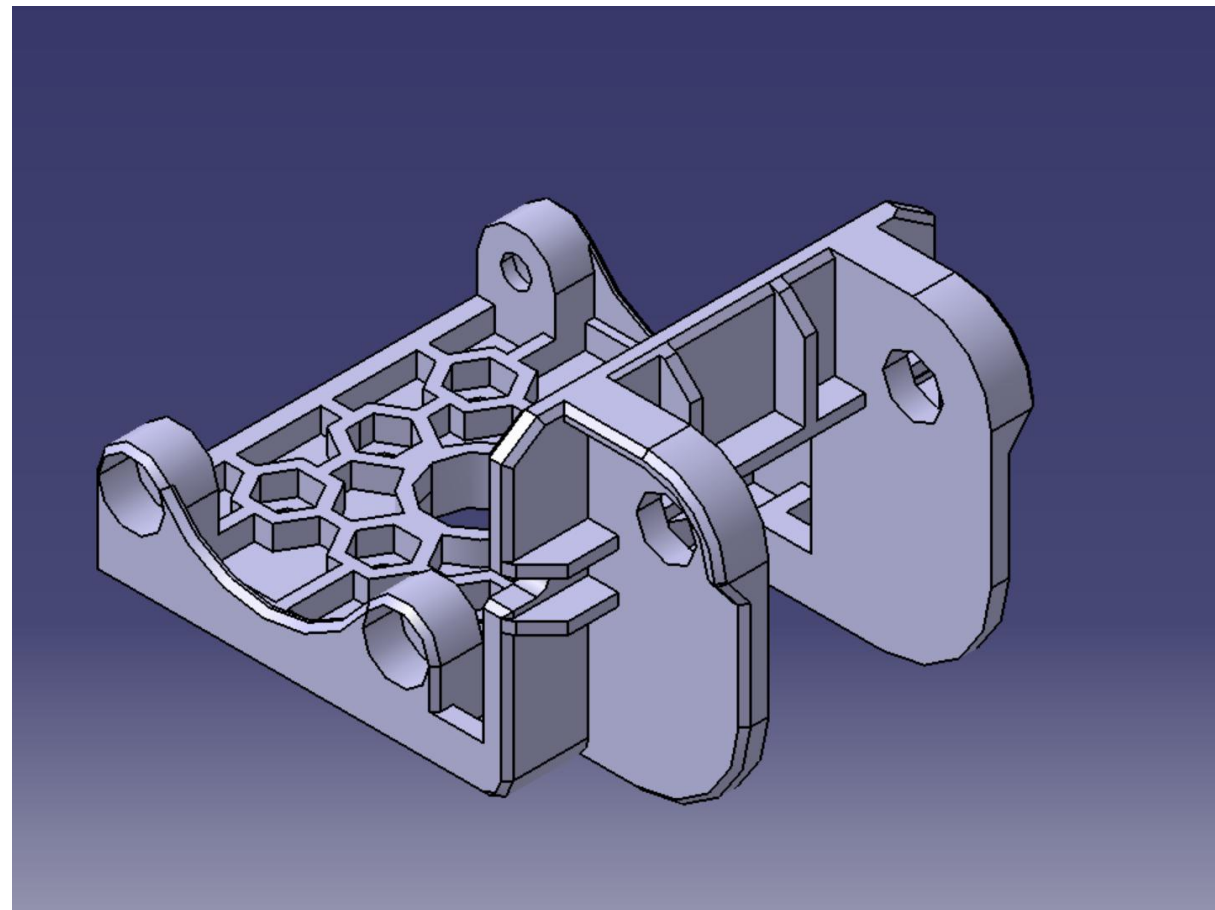
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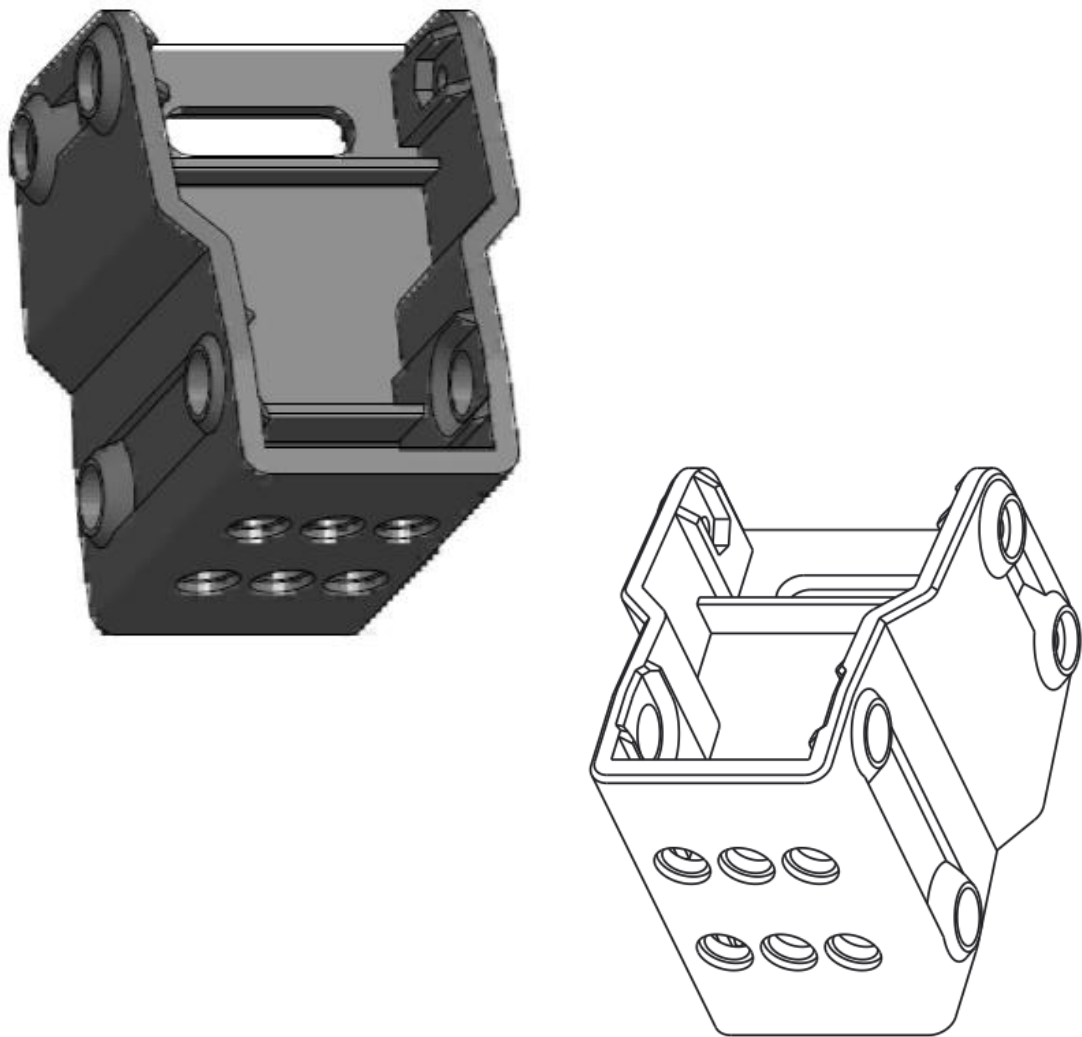
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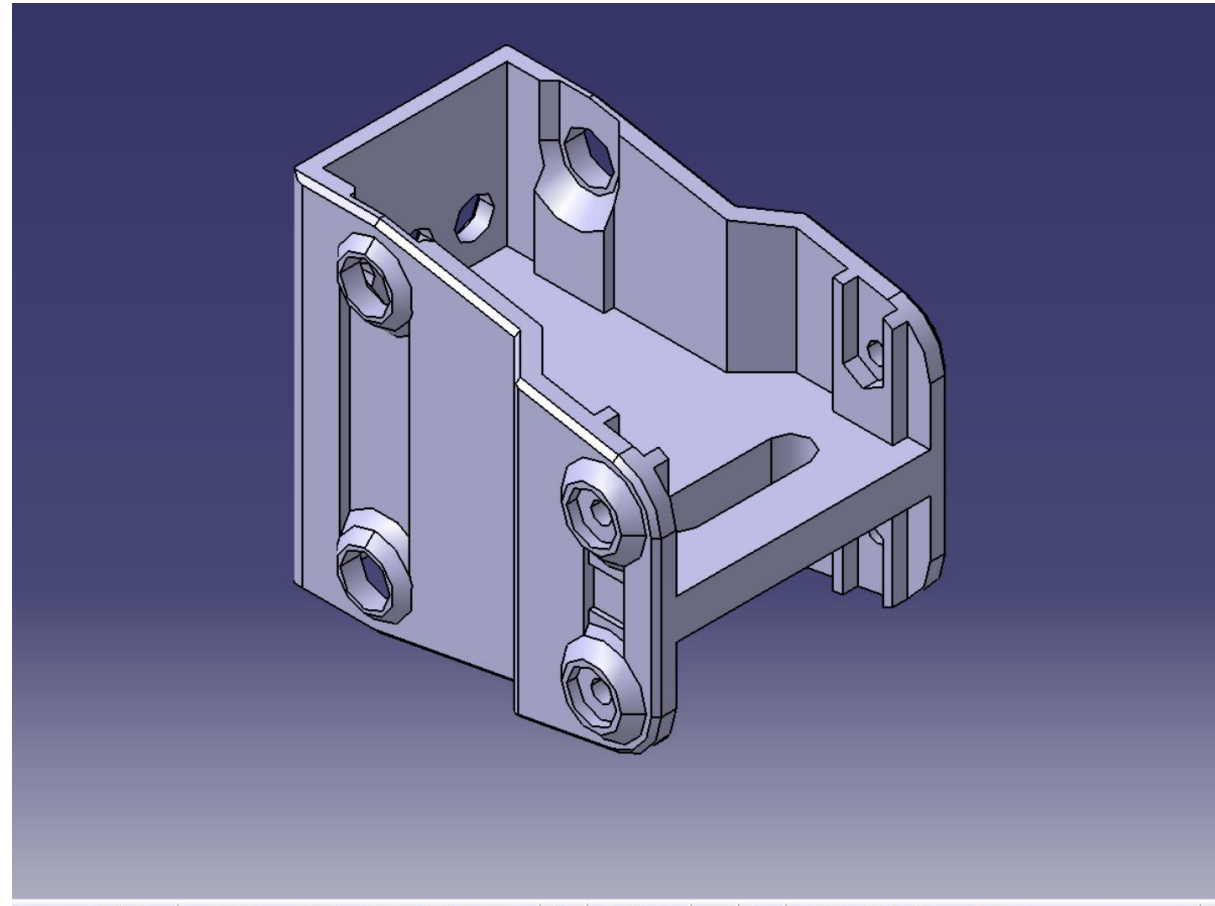
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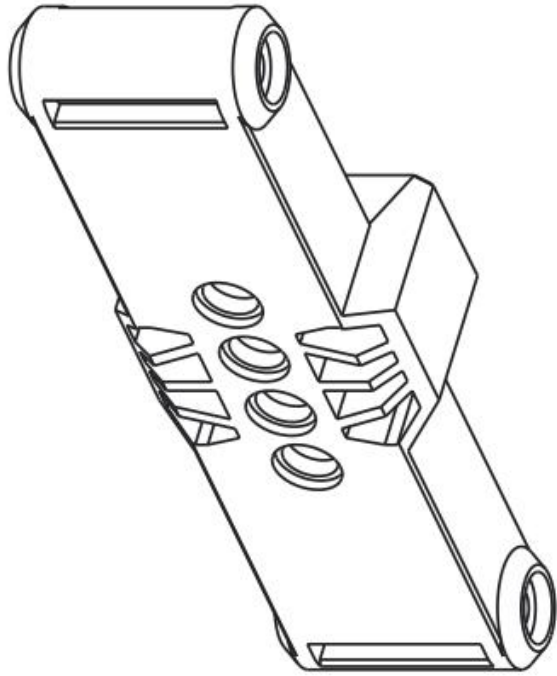
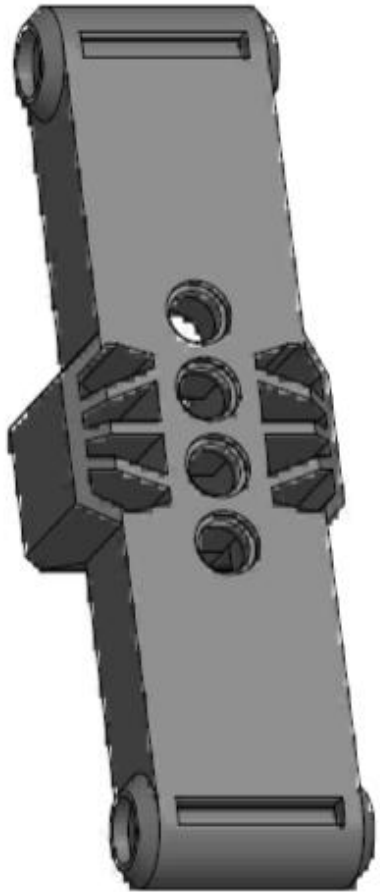
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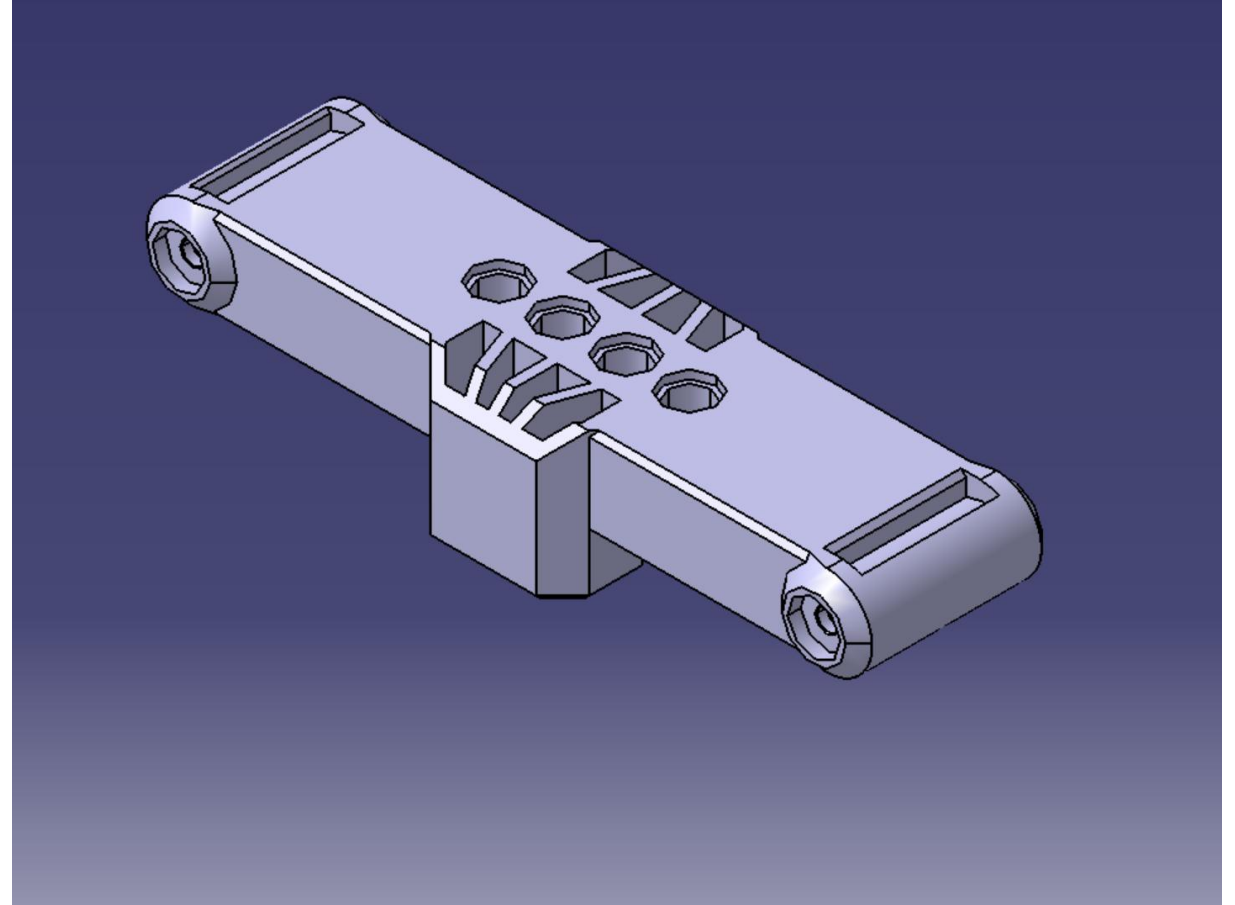
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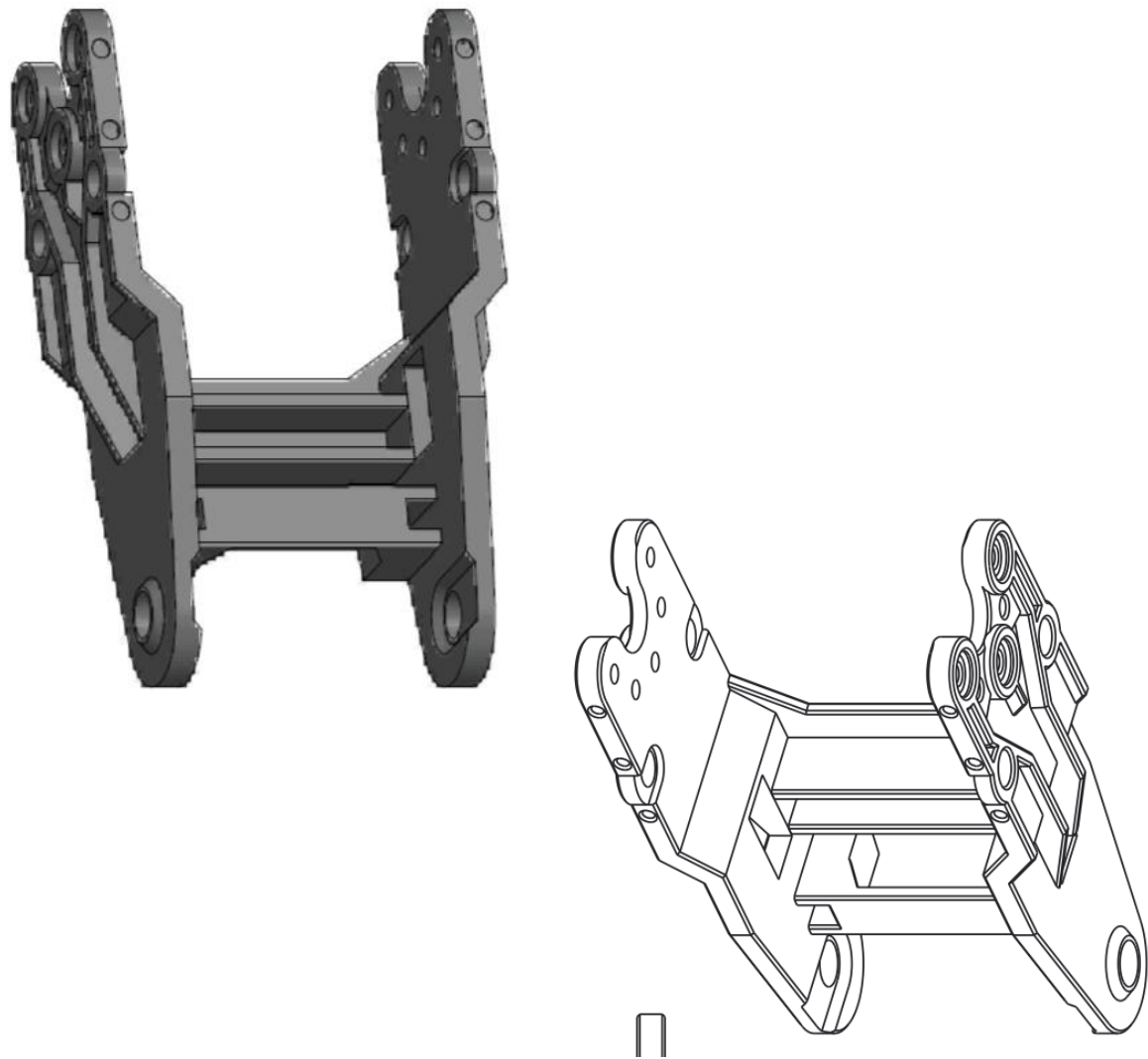
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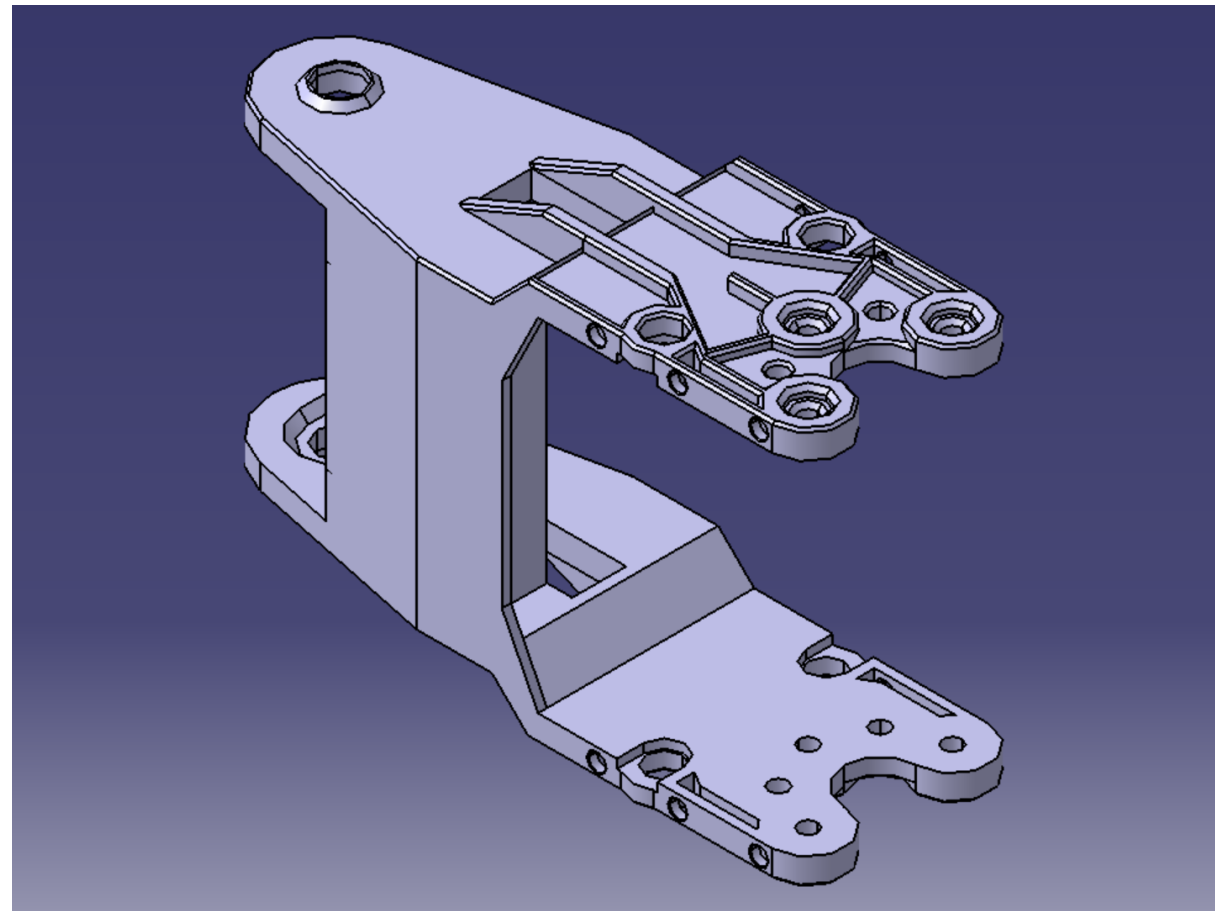
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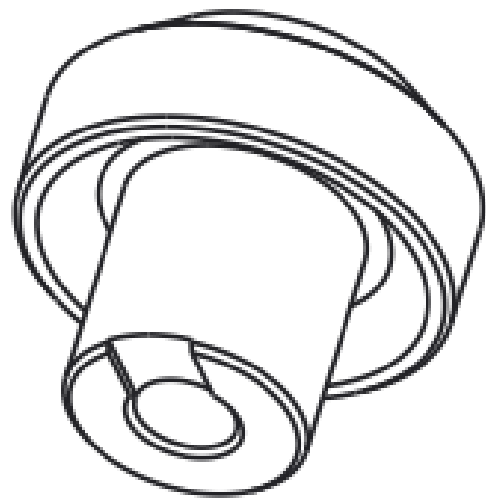
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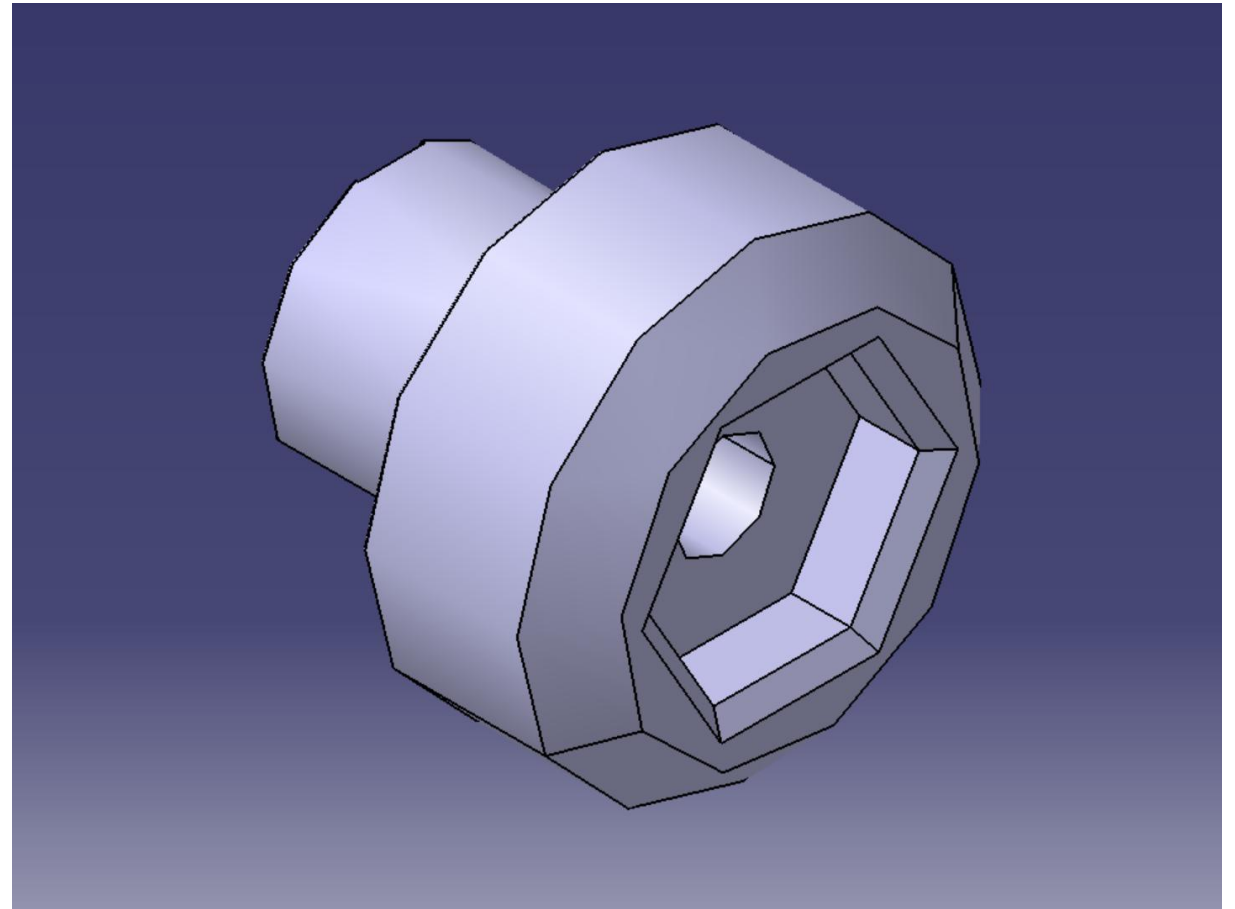
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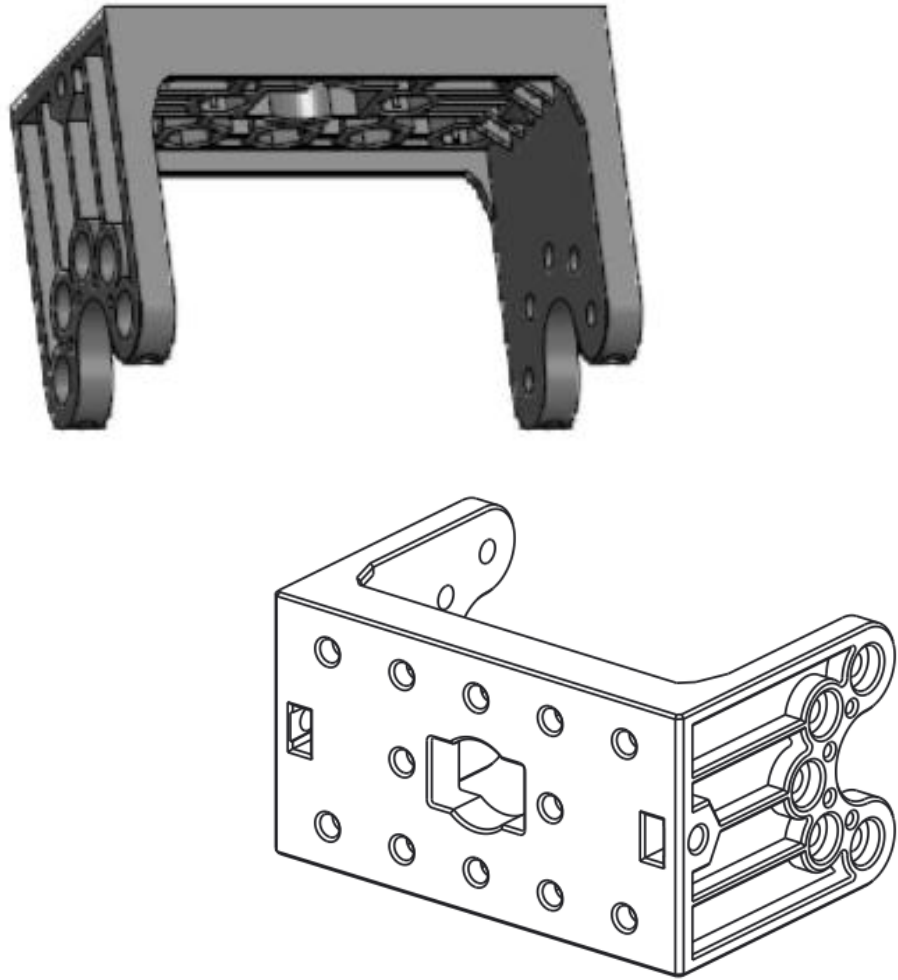
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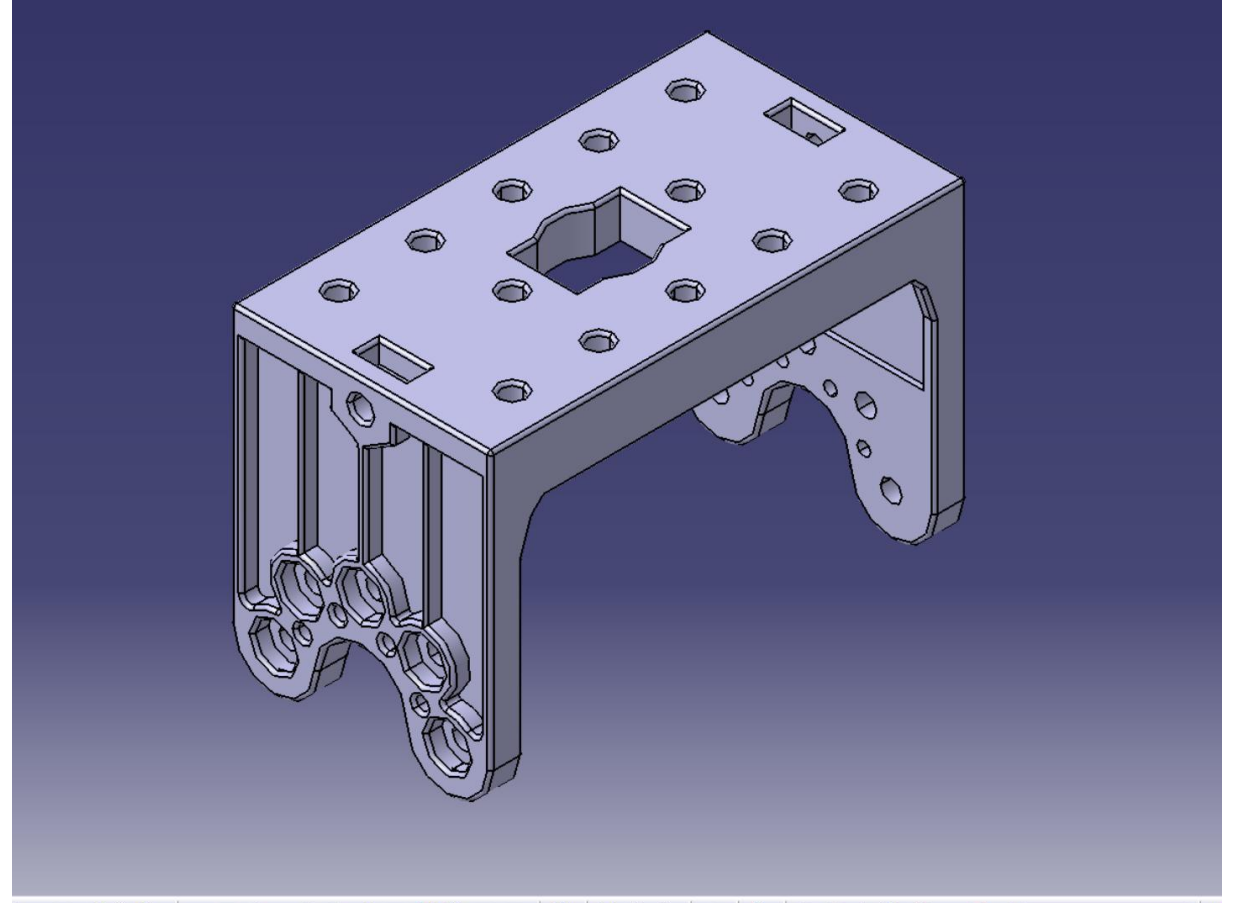
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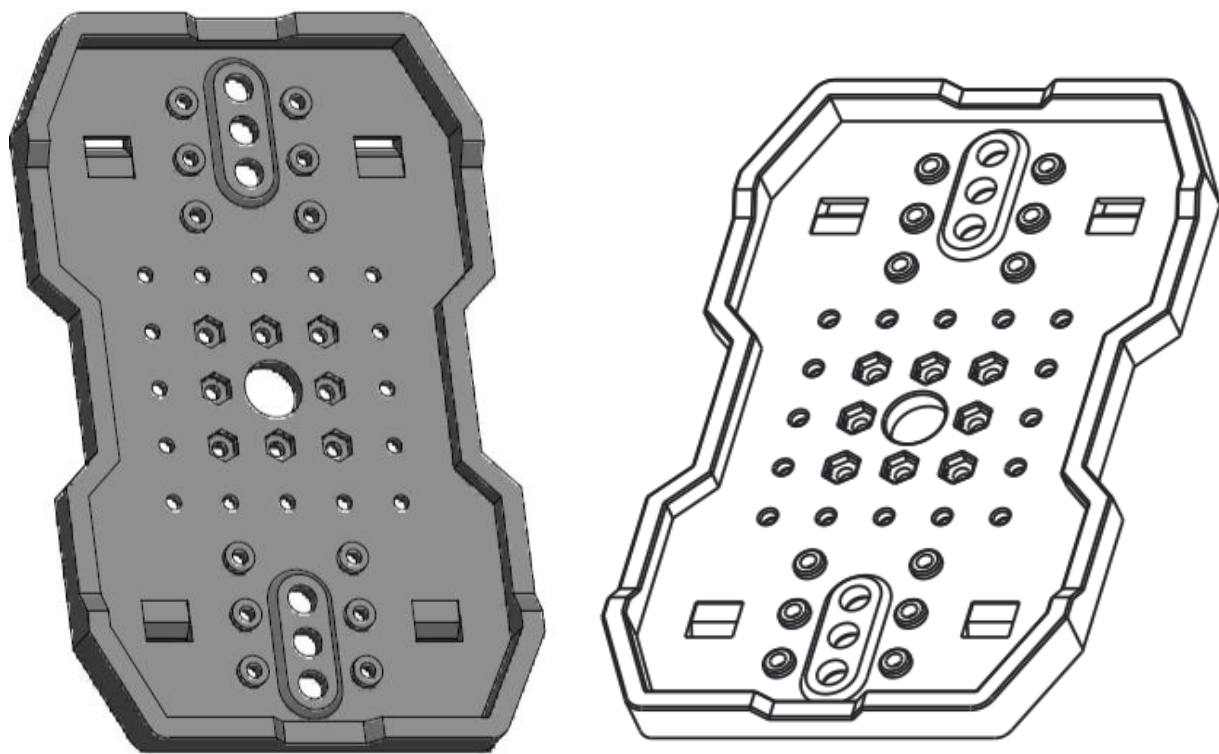
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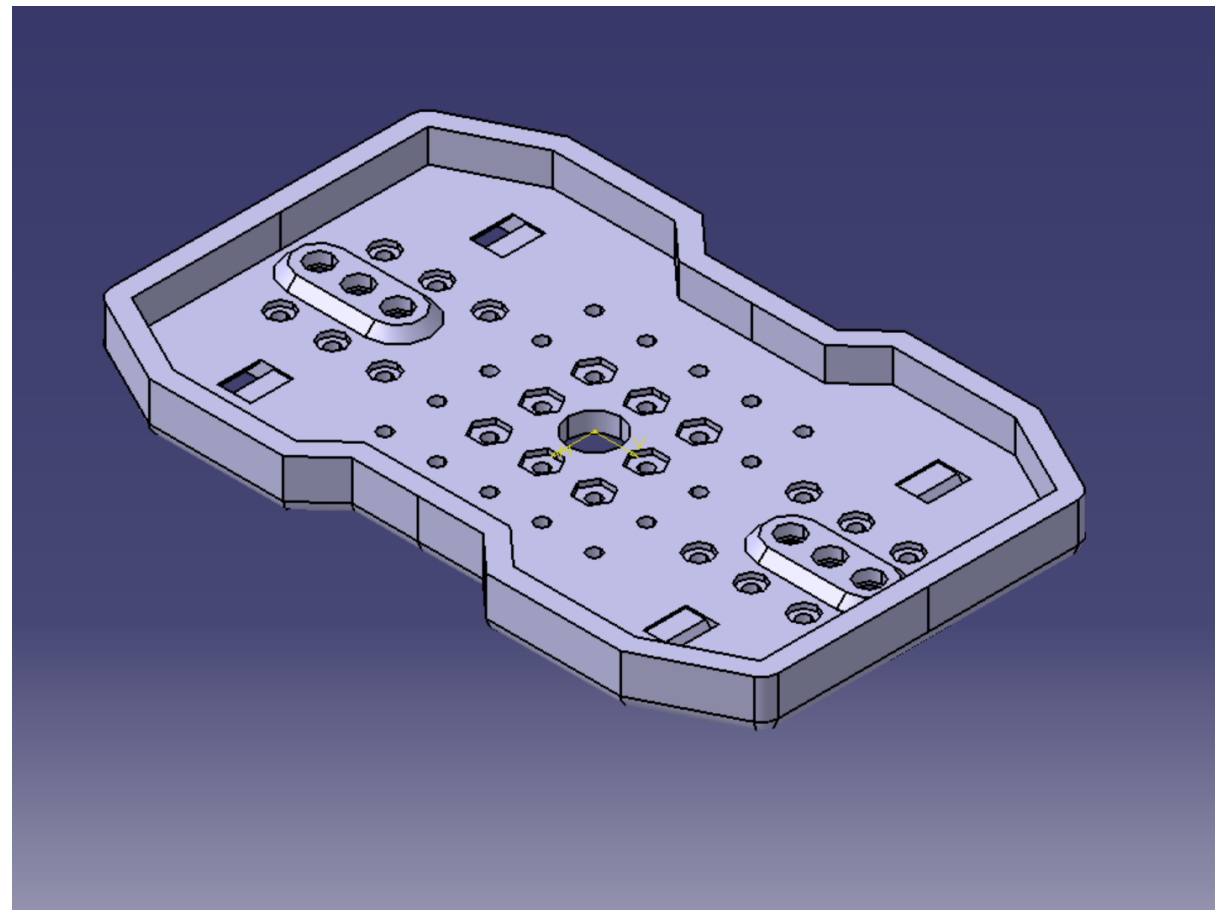
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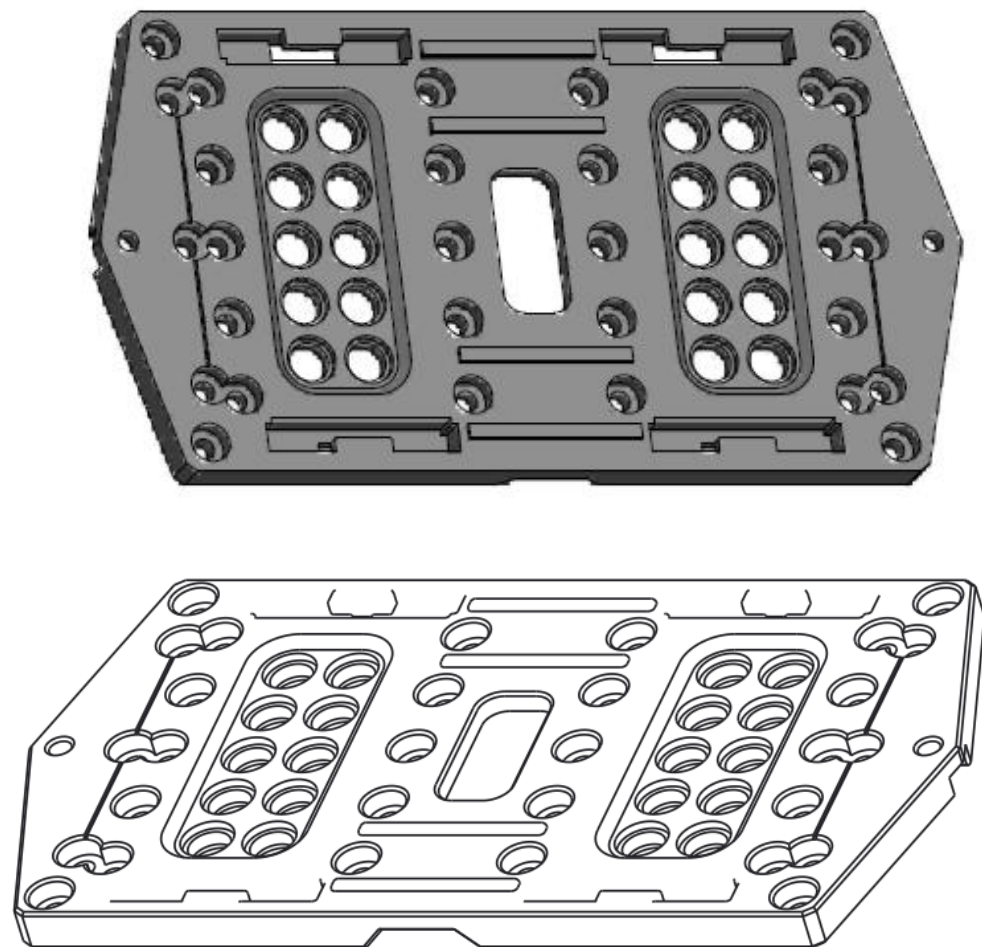
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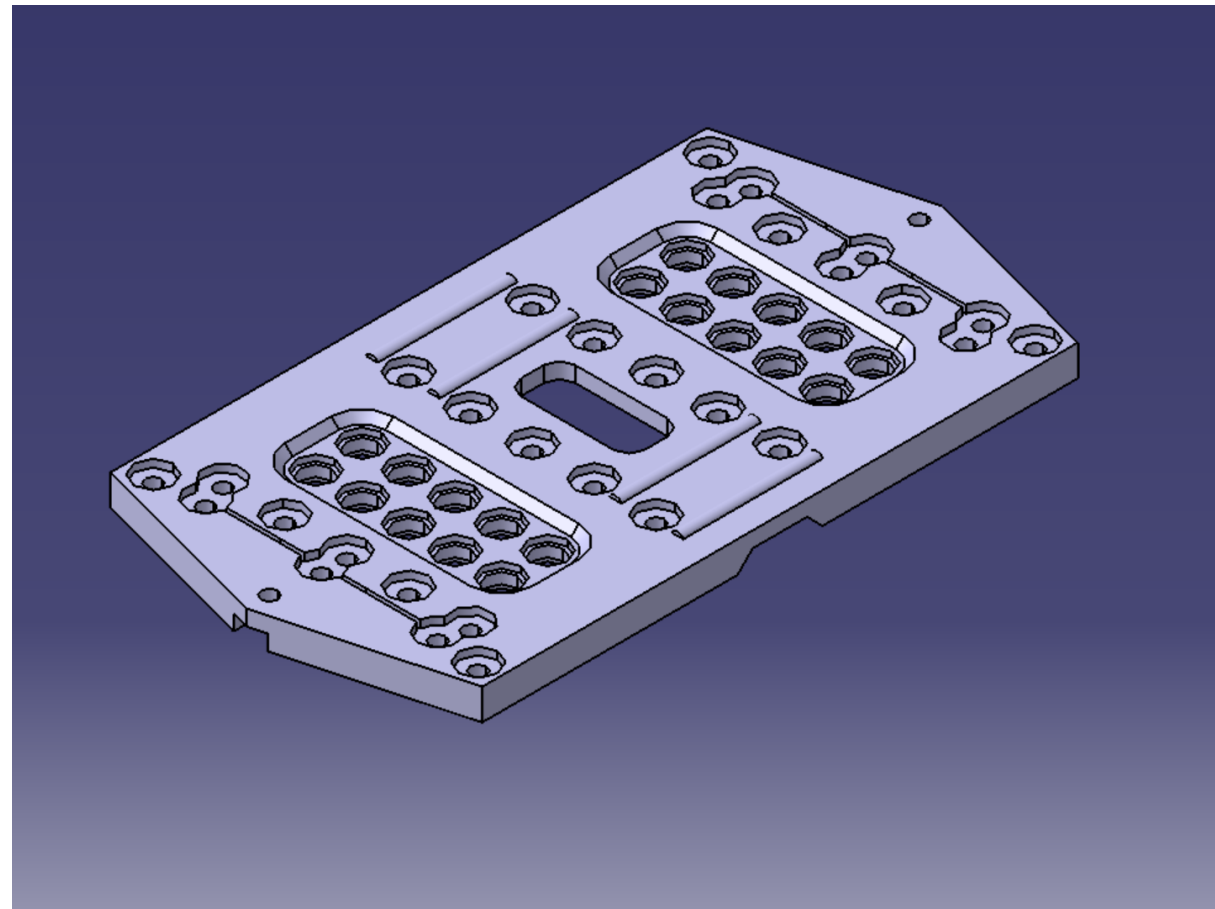
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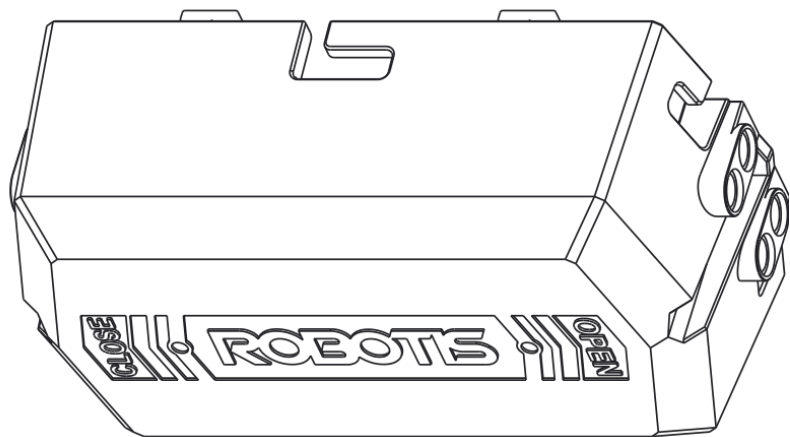
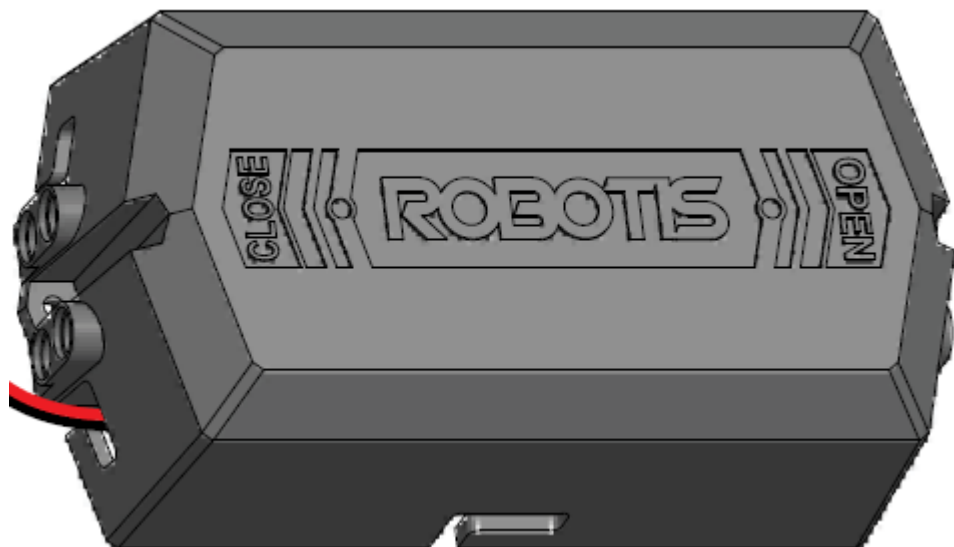
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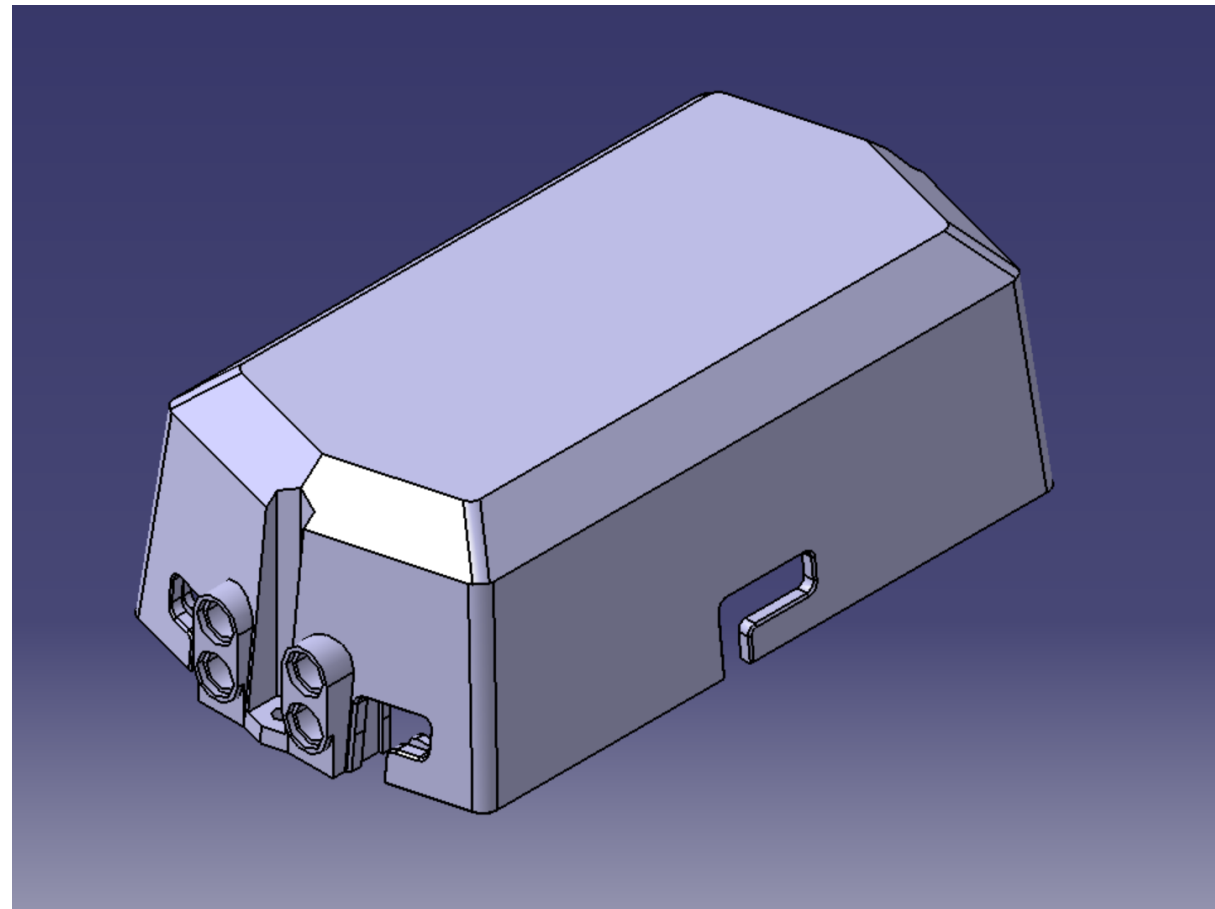
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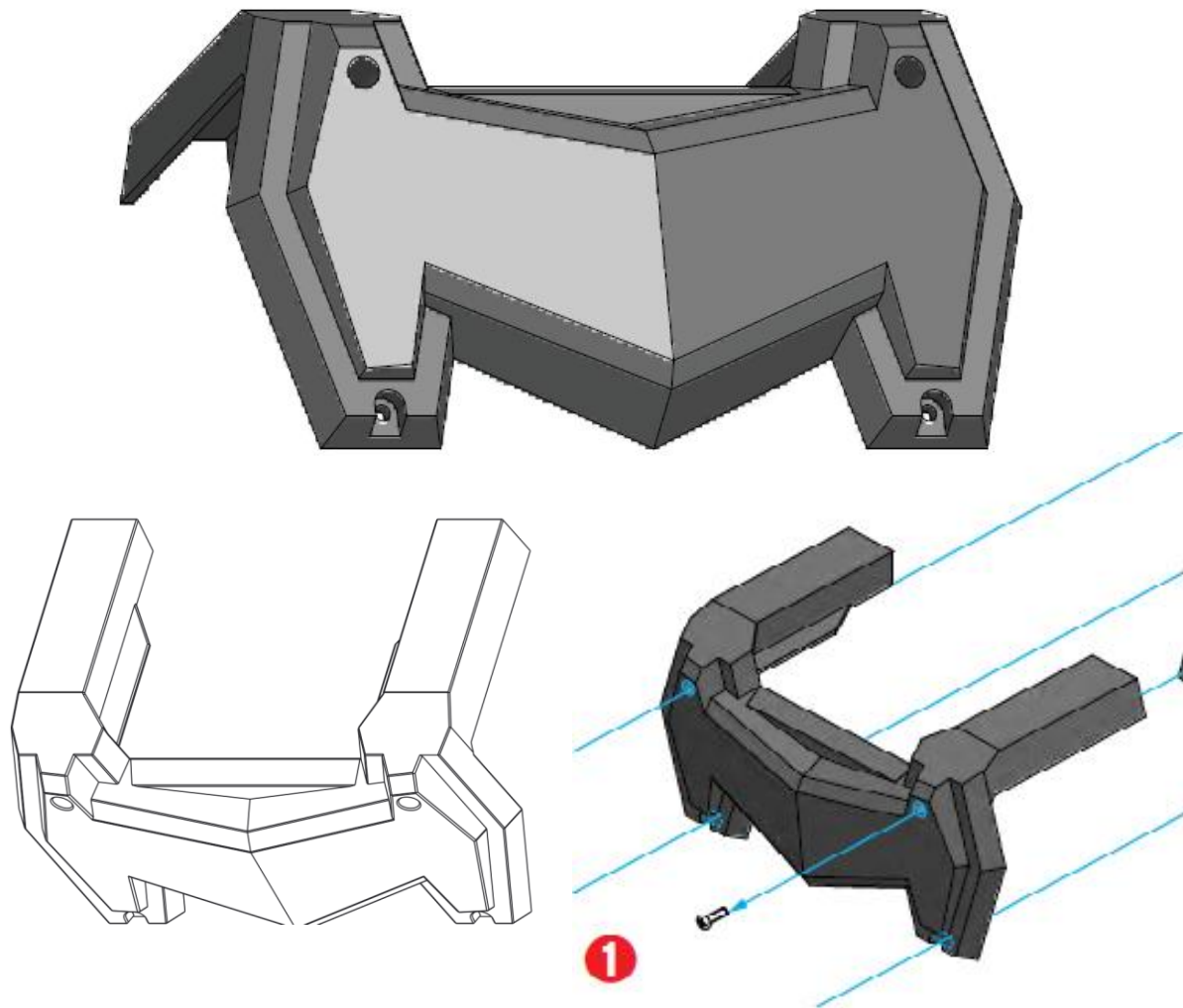
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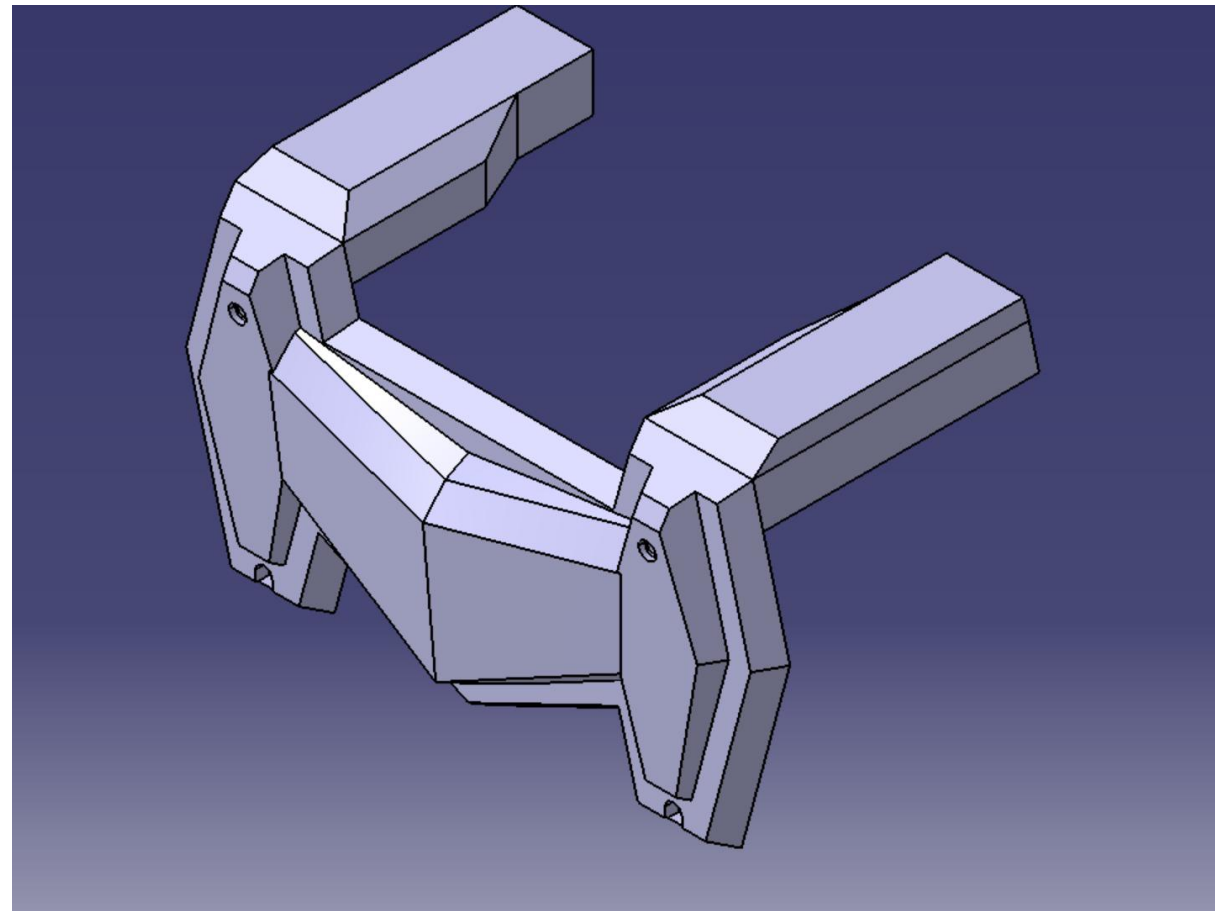
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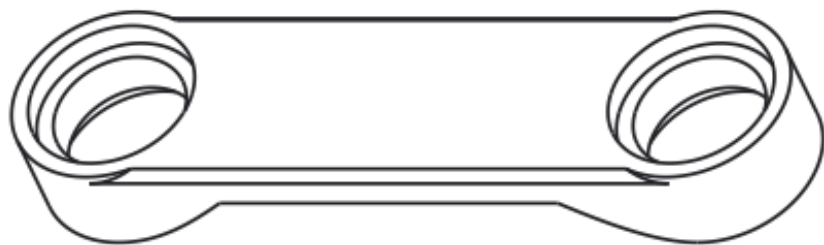
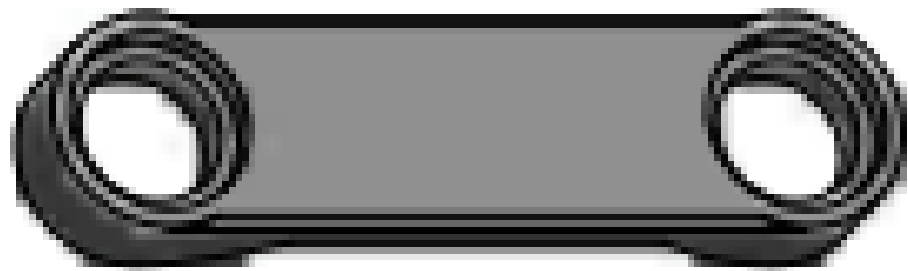
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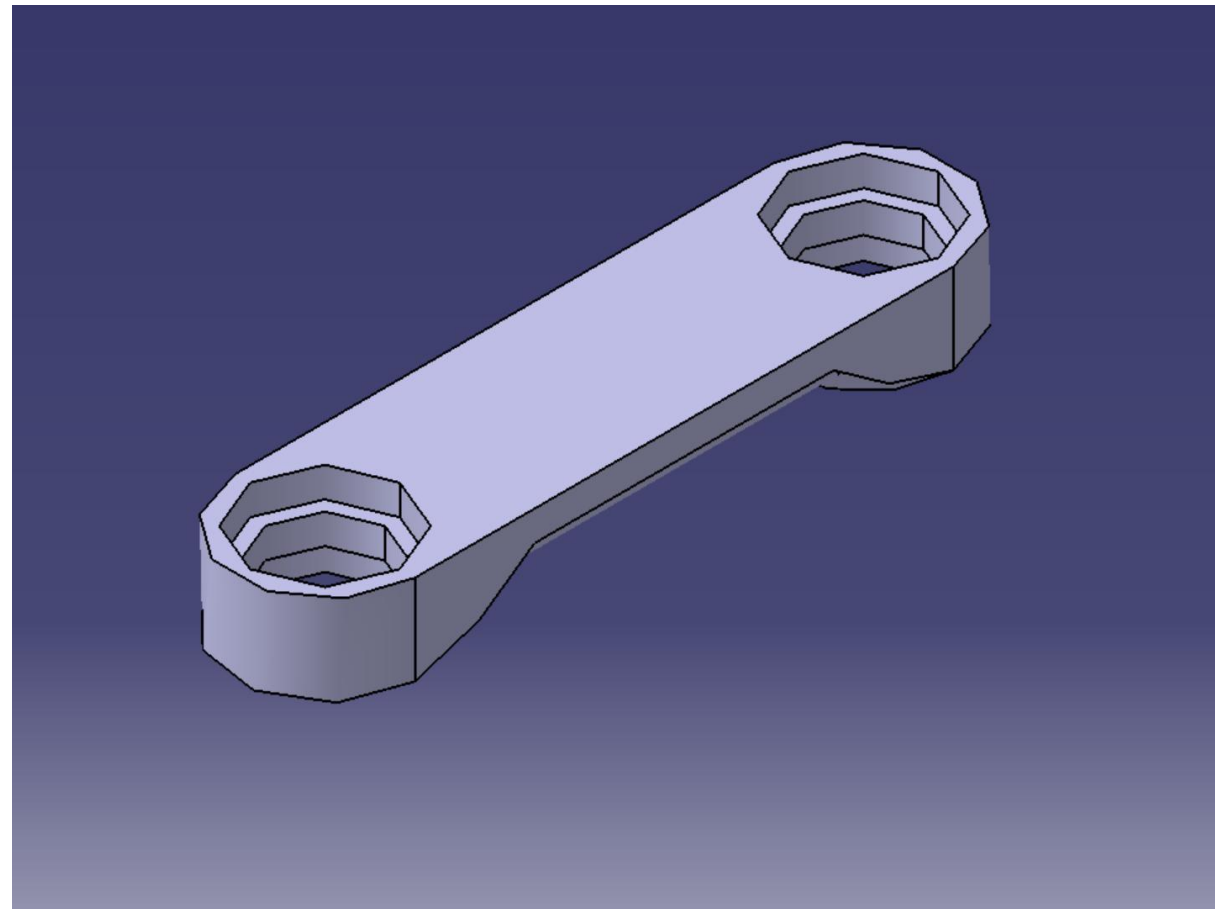
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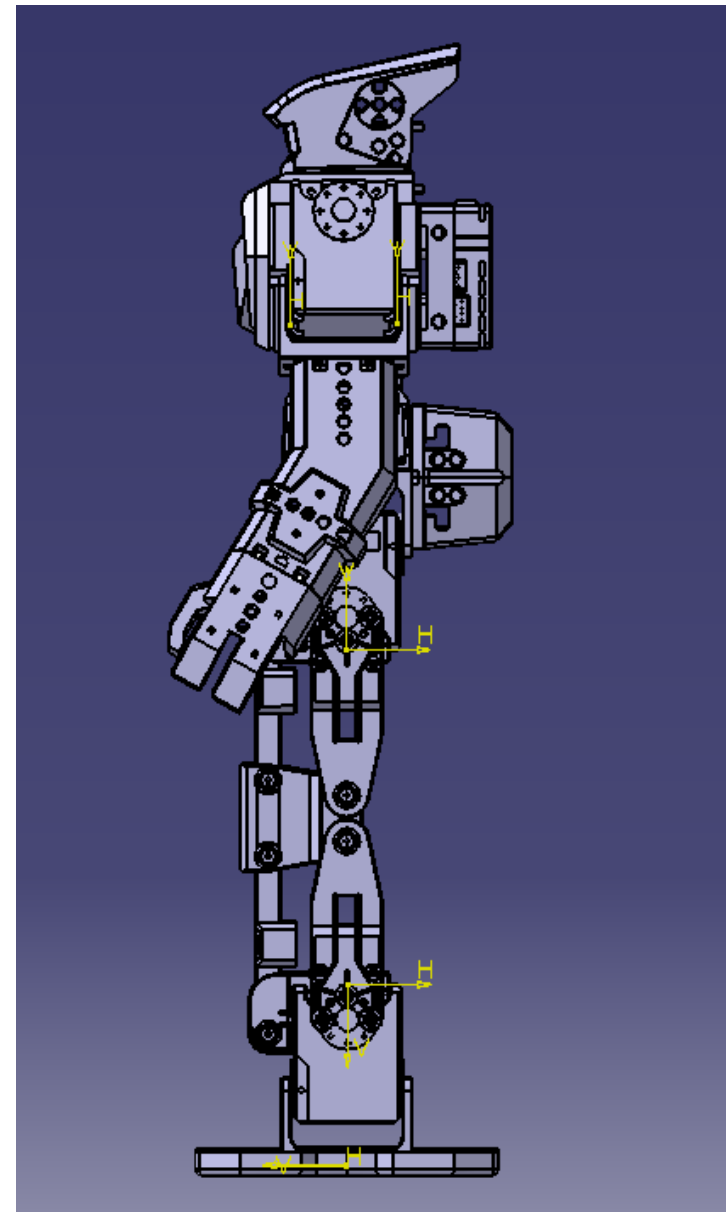
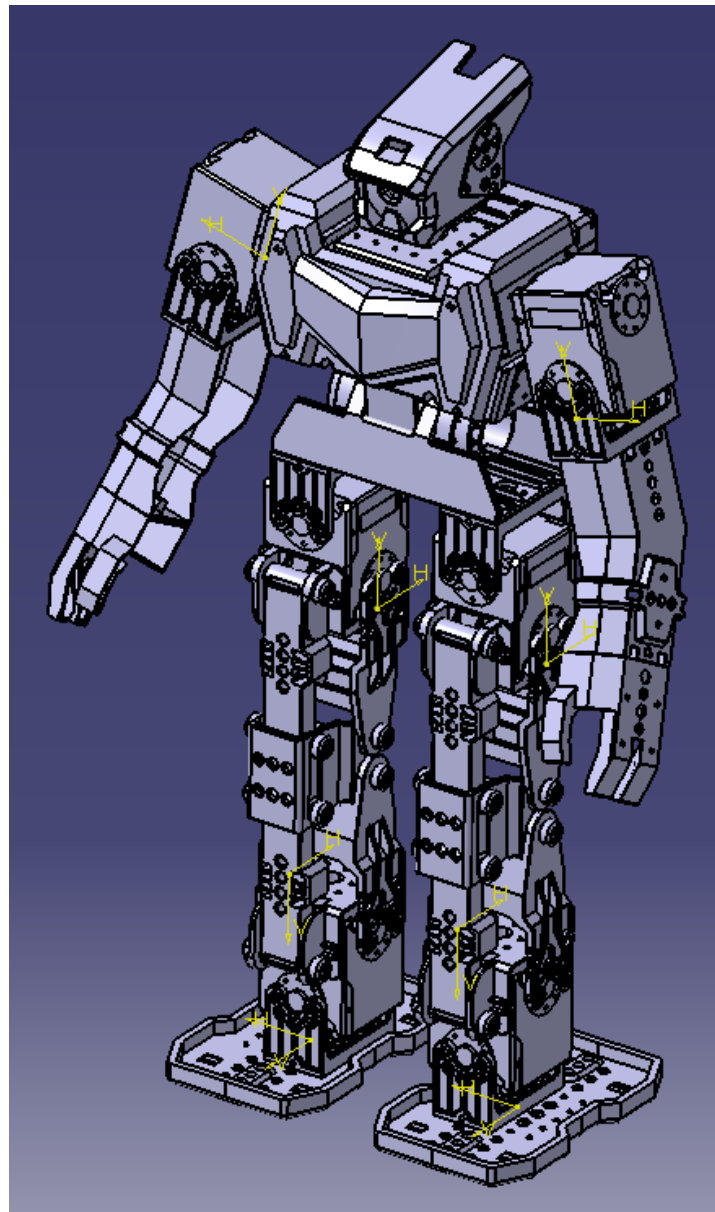
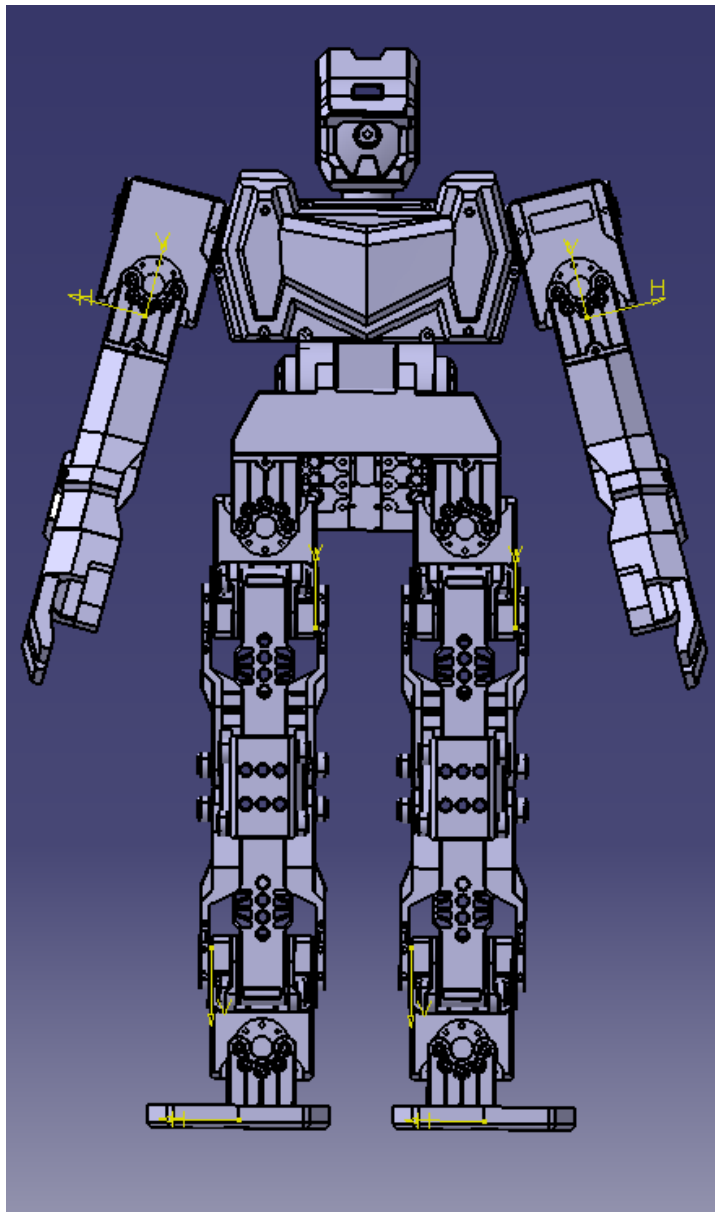
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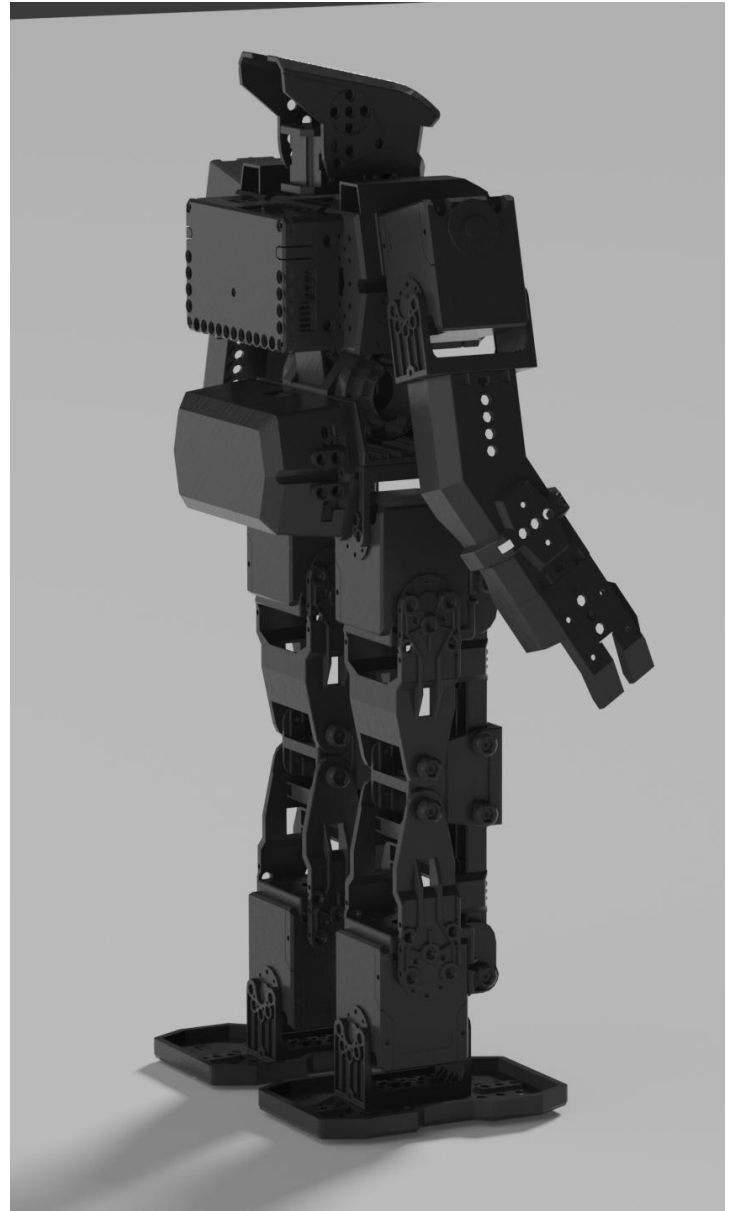
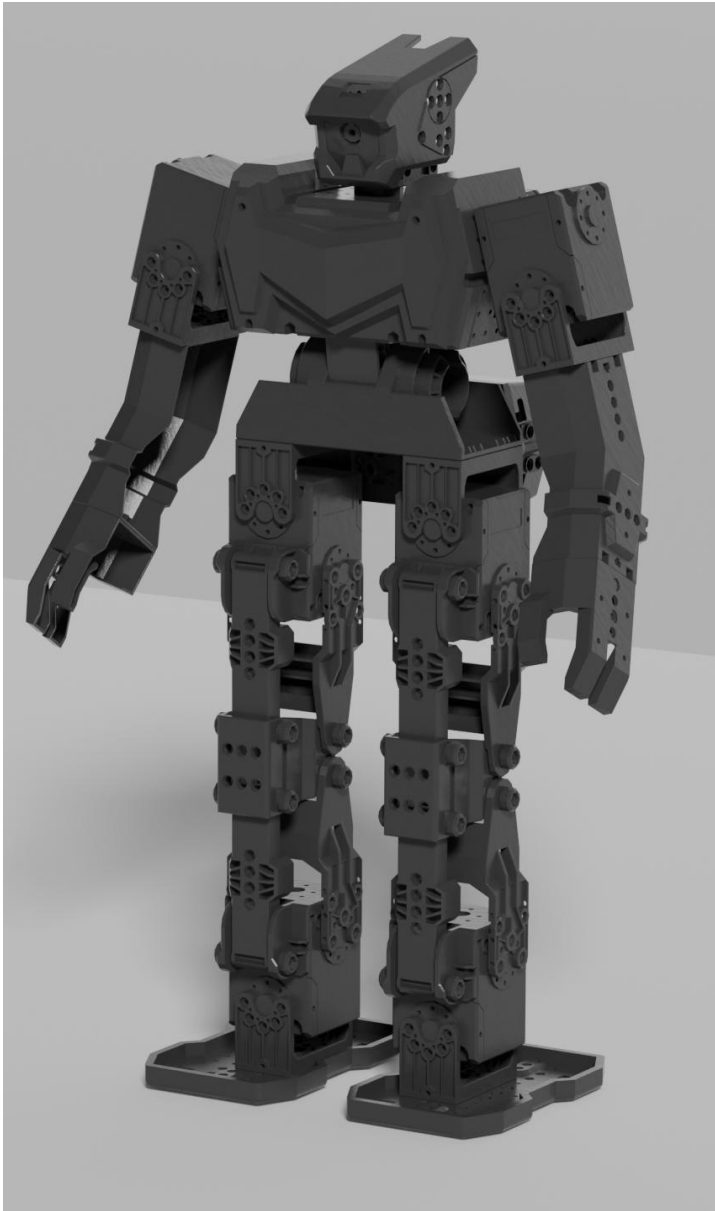
역설계



3. Results (CATIA & Rendering)



[illegible]



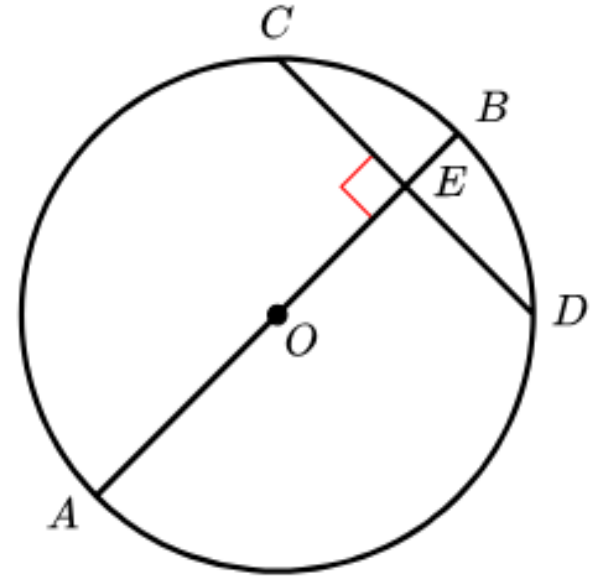
4. Crisis

Crisis

- 라운드 처리된 외곽에서 호의 중점 위치
- 복잡하고 다양한 형상
- 경사면의 각도 계산

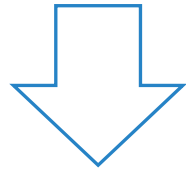
Crisis - 라운드 처리된 외곽에서 호의 중점 위치

- 현의 수직이등분선이 교차하는 점은 원의 중점임을 이용하여 호의 중점을 설정
- 너무 작아서 측정하기 어려운 경우에는 임의의 값을 준 후, 도면과 비교하여 적절히 수정

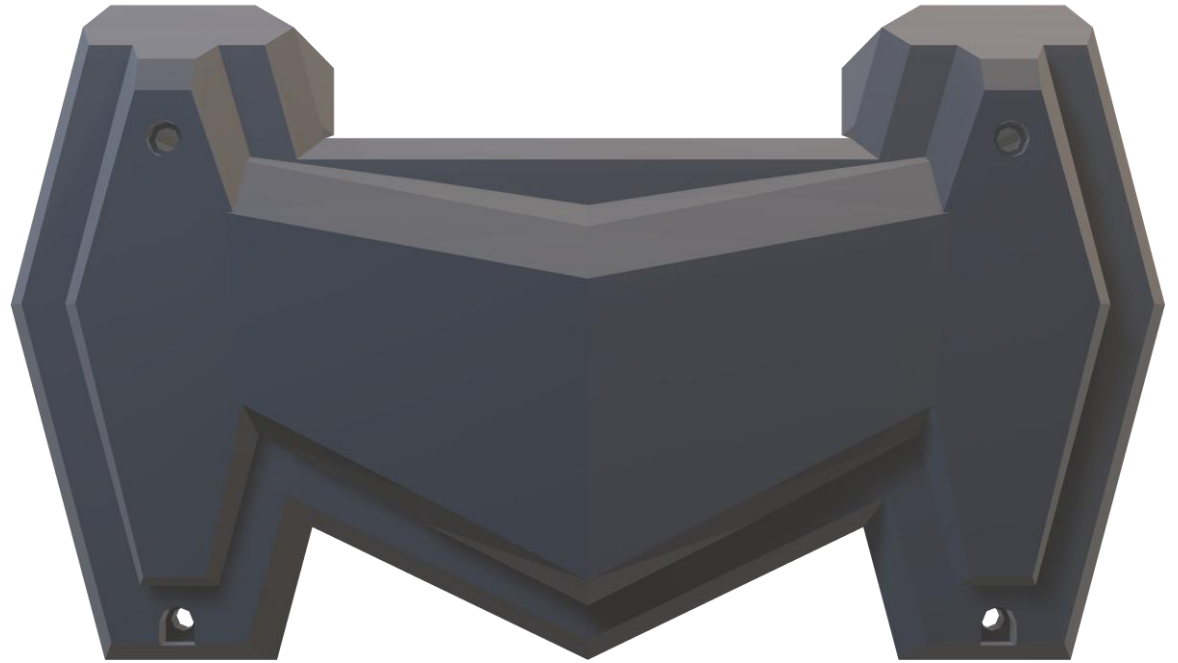


Crisis - 복잡하고 다양한 형상

형상 제작에 어려움이 예상되고 많은
시간이 소요될 것으로 예상

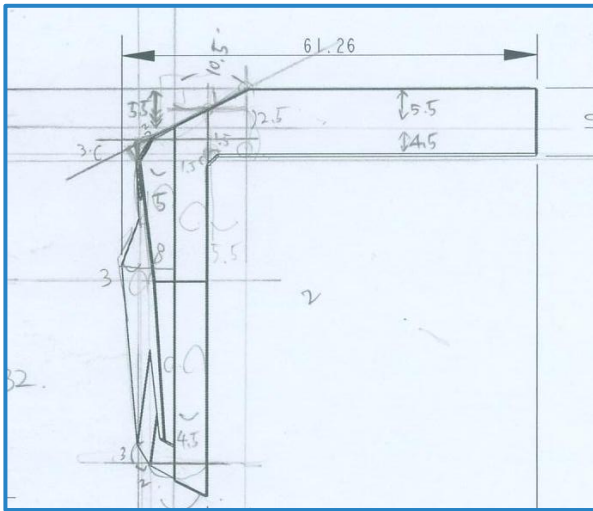


주제선정이후 바로 제작 시작

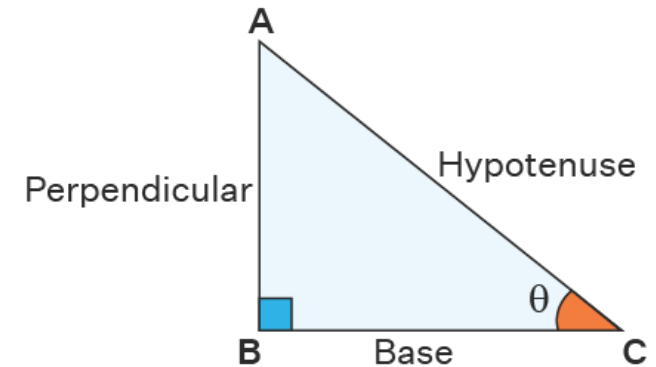


Crisis - 경사면의 각도 계산

밑변과 높이를 측정 후, 공학계산기를
이용해 기울기를 계산함



Arctan



$$\tan \theta = \frac{\text{Perpendicular}}{\text{Base}}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{\text{Perpendicular}}{\text{Base}} \right)$$

5. Takeaway

Takeaway



3D CAD 실력향상



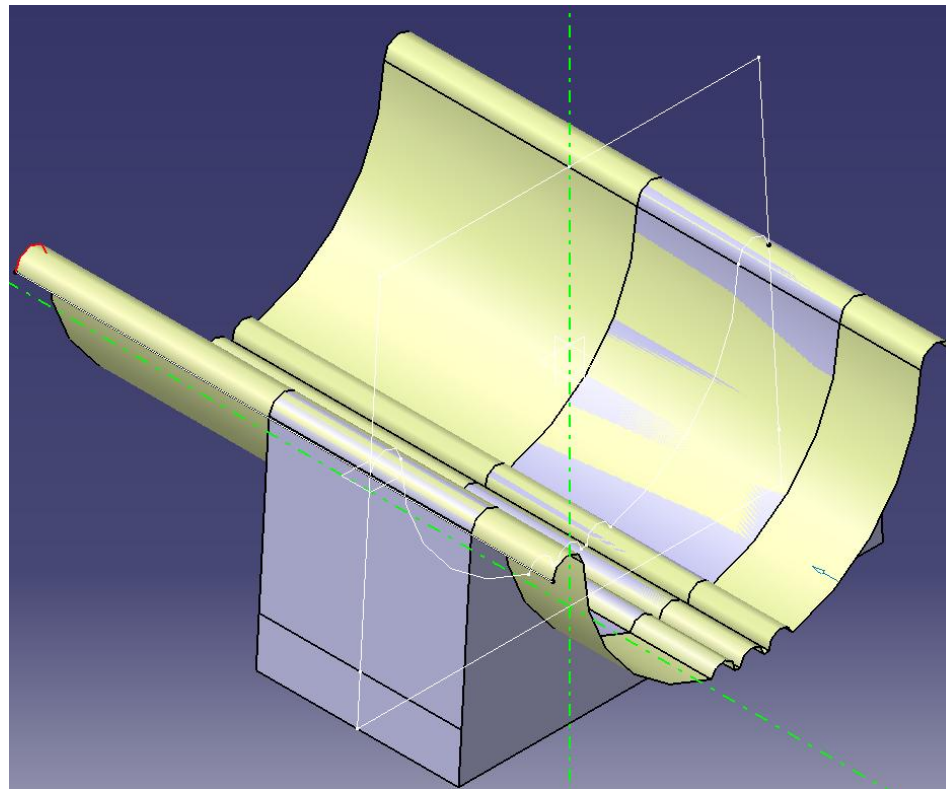
도면에 대한 이해력 향상



각 부품들 간의 긴밀한 유기성

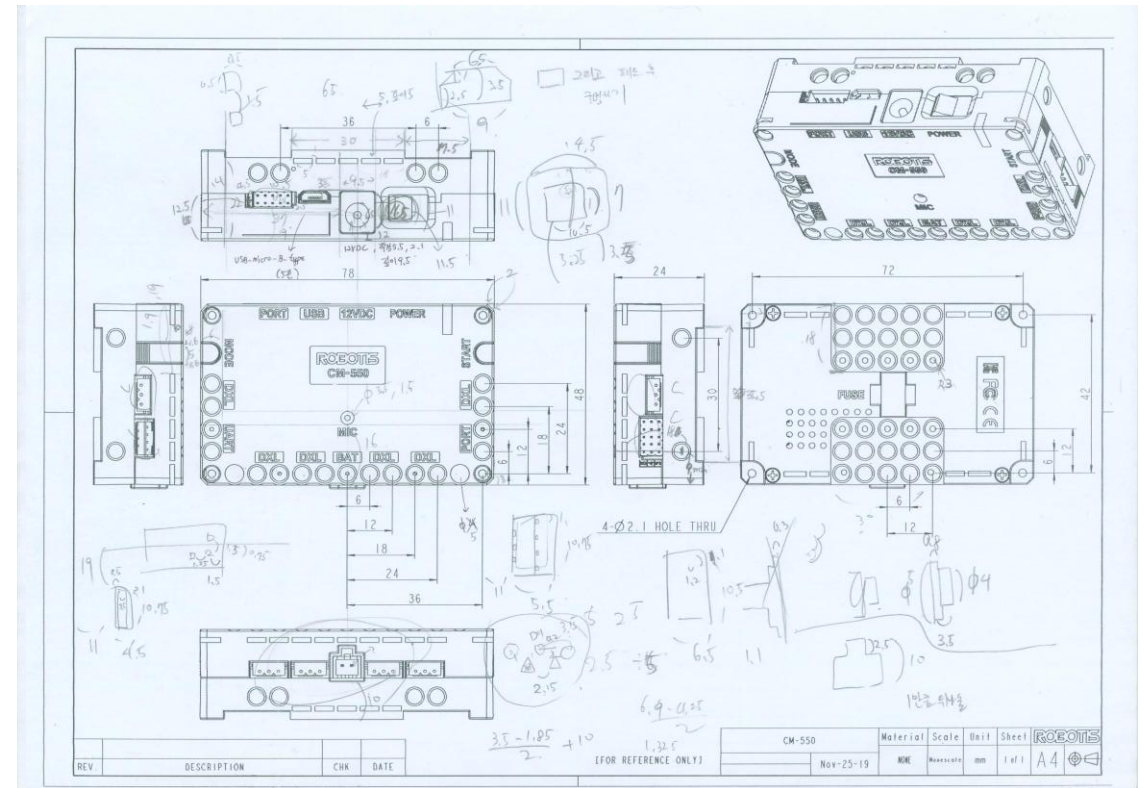
Takeaway - 3D CAD 실력향상

수업시간에 배운
모든 기능을 사용



Takeaway - 도면에 대한 이해력 향상

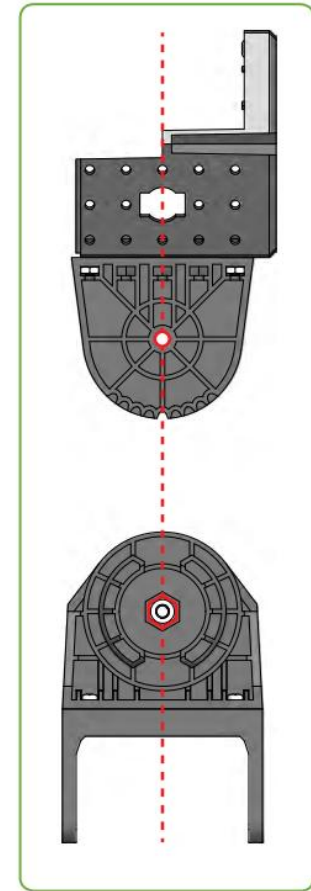
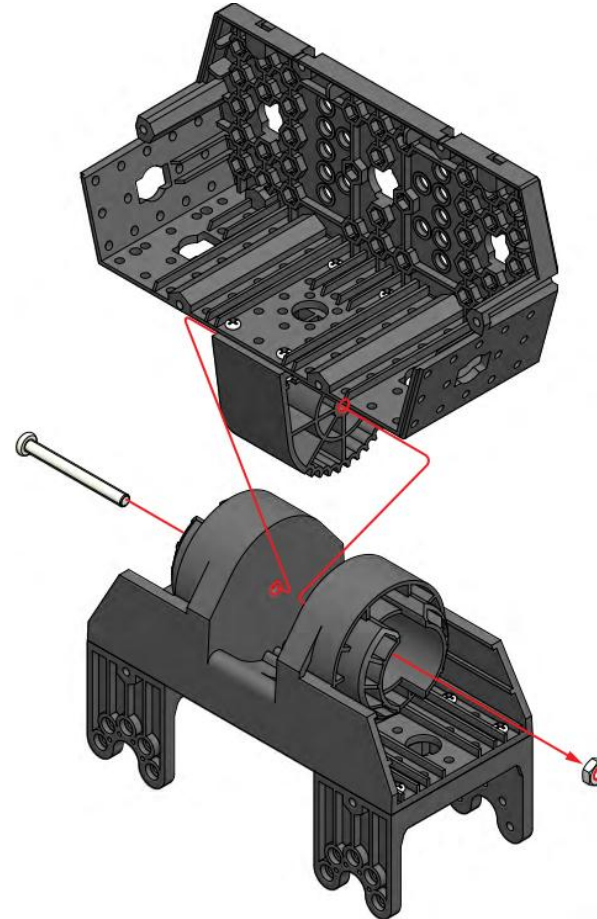
역설계를 하고, 도면을 제작하면서
도면에 대한 이해력 향상



Takeaway - 각 부품들 간의 긴밀한 유기성

모든 형상이 다른 부품들과 긴밀하게
연결되어 있음

볼트구멍과 같은 **연결부위는 제작에
주의가 필요함**을 느낌



6. Conclusion

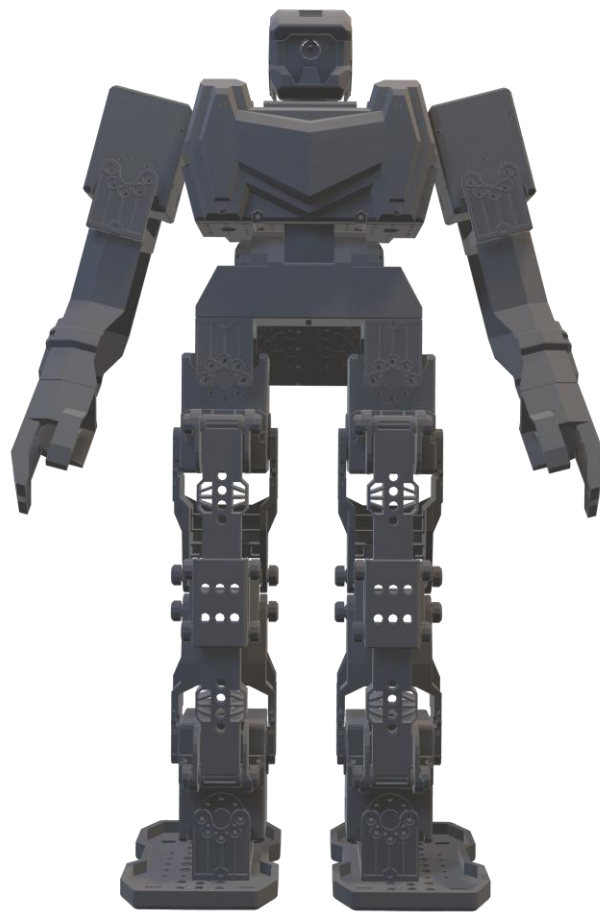
Conclusion - 유찬영

- 다양한 기능을 사용하면서 캐드실력이 많이 향상된 것이 체감되어 유익한 경험이었다고 생각한다
- 생각보다 치수들이 깔끔하게 떨어져서 놀랐다
- 전체적으로 형상이 각진 모양이라 곡면의 형상을 연습하기에는 부족했다고 느꼈다

Conclusion - 양민석

- 카티아 프로그램의 장점을 이번 프로젝트를 통해 체감할 수 있었다
- 역설계를 진행하면서 도면에 대한 이해도가 향상되었다
- 부족한 실력이었지만 프로젝트를 진행하며 실력향상이 되어 성취감을 느꼈다

7. Q&A



Thank you